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Dockery

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(54) **AIRFIELD SYSTEMS, DEVICES, AND METHODS**

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Related U.S. Application Data

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(60) Provisional application No. 63/263,843, filed on Nov. 10, 2021.

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F24F 9/00 (2006.01)
F24F 8/10 (2021.01)

(52) **U.S. Cl.**
CPC .. **F24F 9/00** (2013.01); **F24F 8/10** (2021.01)

(58) **Field of Classification Search**
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F24F 9/00; F24F 8/10; F24F 8/108; F24F
13/28
USPC 55/385.1, 385.2, 471-473
See application file for complete search history.

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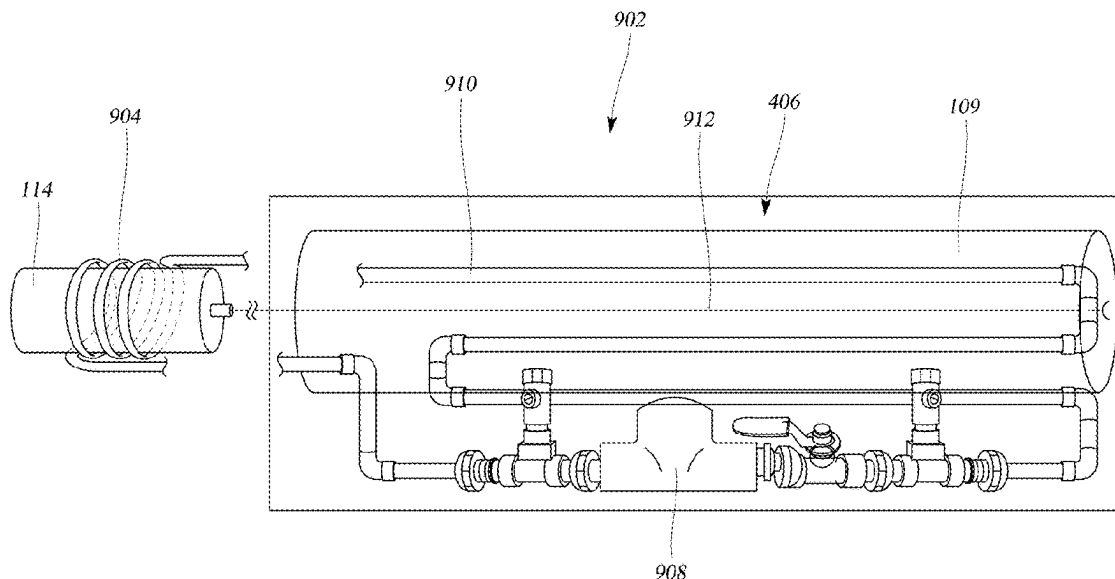
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(57) **ABSTRACT**

Various systems, devices, and methods disclosed herein relate to airfield systems are disclosed. Some embodiments relate to generating airfield barriers with purified air, devices for generating air fields, methods of using such devices, and methods for manufacturing such devices.

20 Claims, 33 Drawing Sheets



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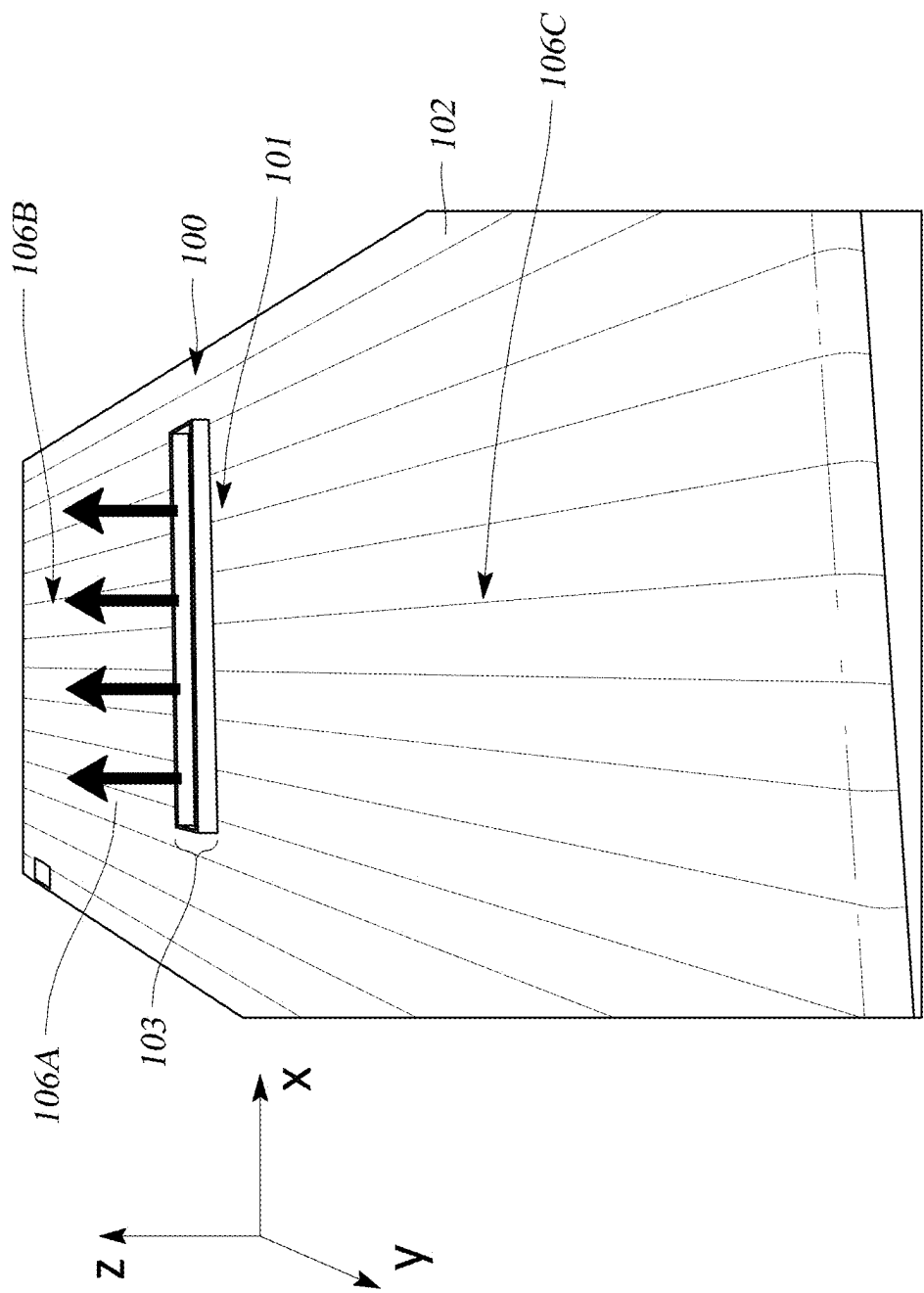


FIG. 1A

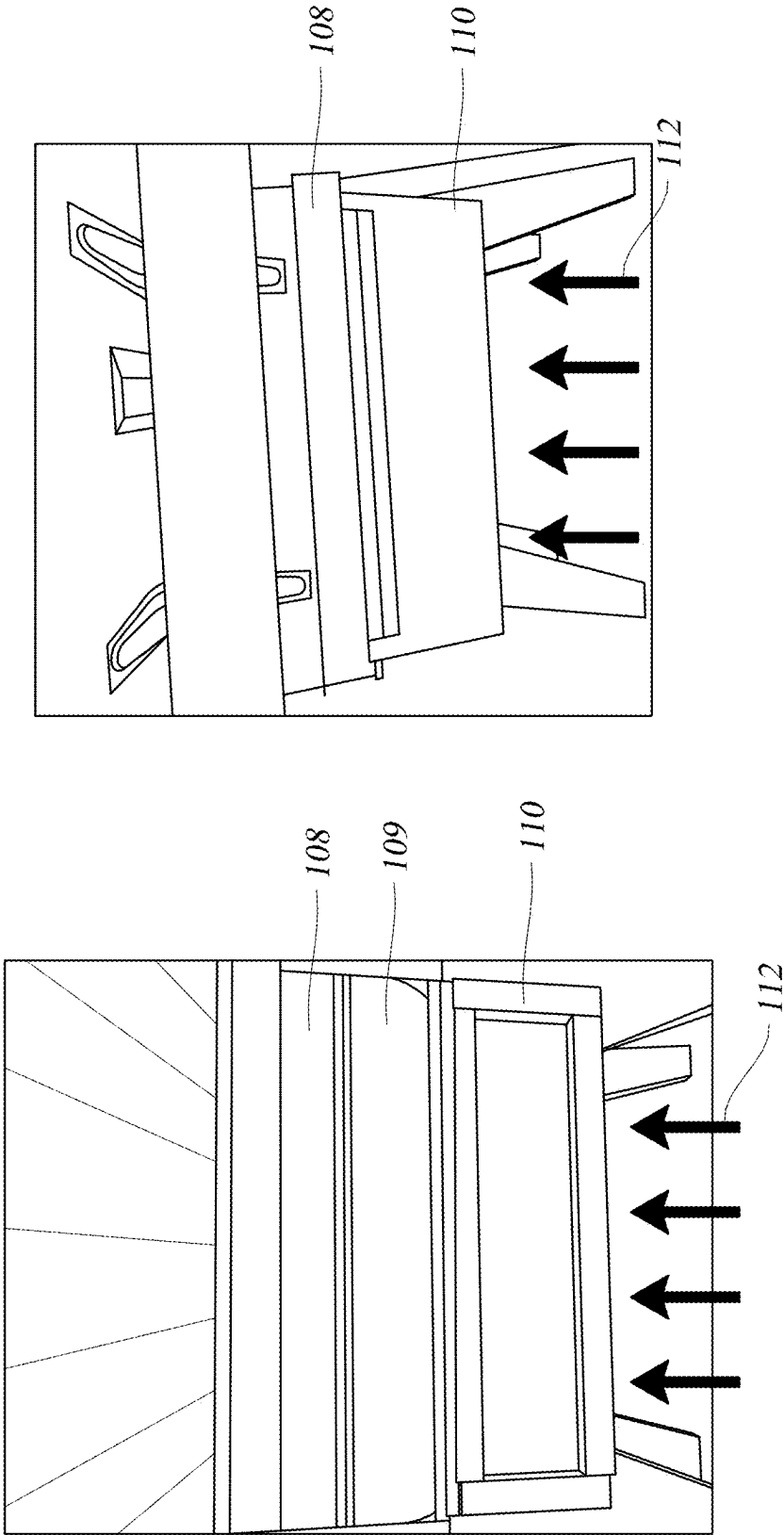


FIG. 1A

FIG. 1B

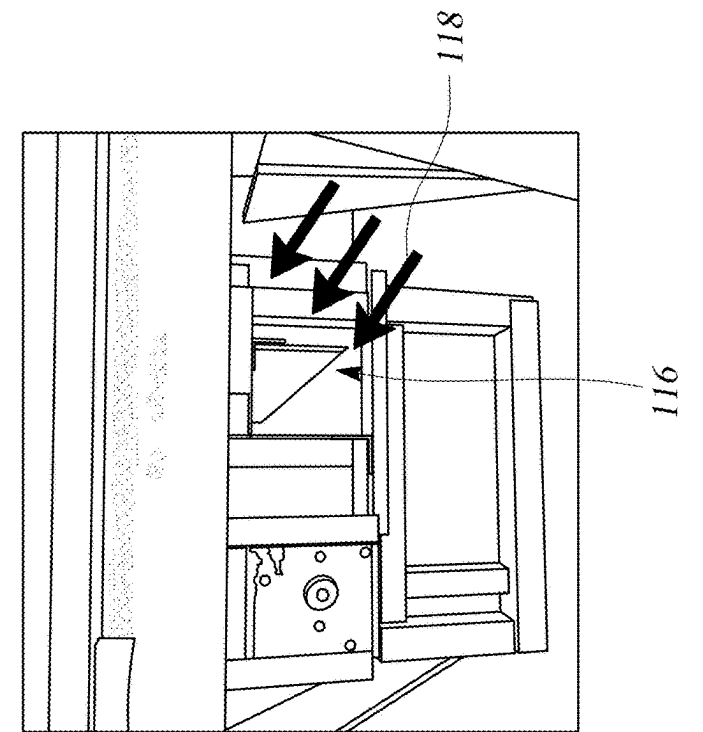
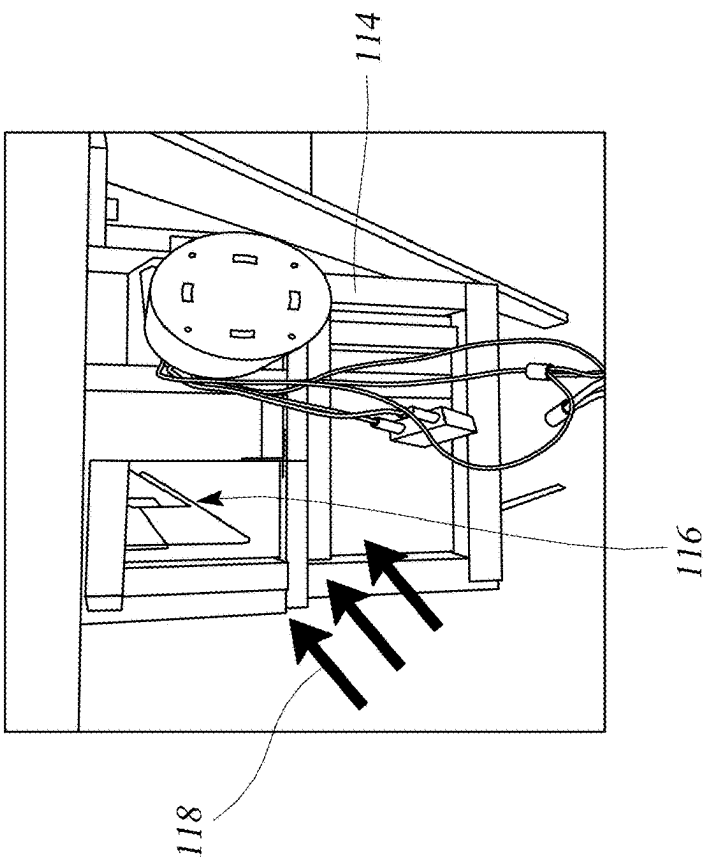


FIG. 1E



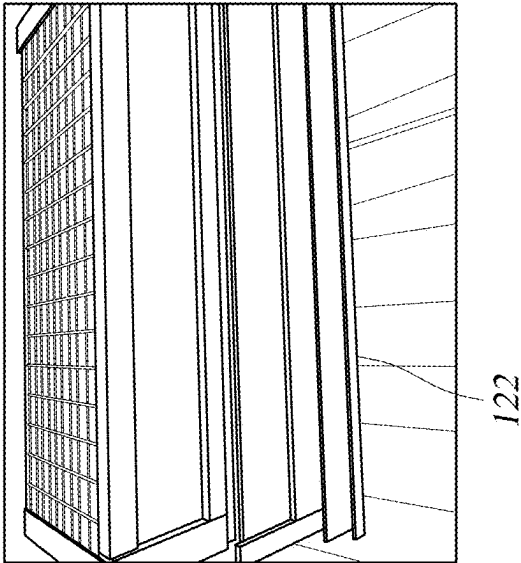


FIG. 1G

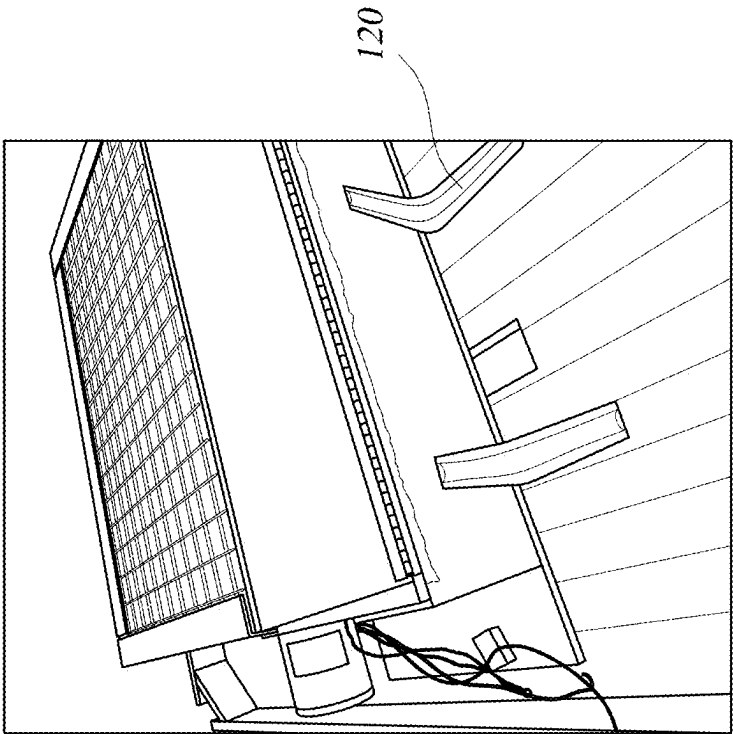
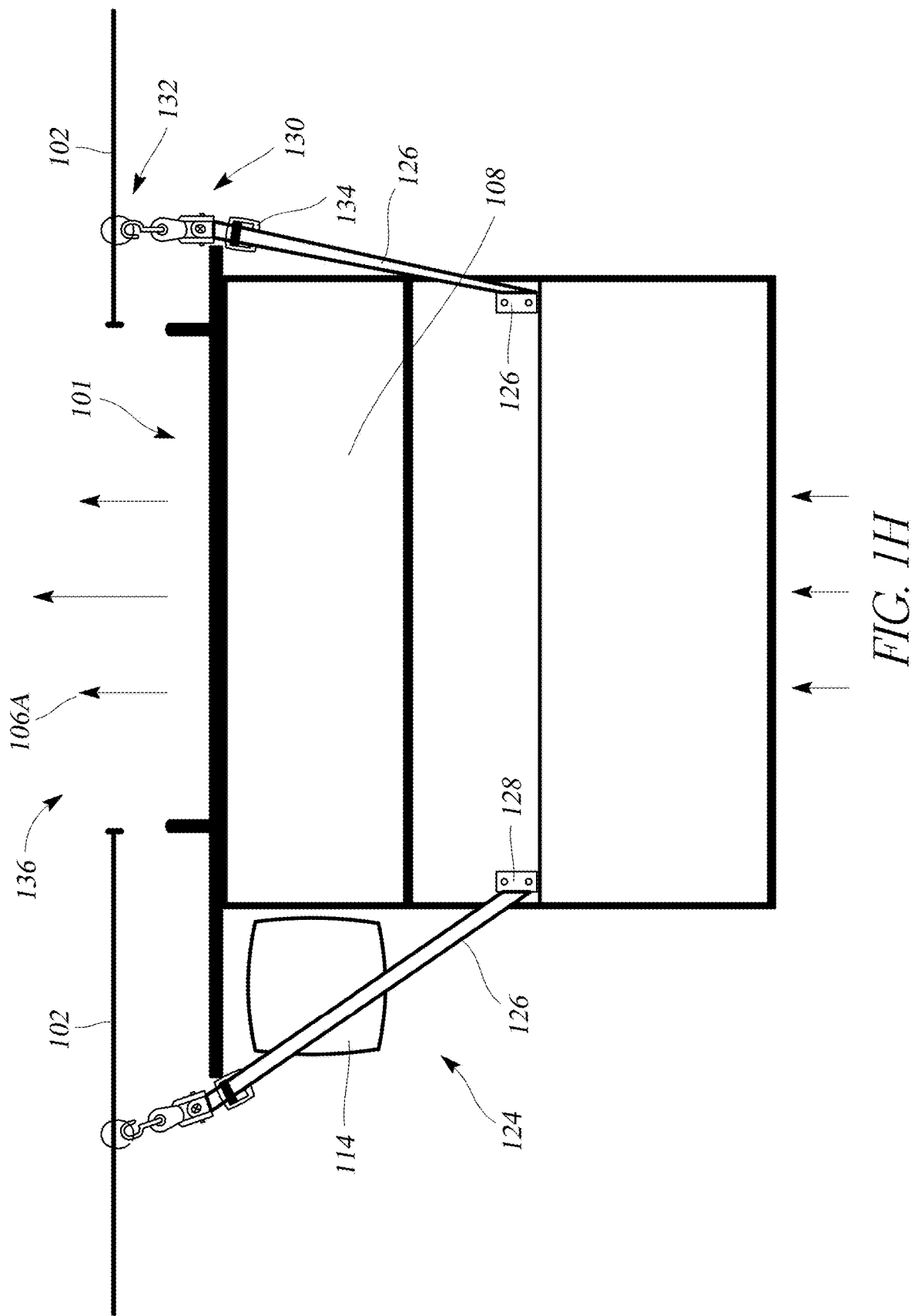


FIG. 1F



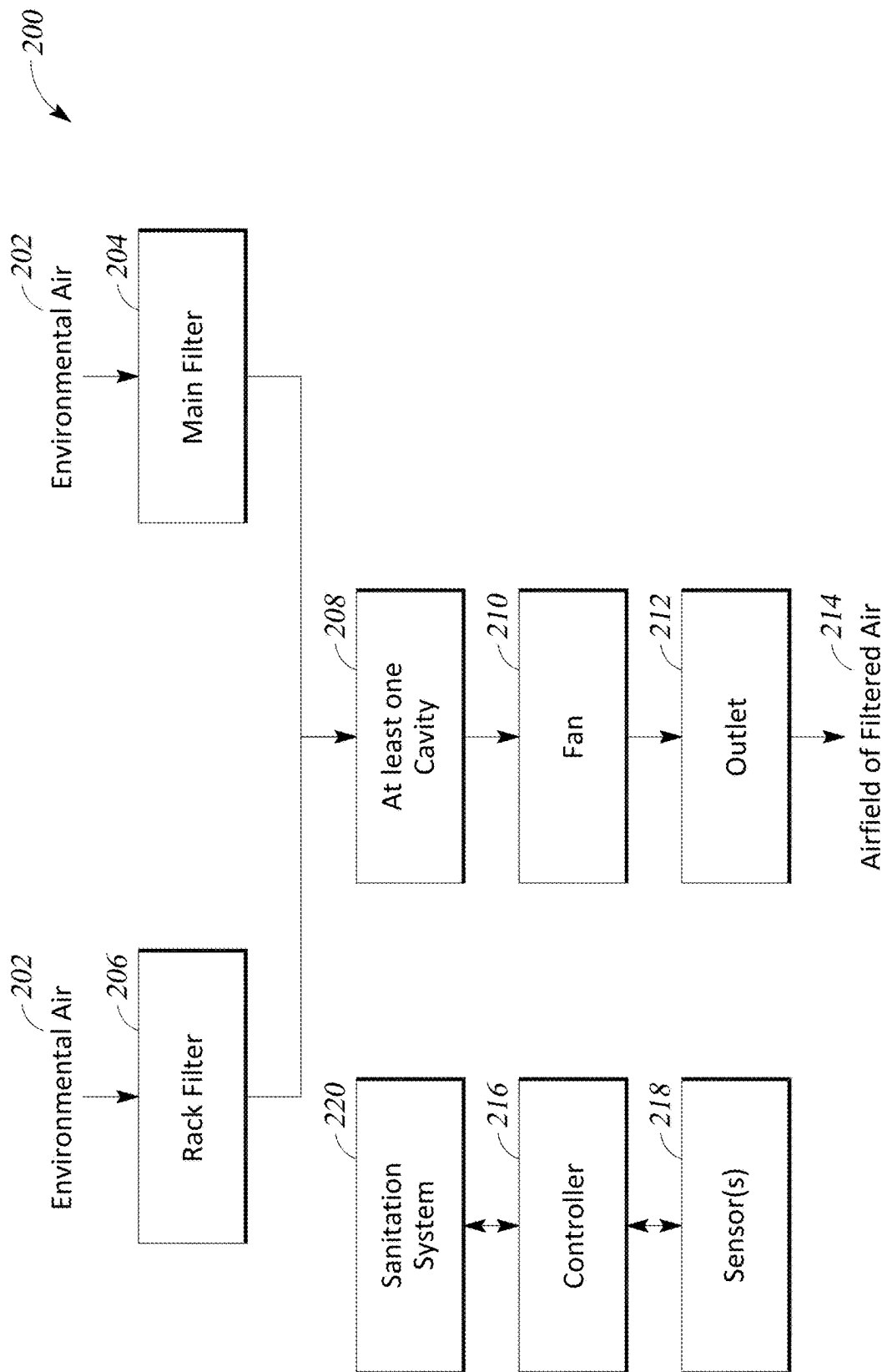


FIG. 2A

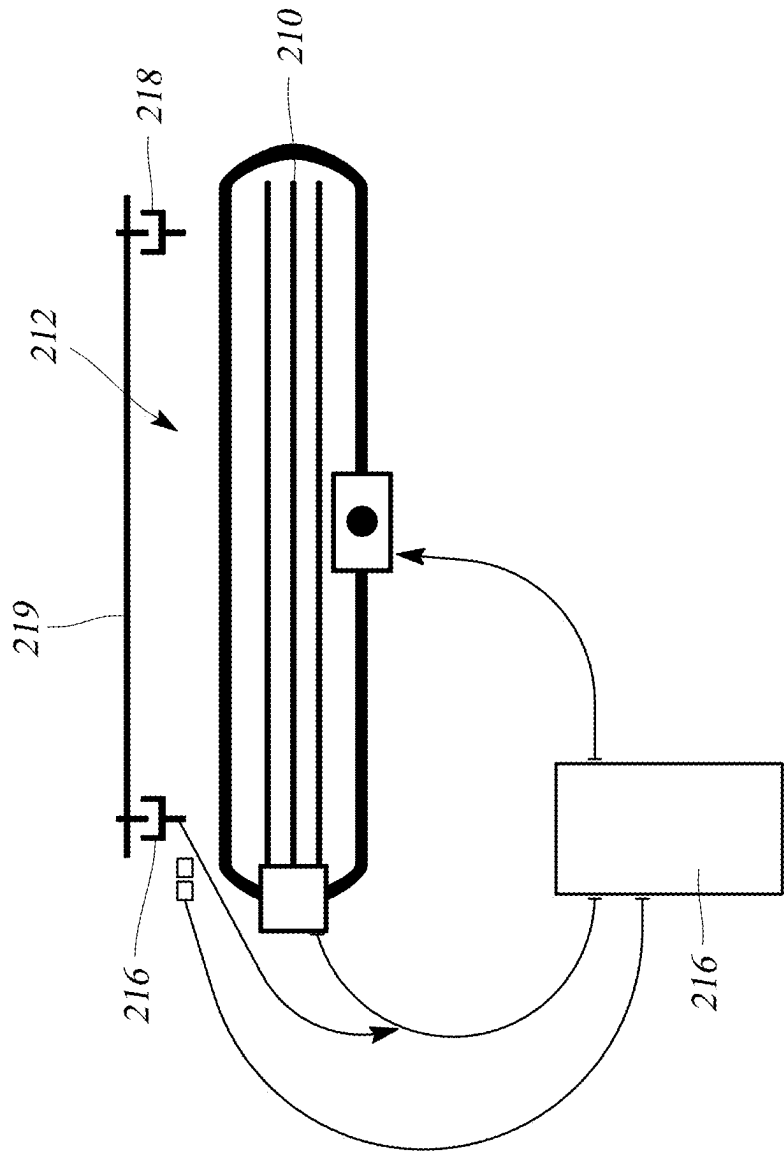


FIG. 2B

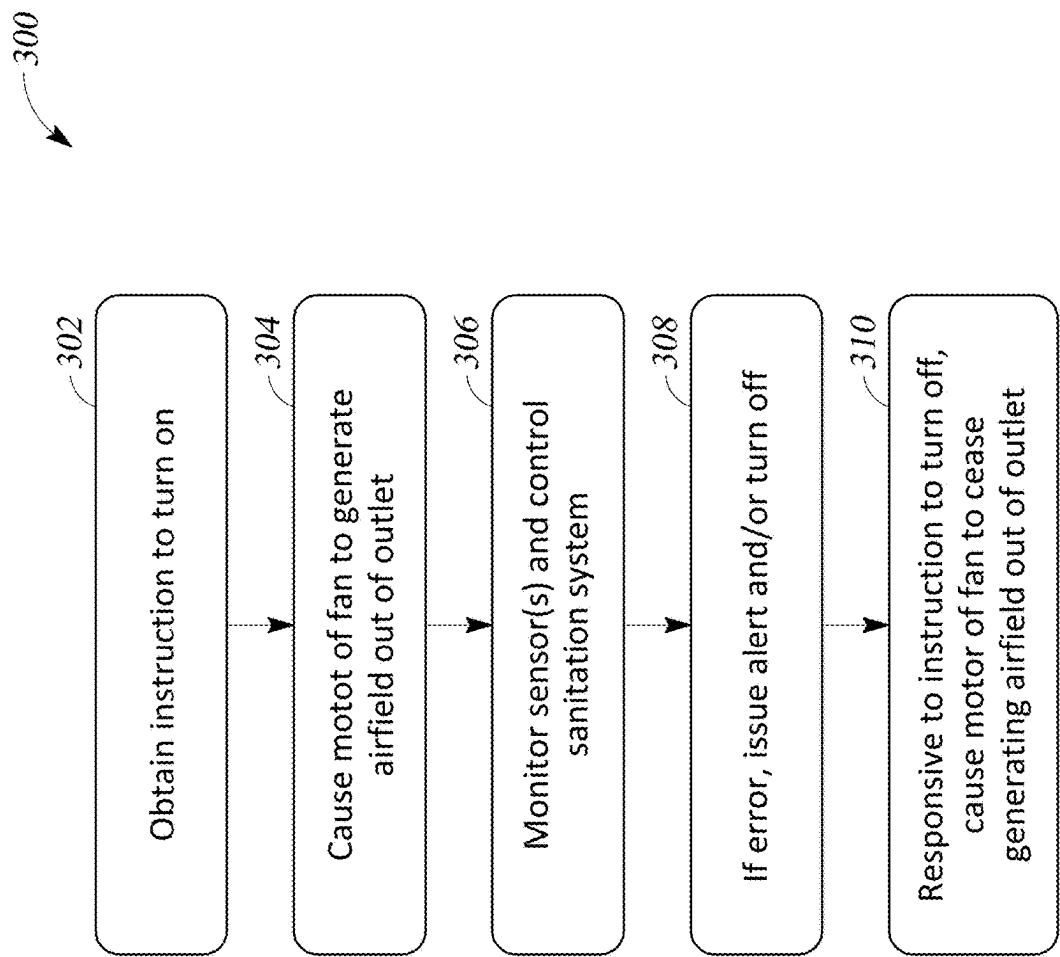


FIG. 3

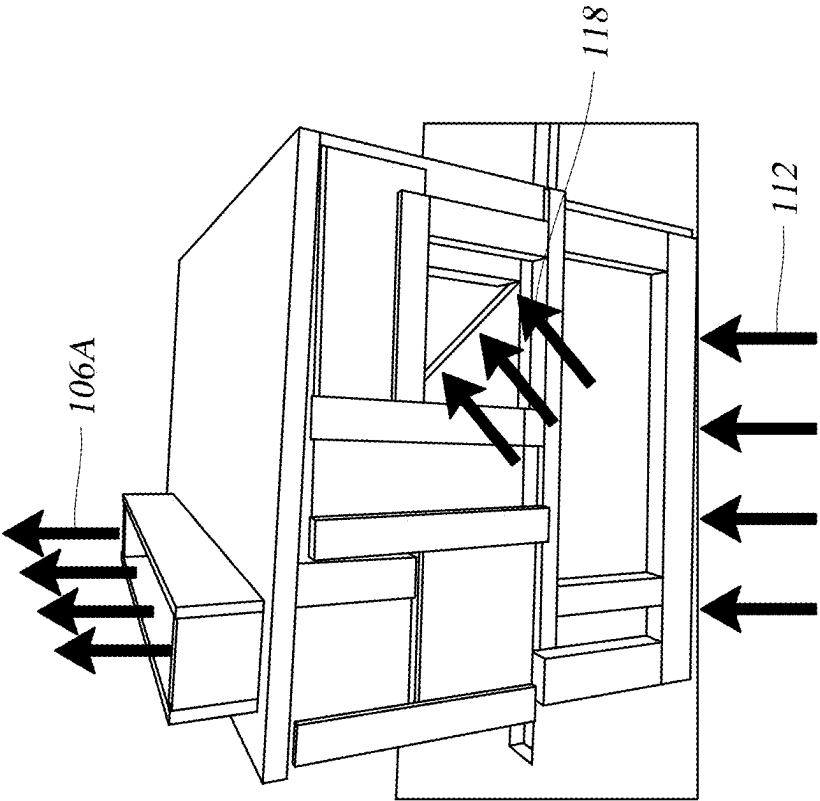
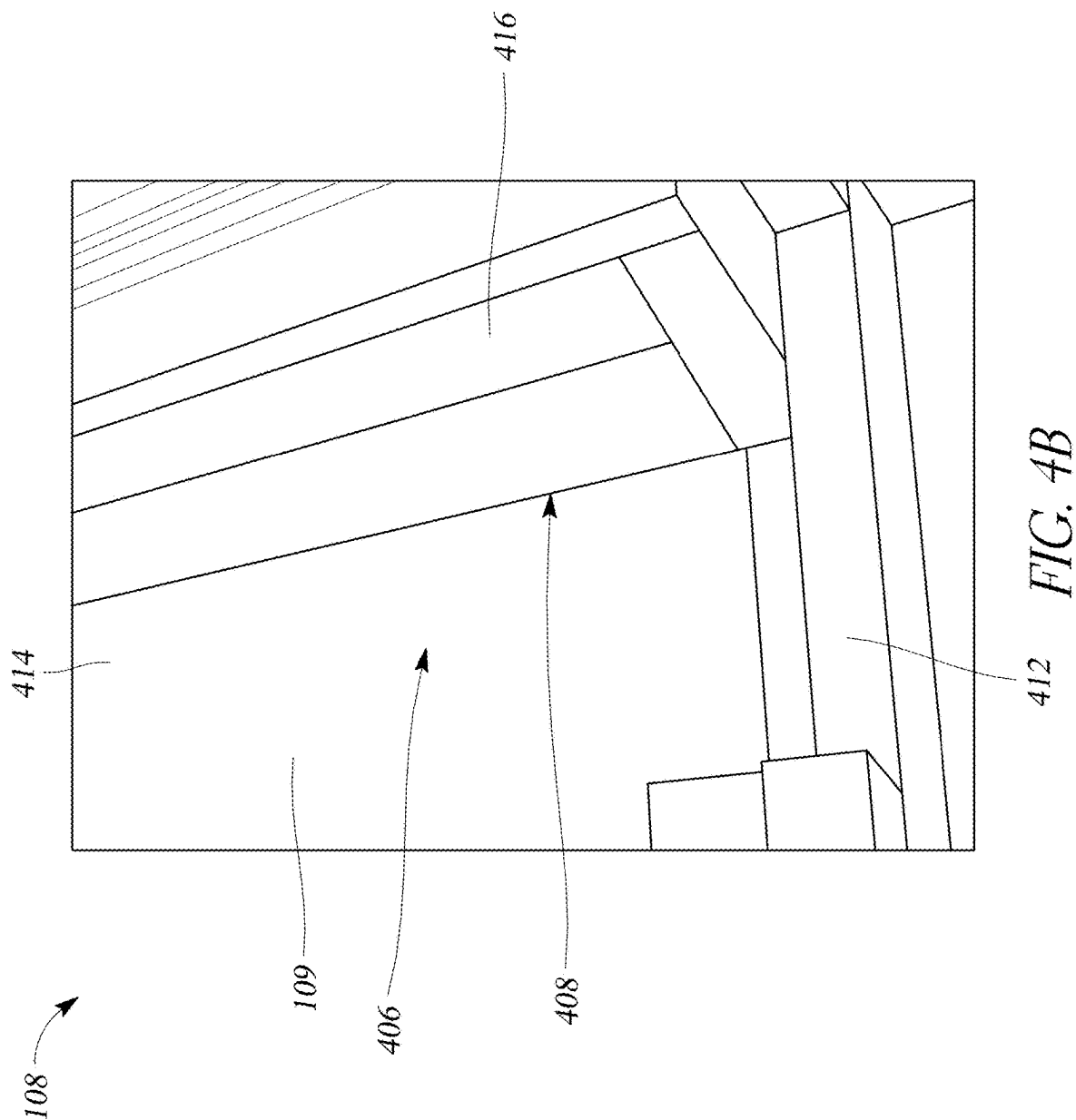


FIG. 4A



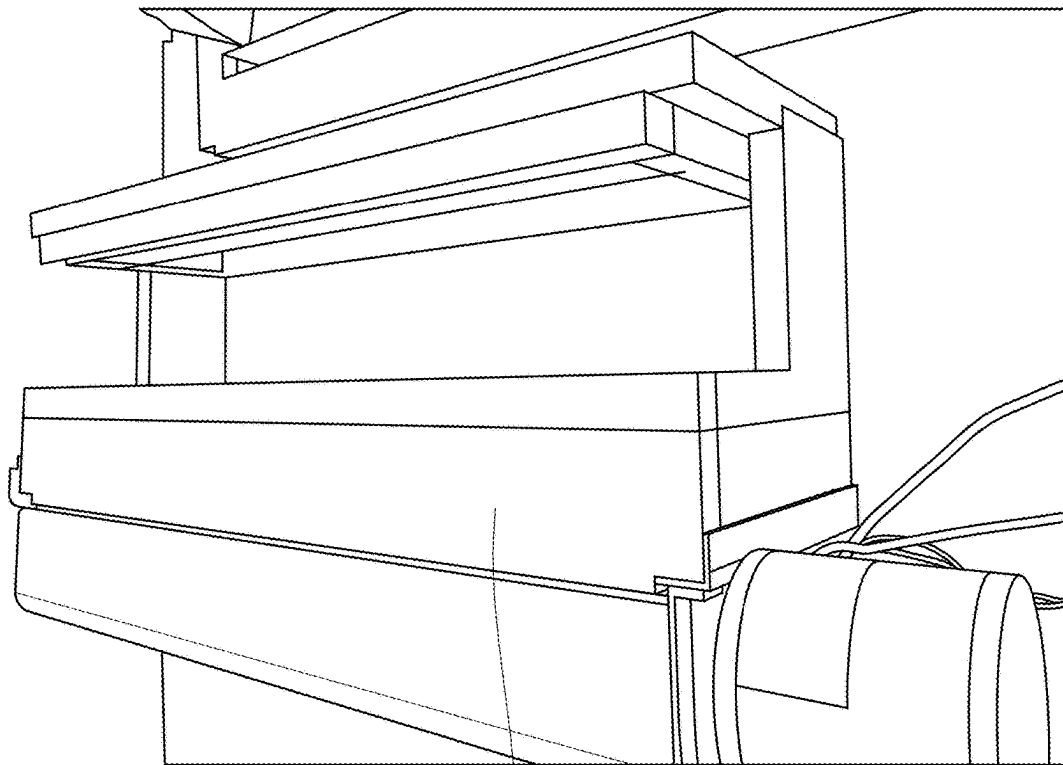


FIG. 4C

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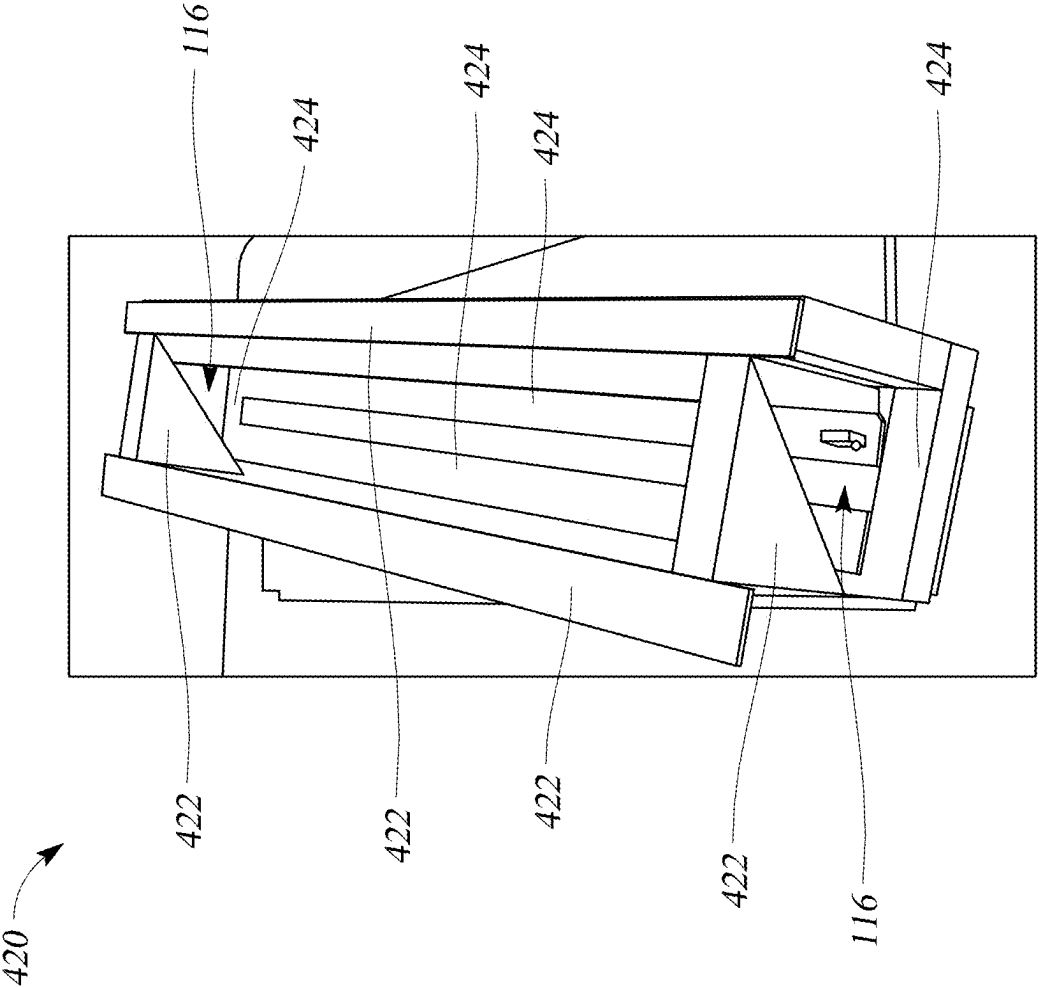


FIG. 4D

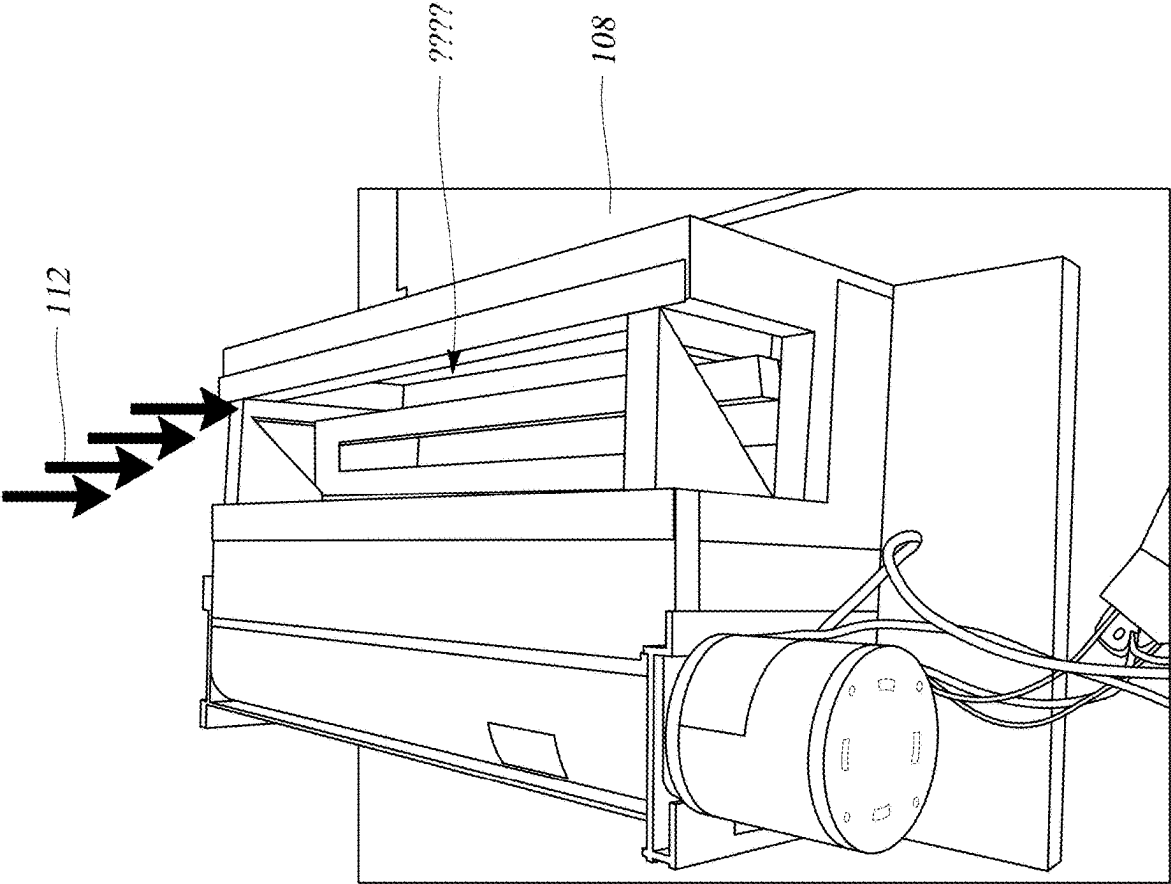


FIG. 4E

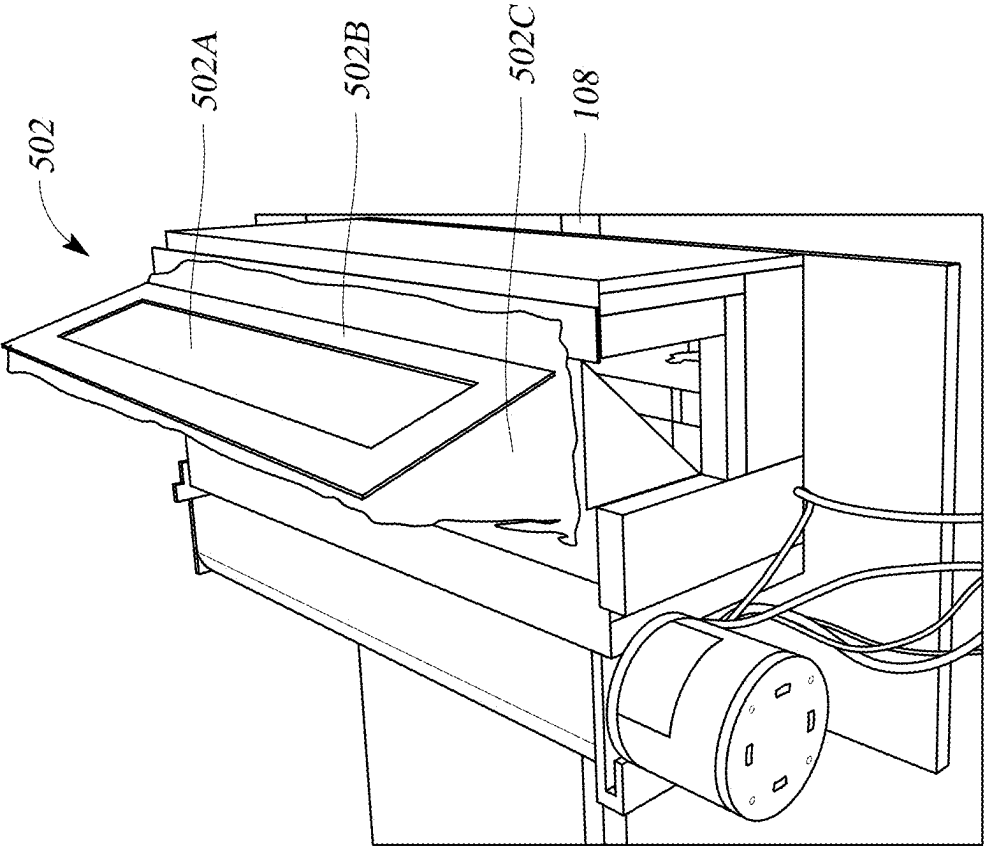


FIG. 5A

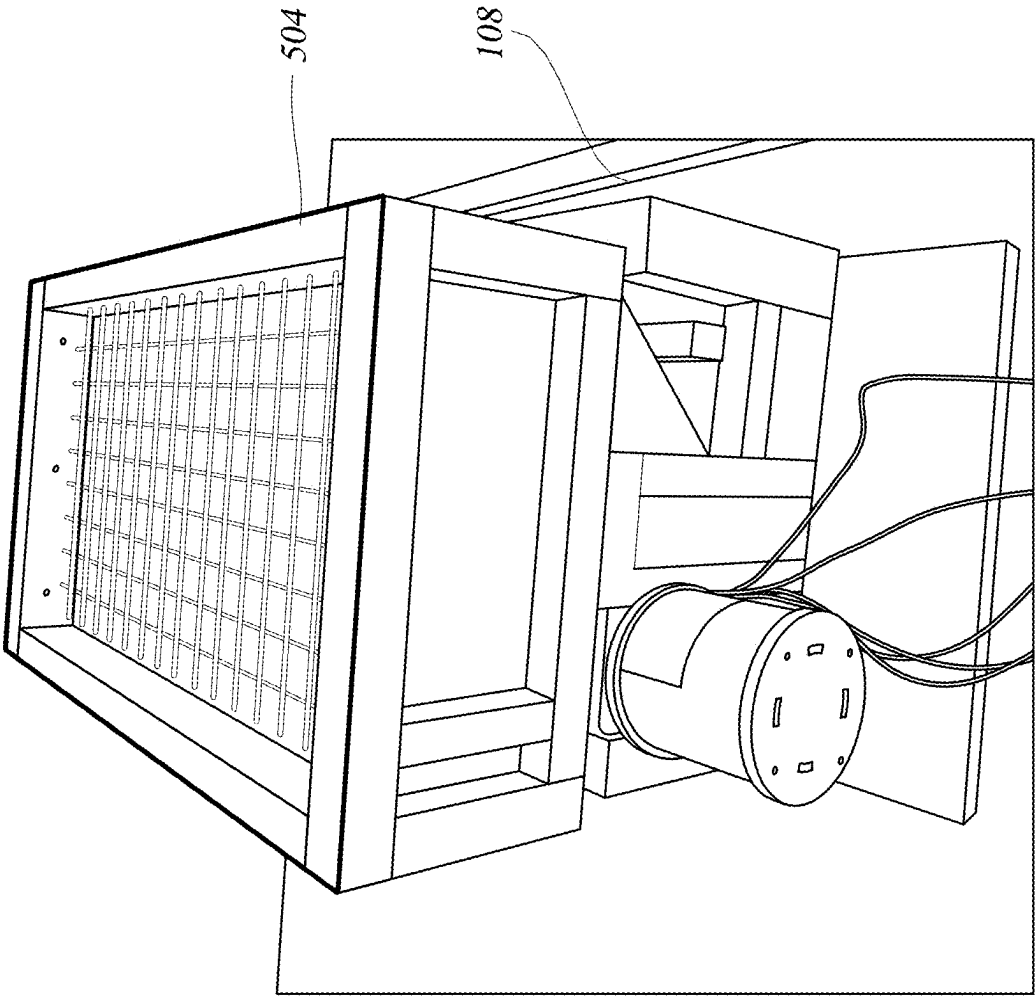


FIG. 5B

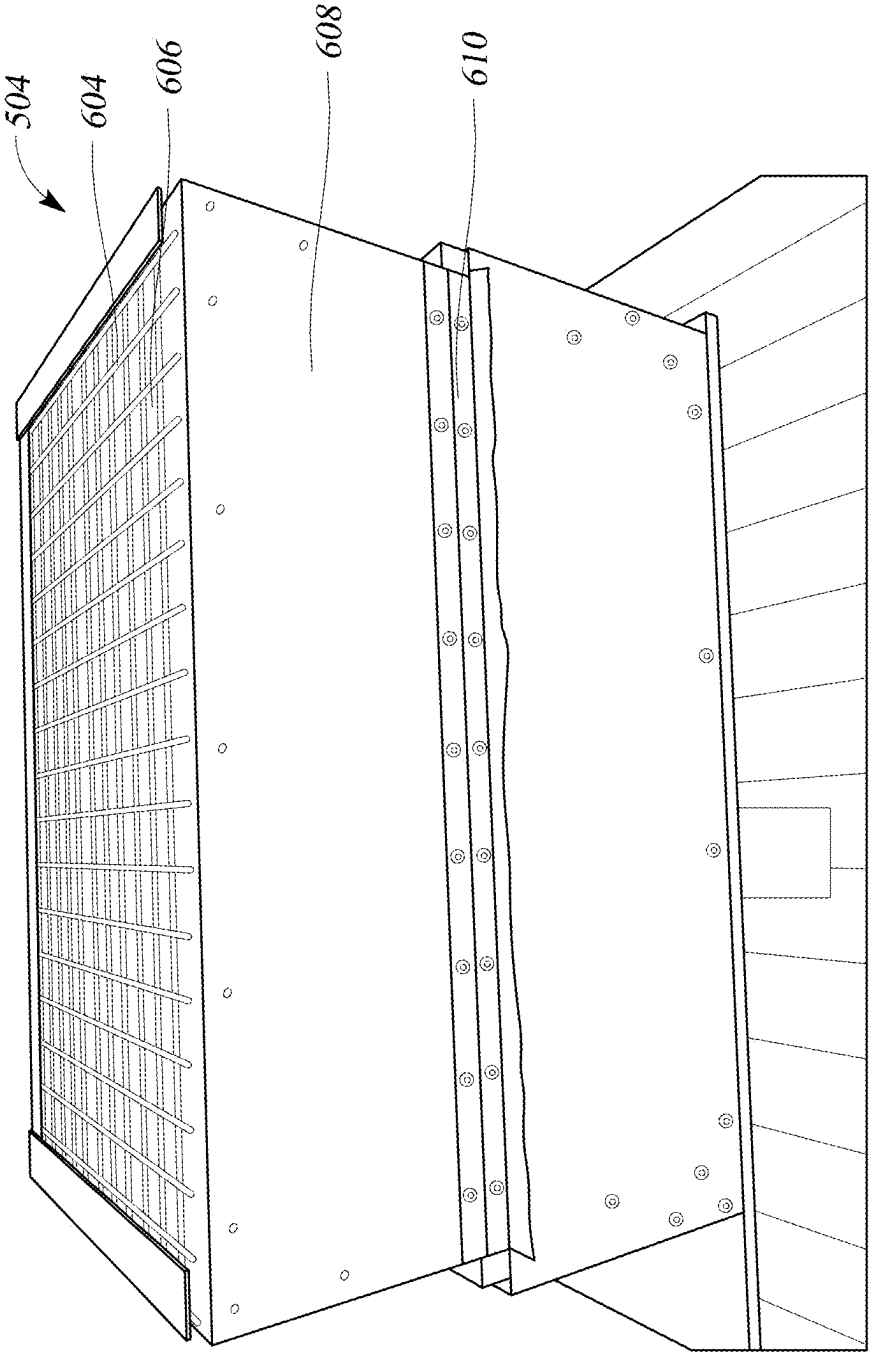


FIG. 6A

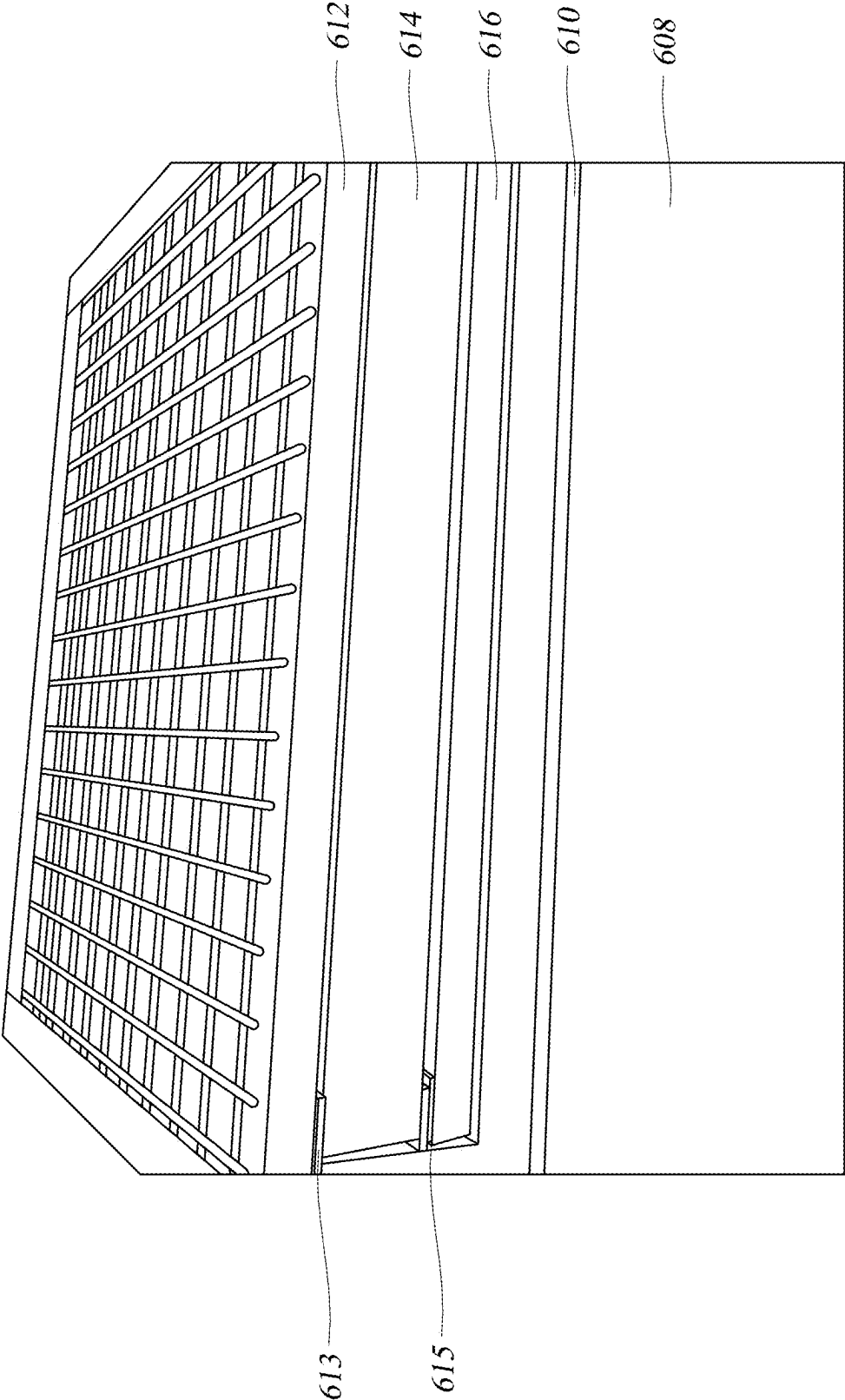


FIG. 6B

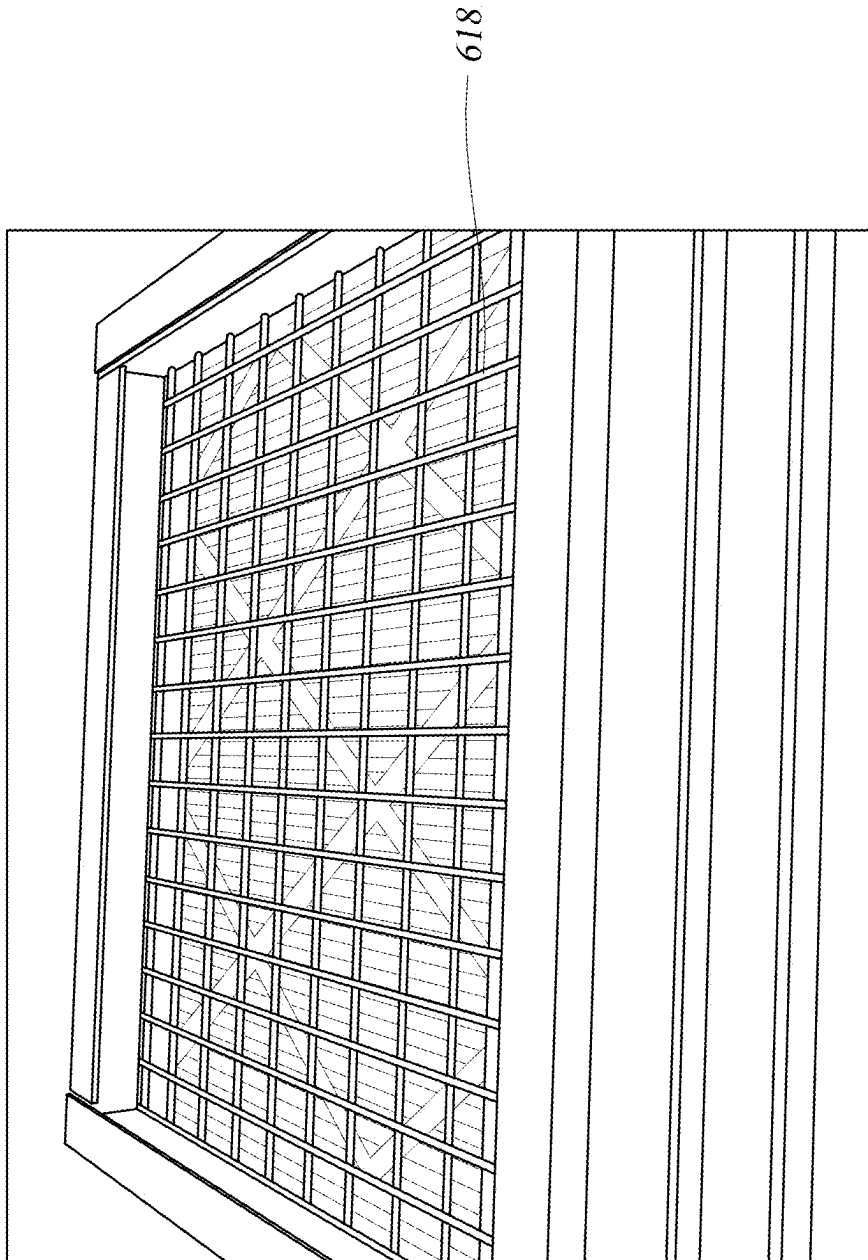


FIG. 6C

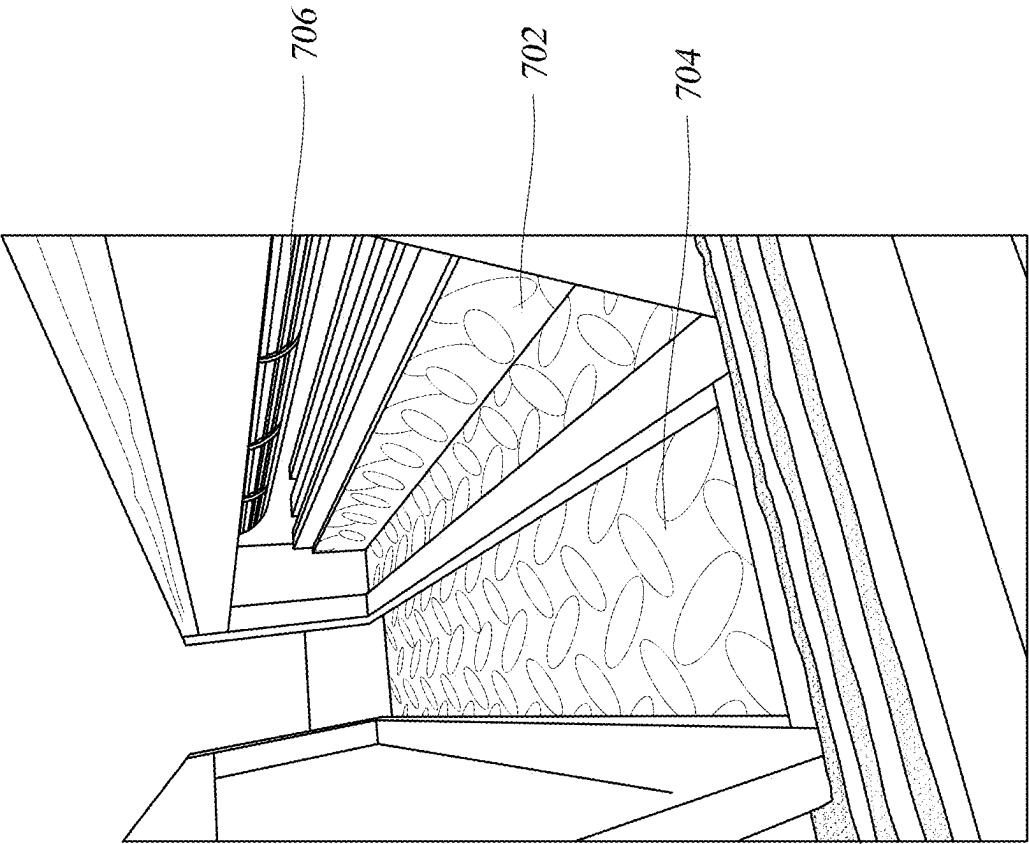
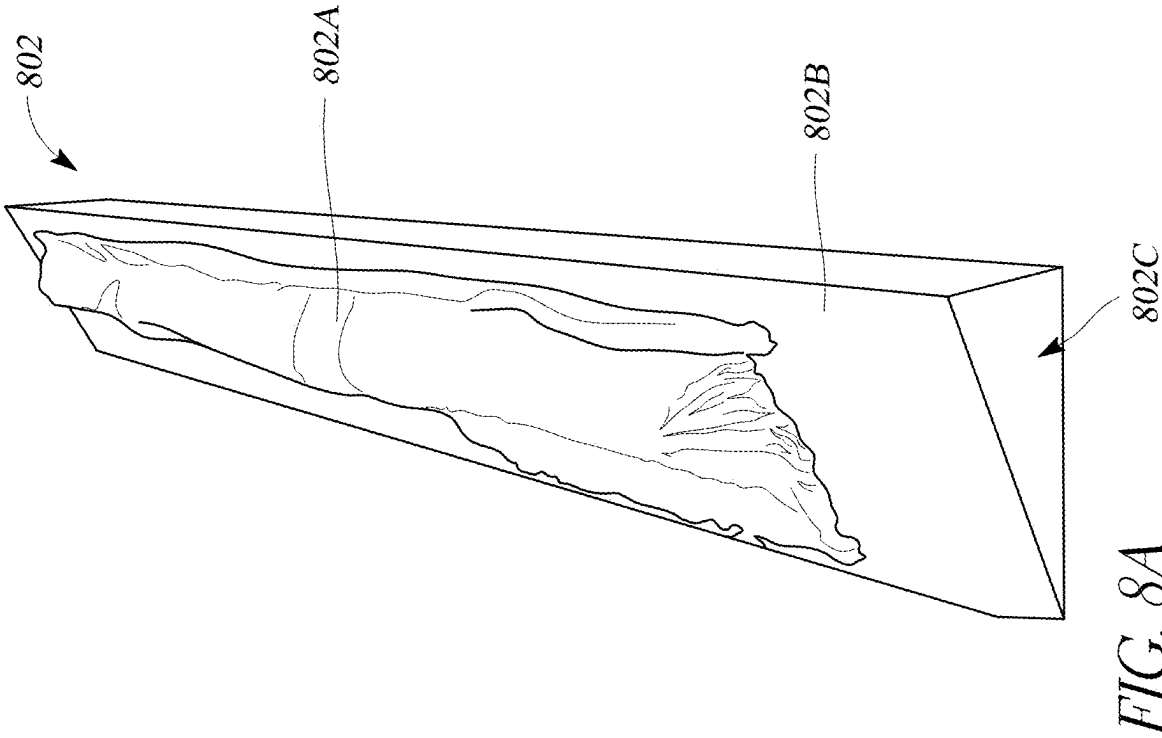
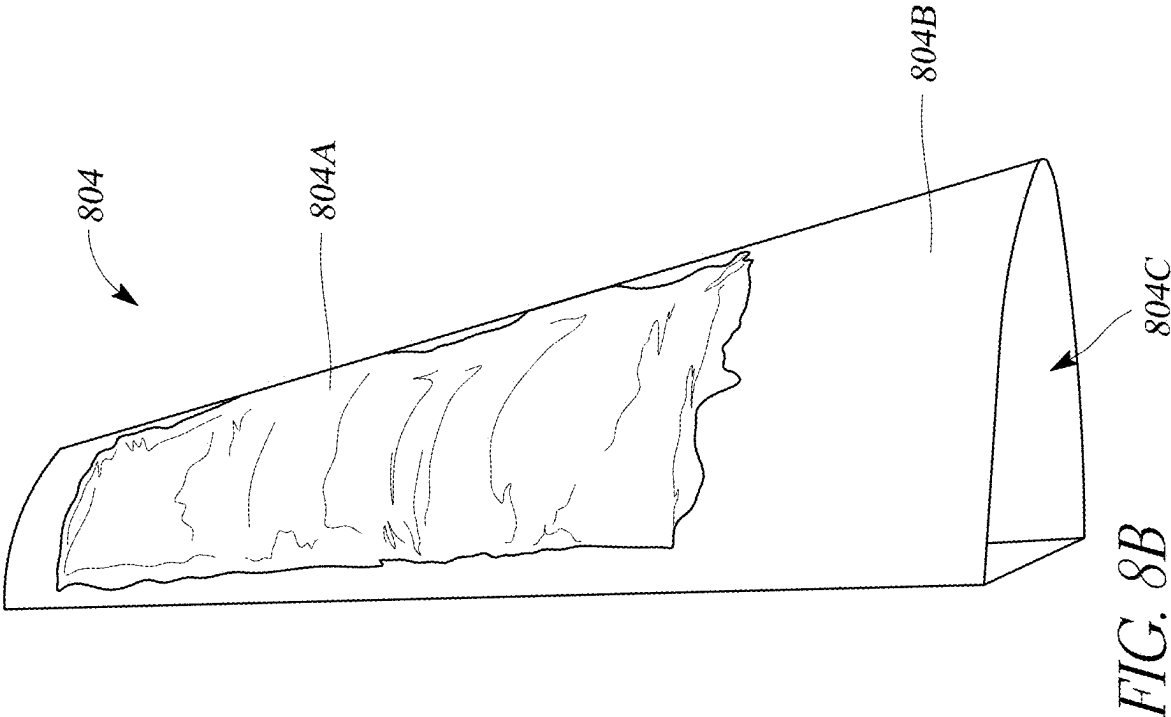
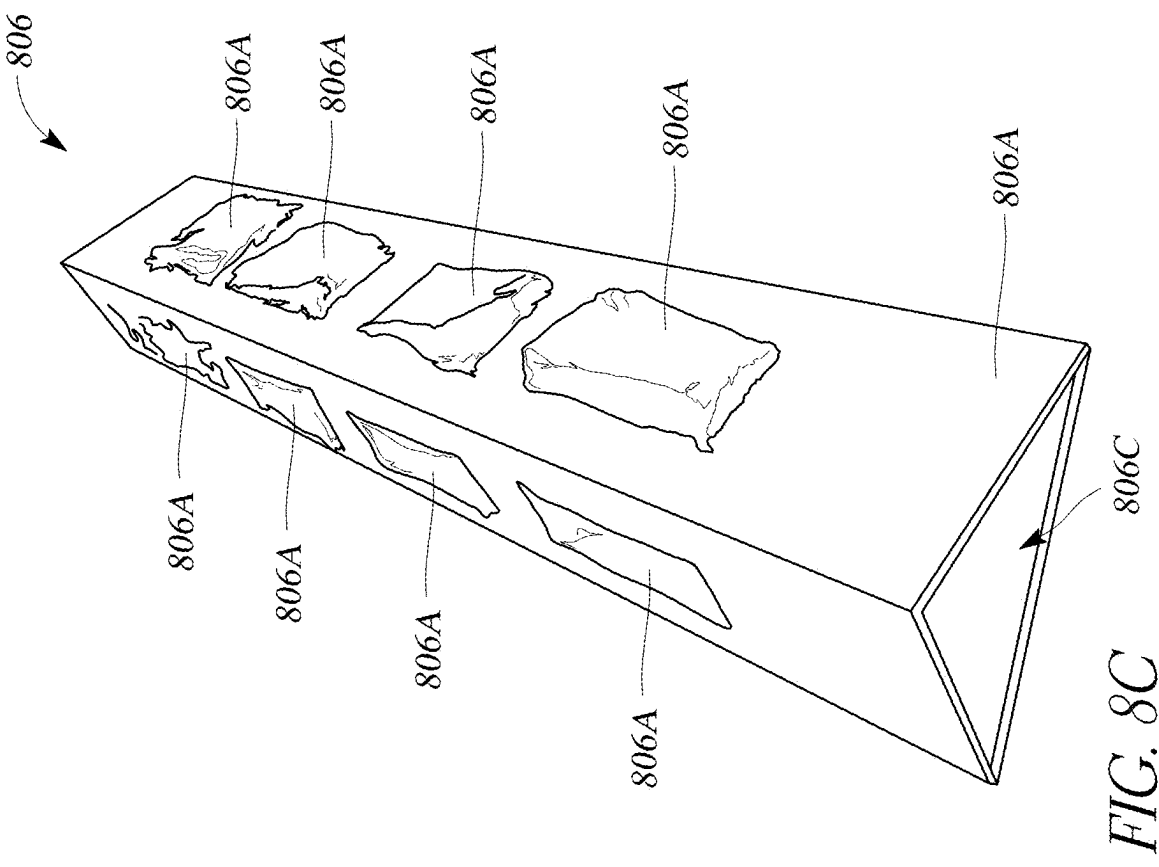
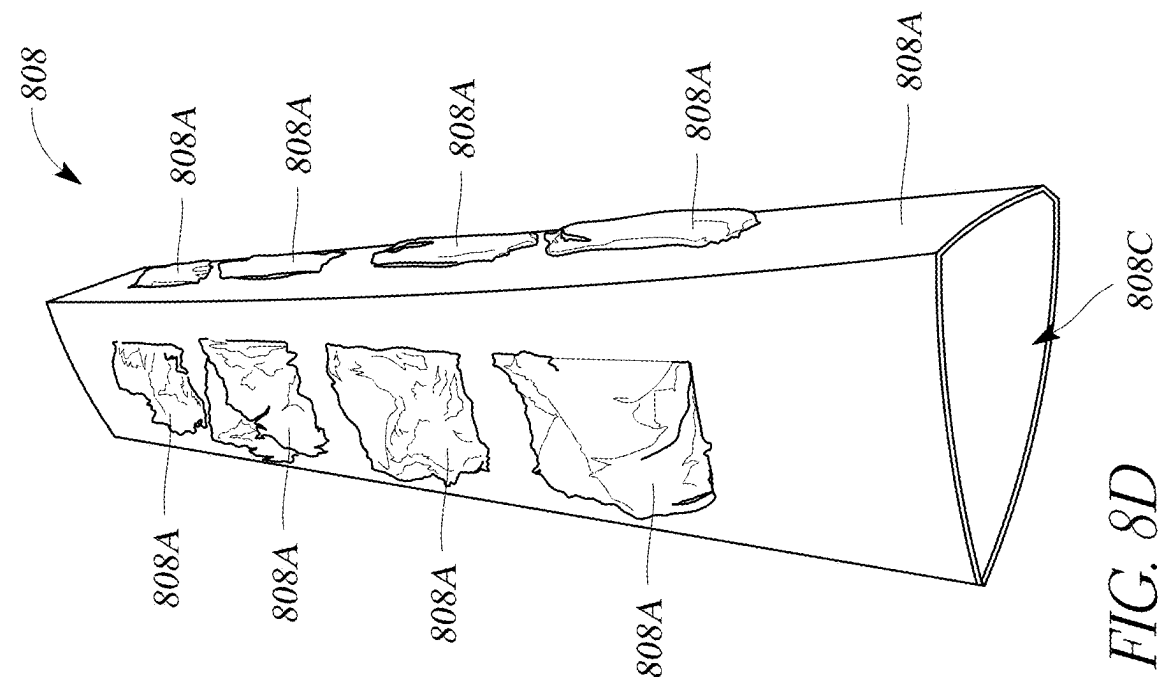


FIG. 7





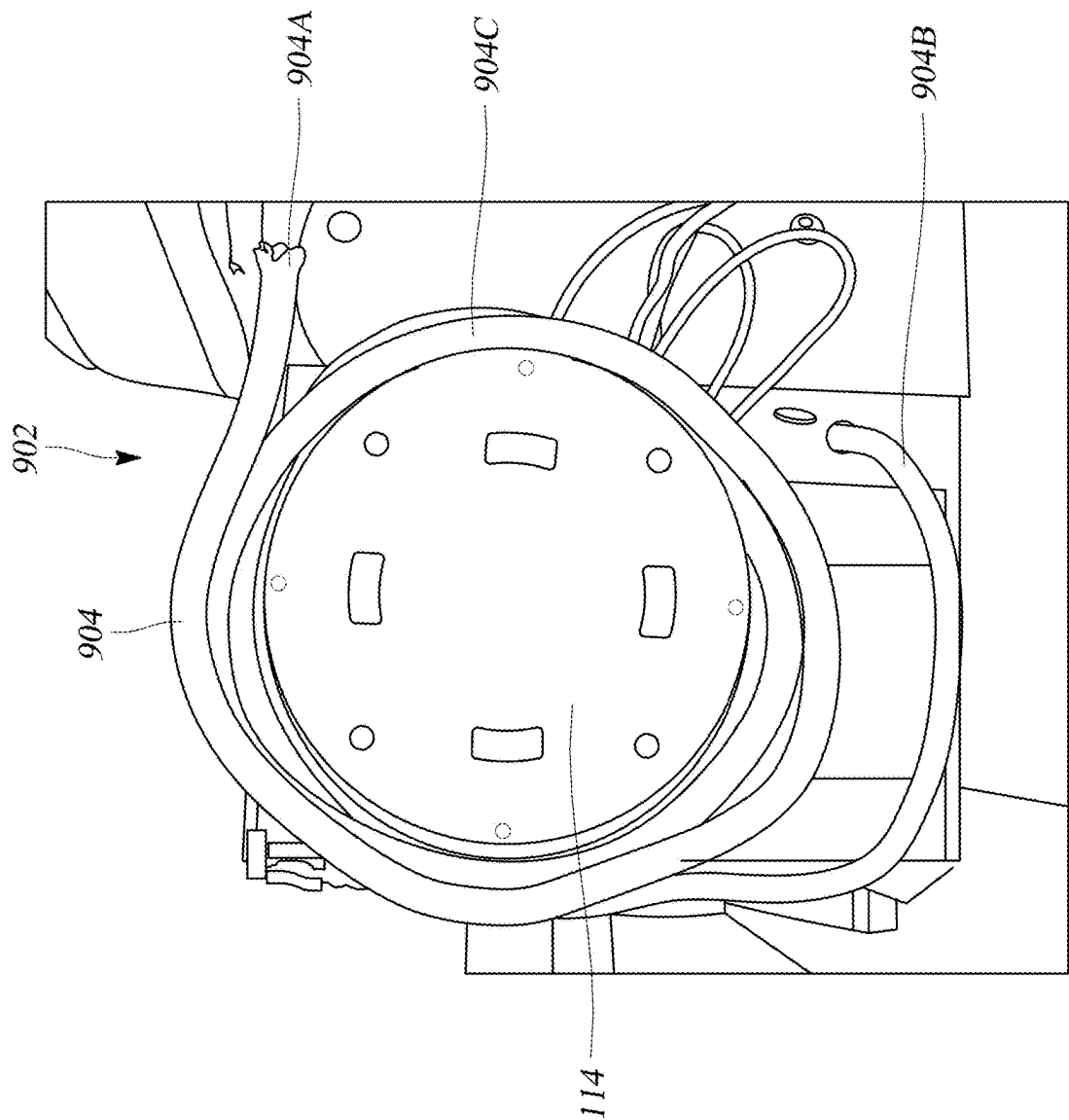


FIG. 9A

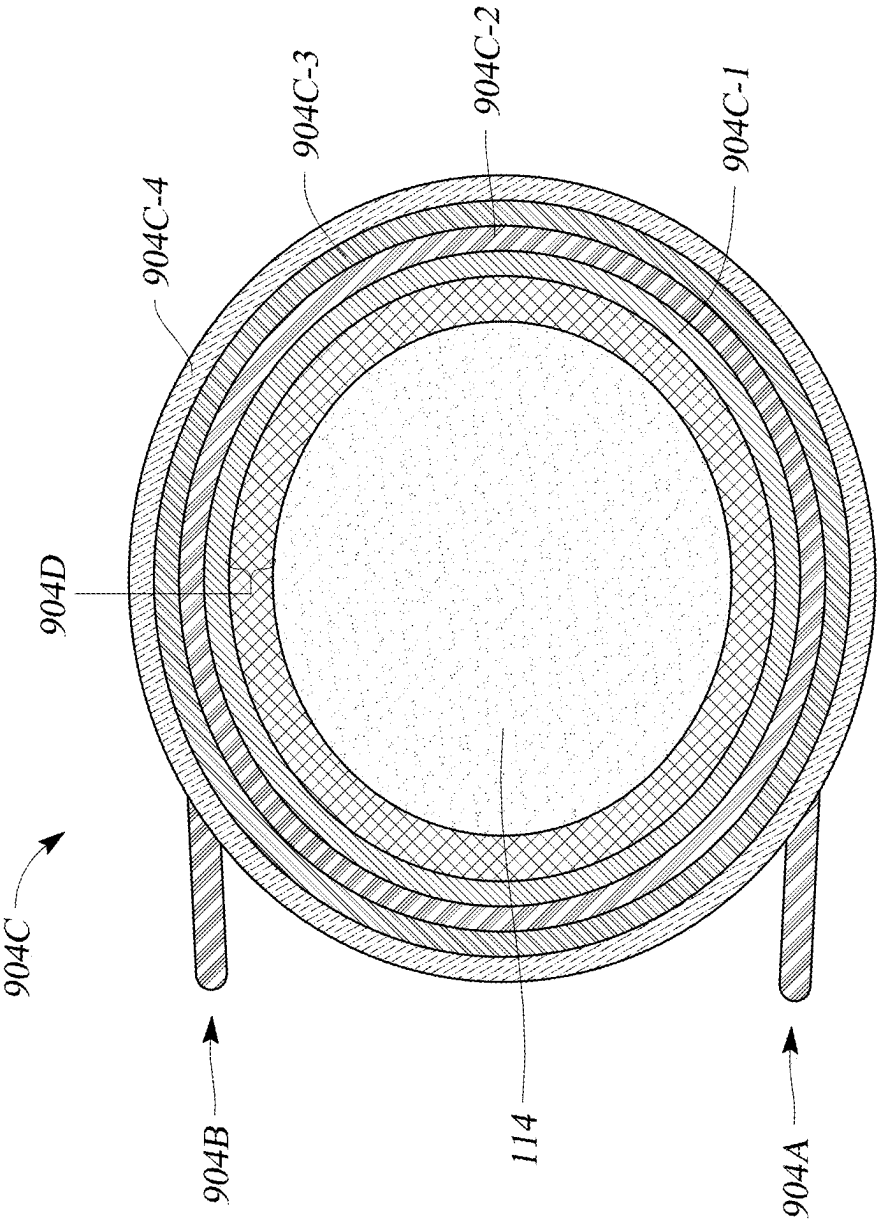
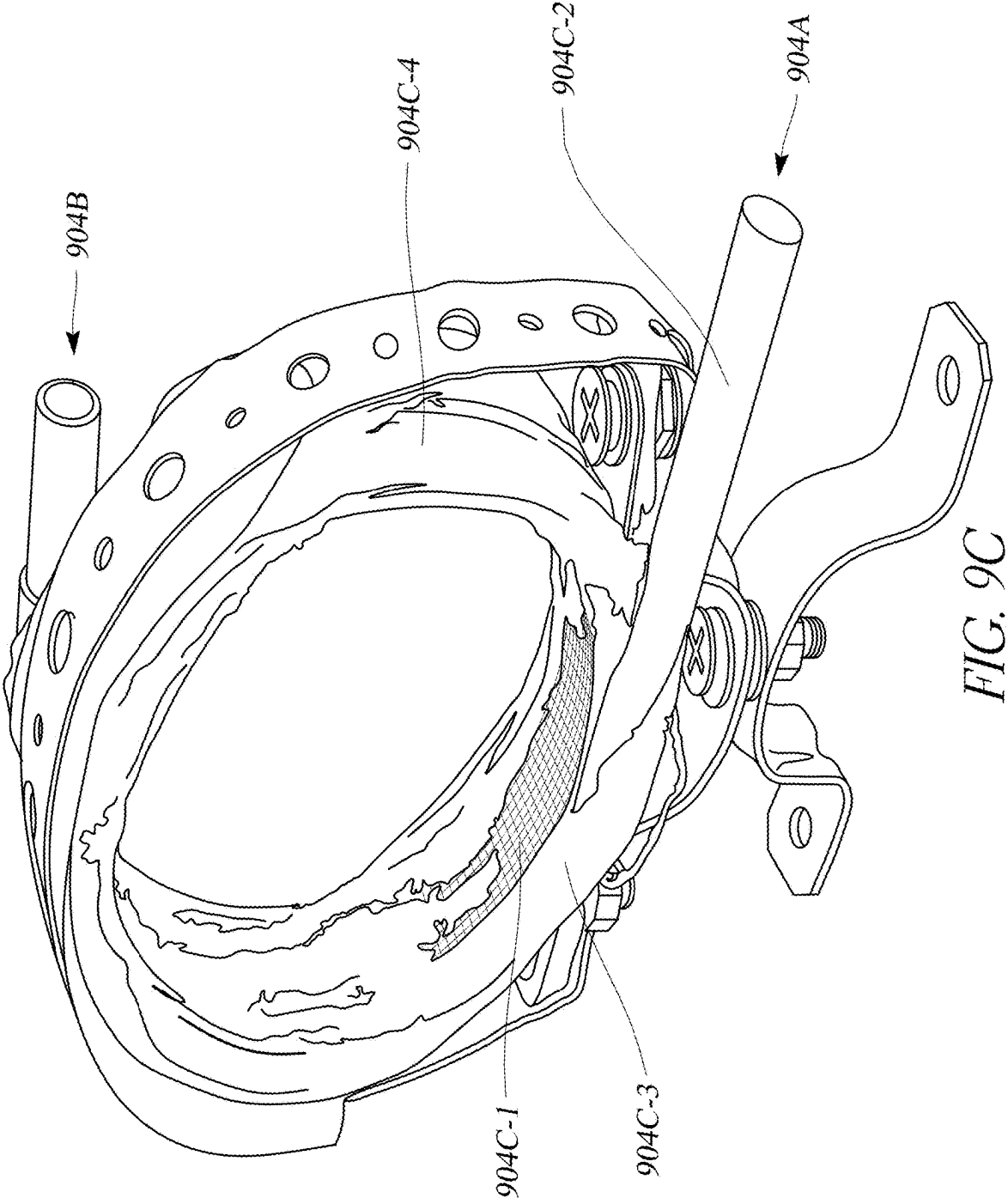


FIG. 9B



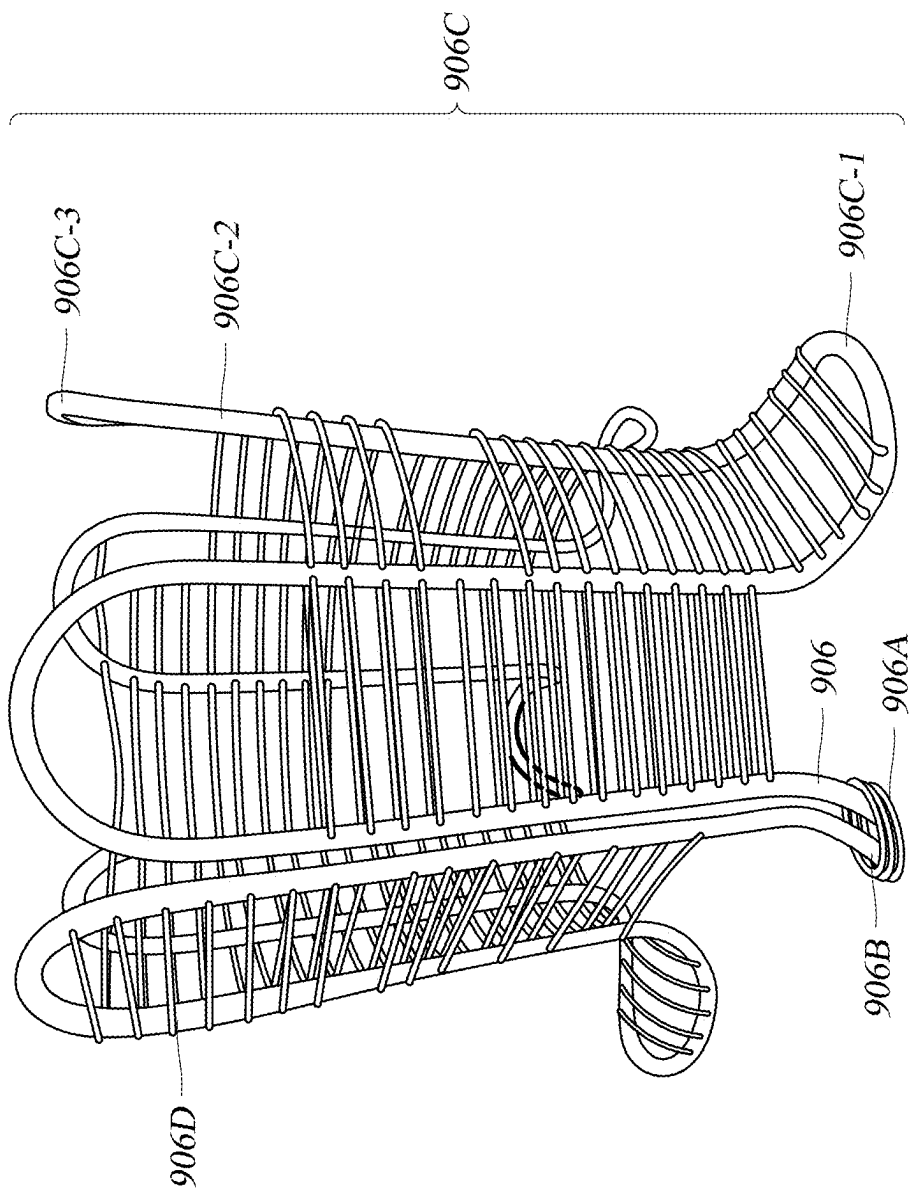


FIG. 9D

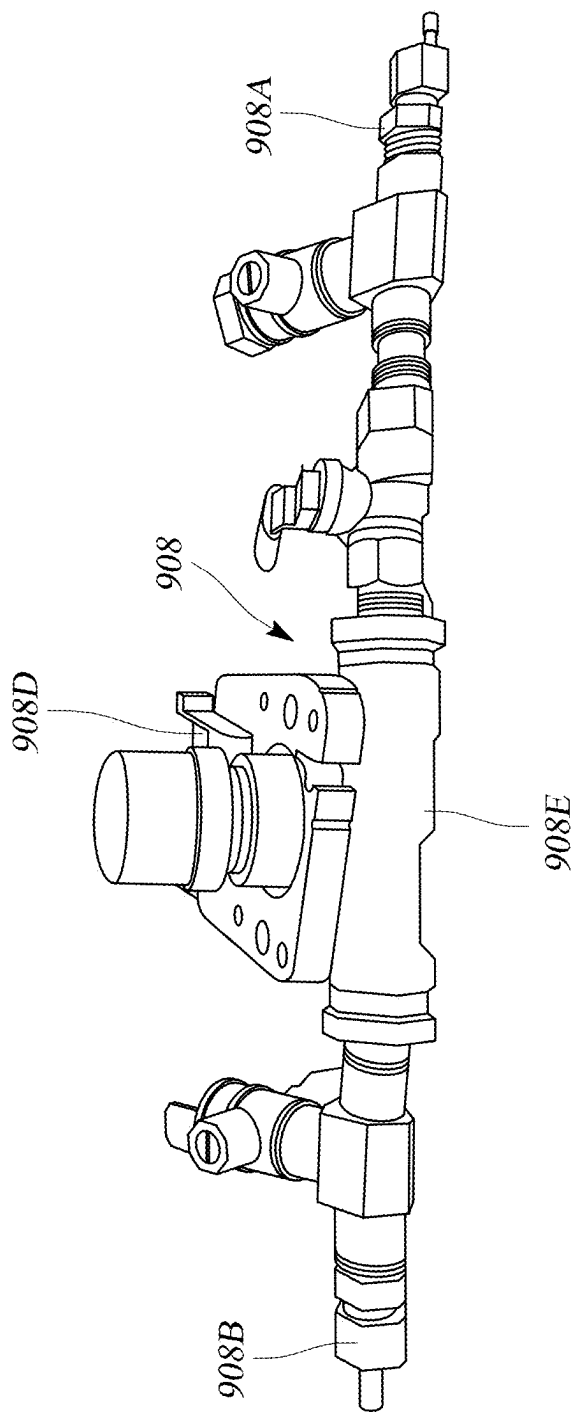
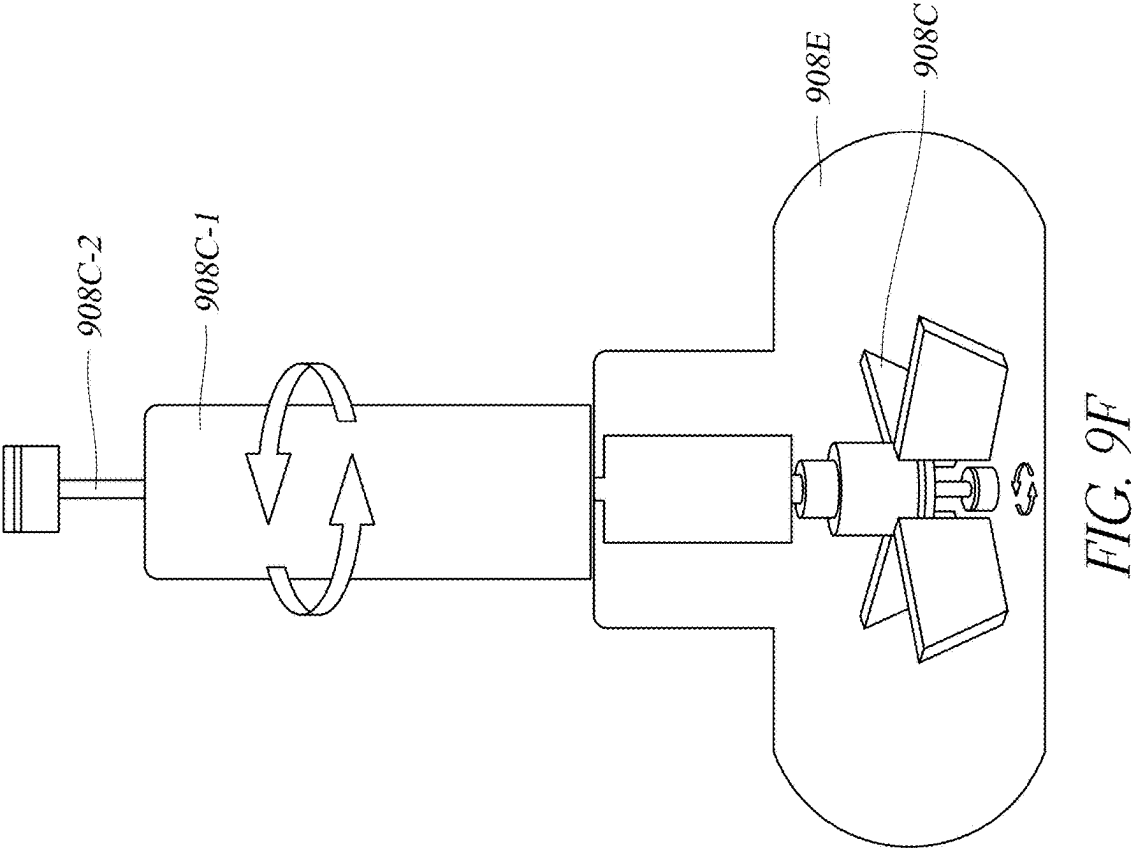
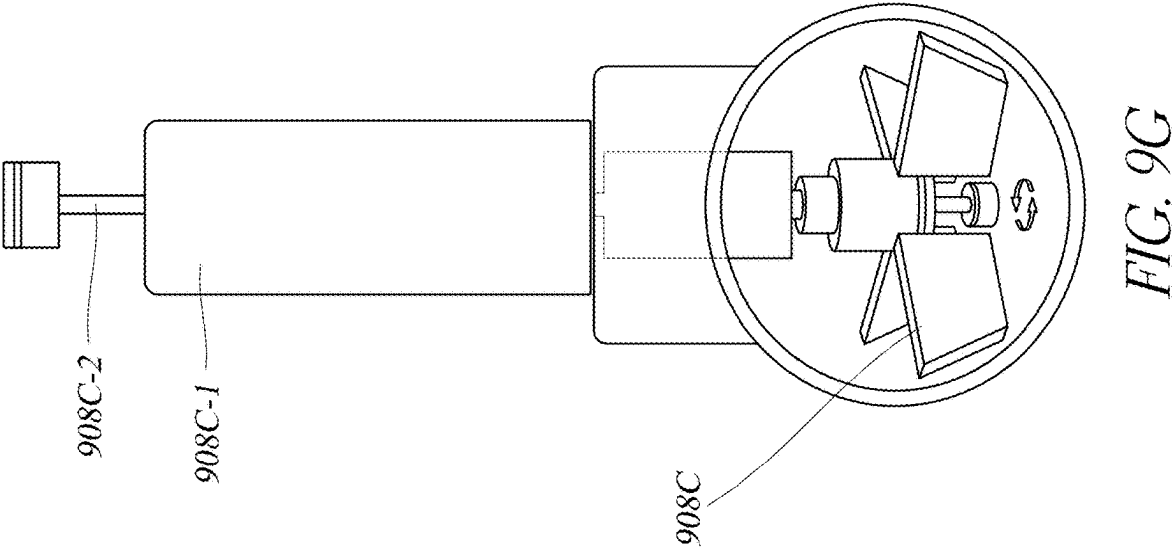


FIG. 9E



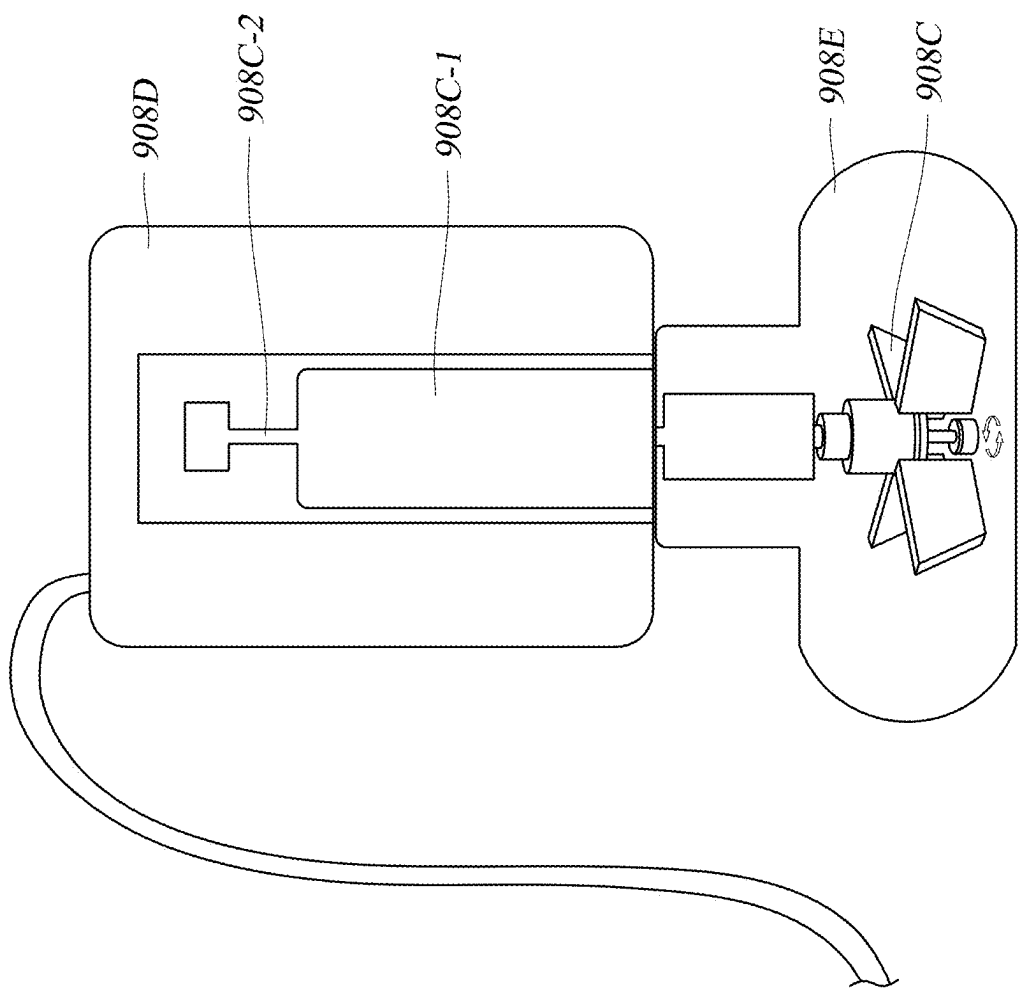


FIG. 9H

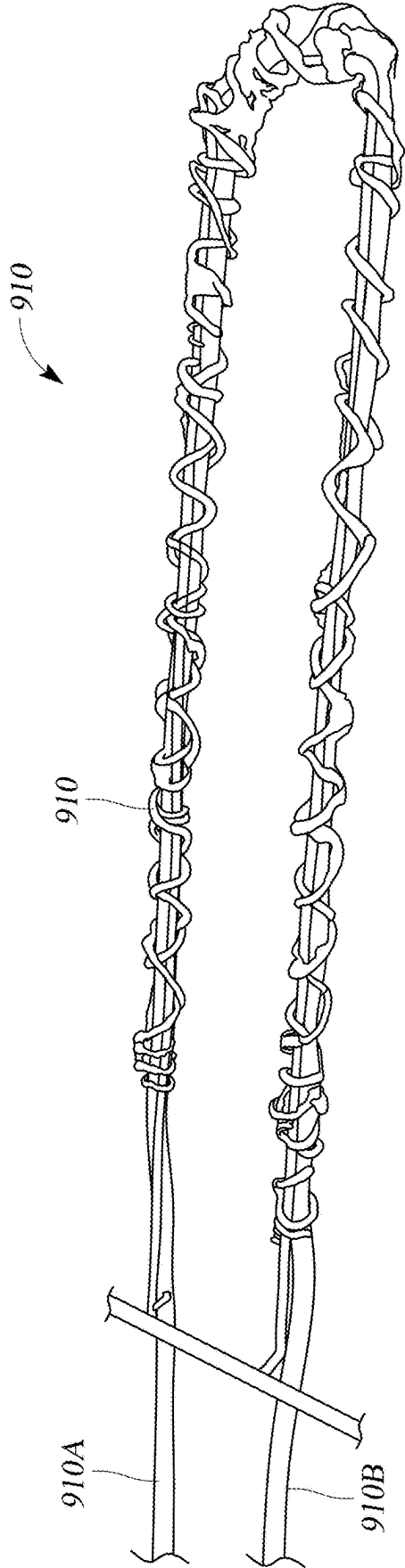
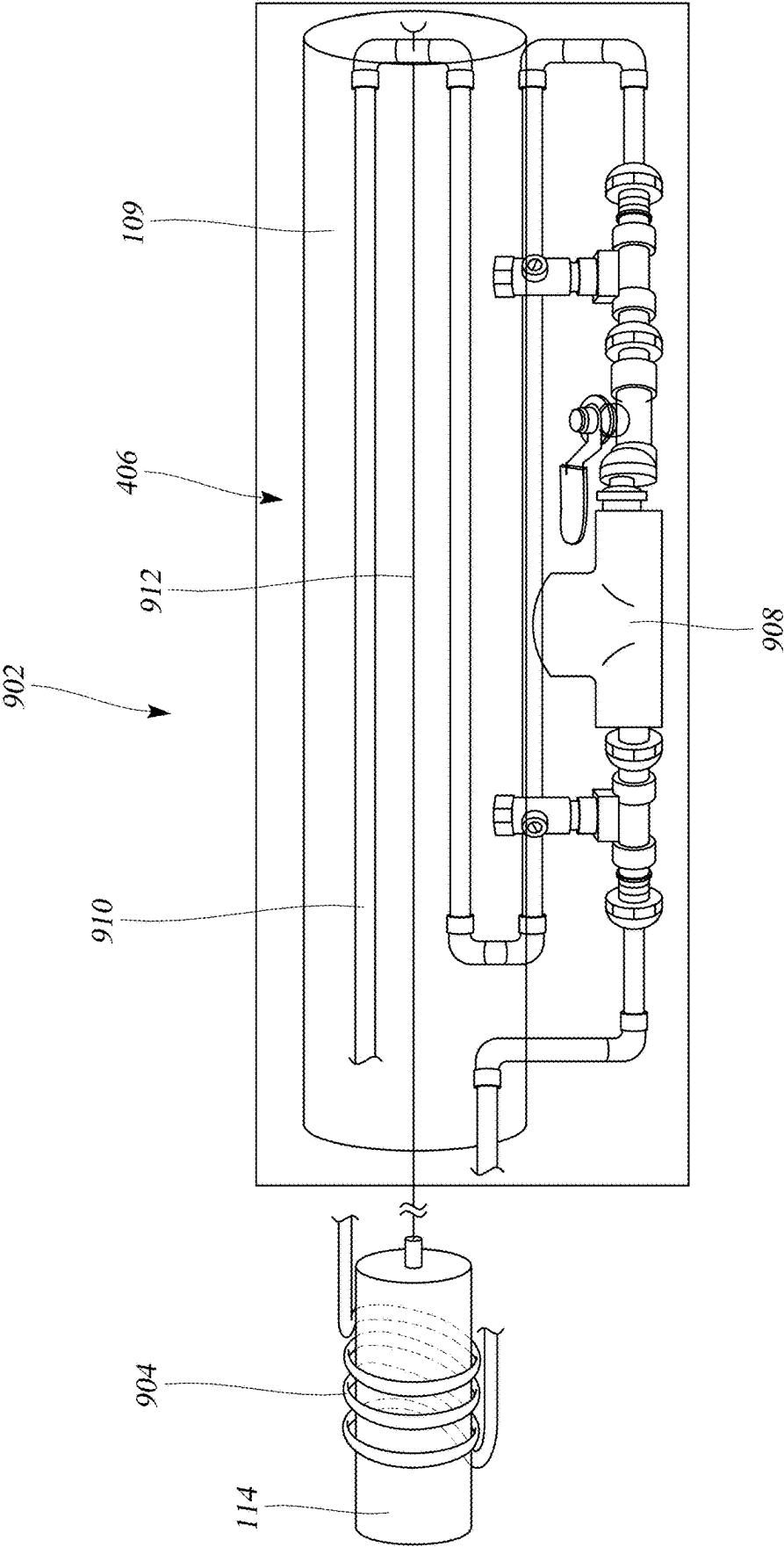


FIG. 91



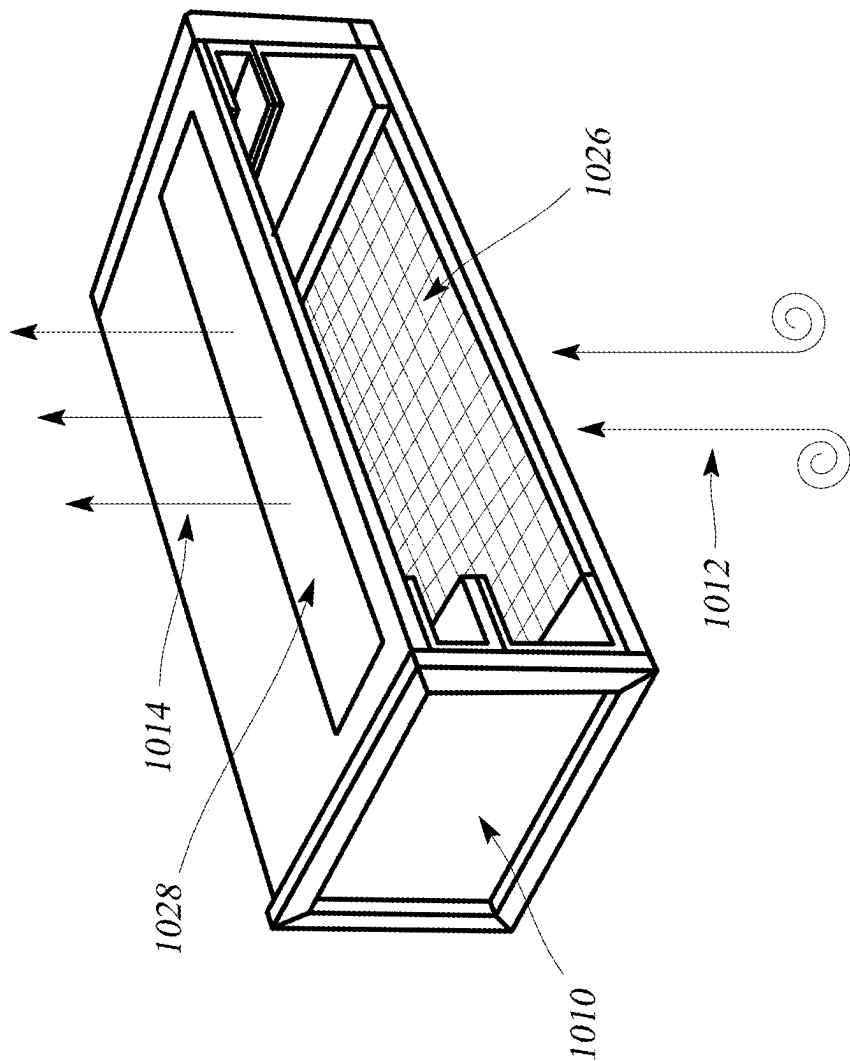


FIG. 10A

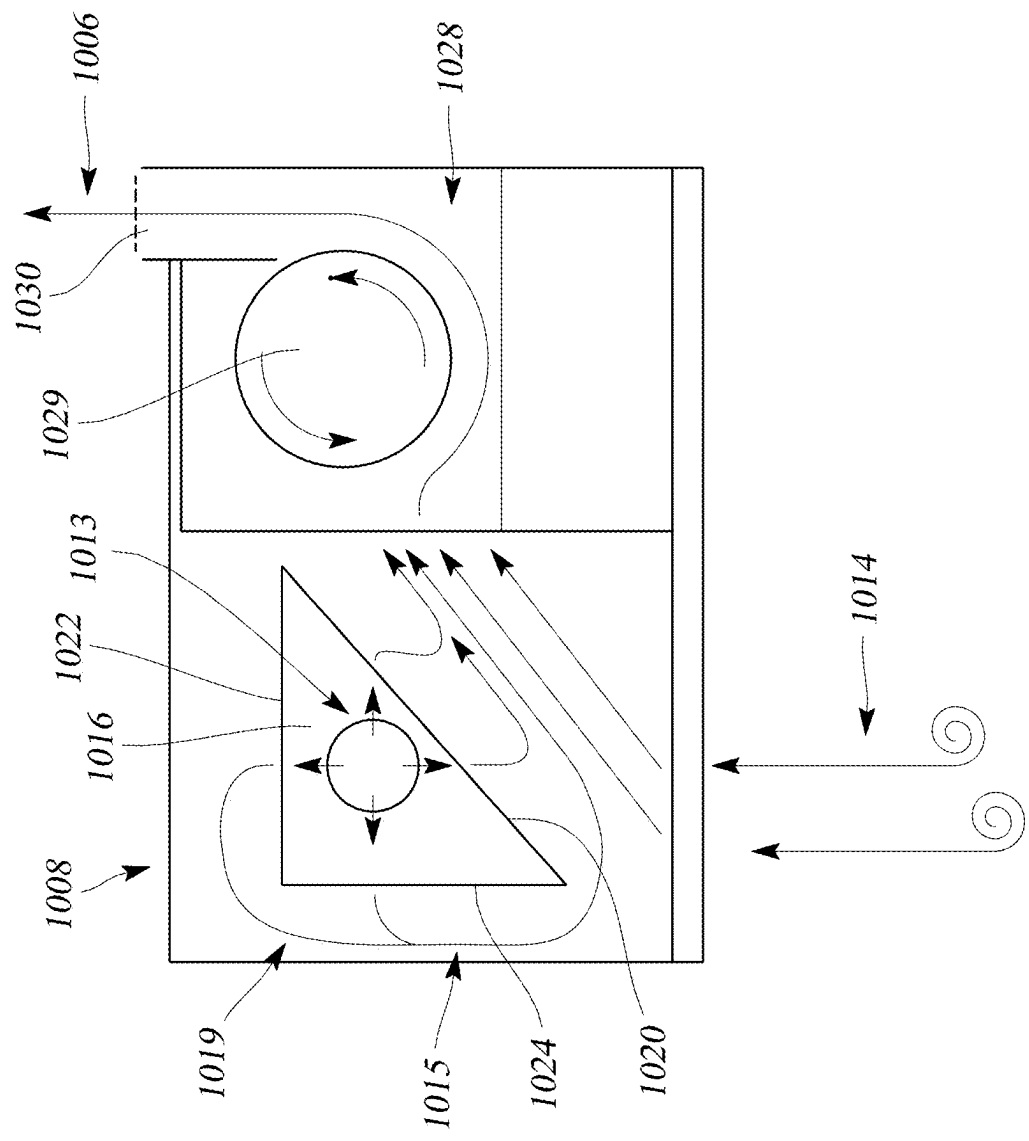


FIG. 10B

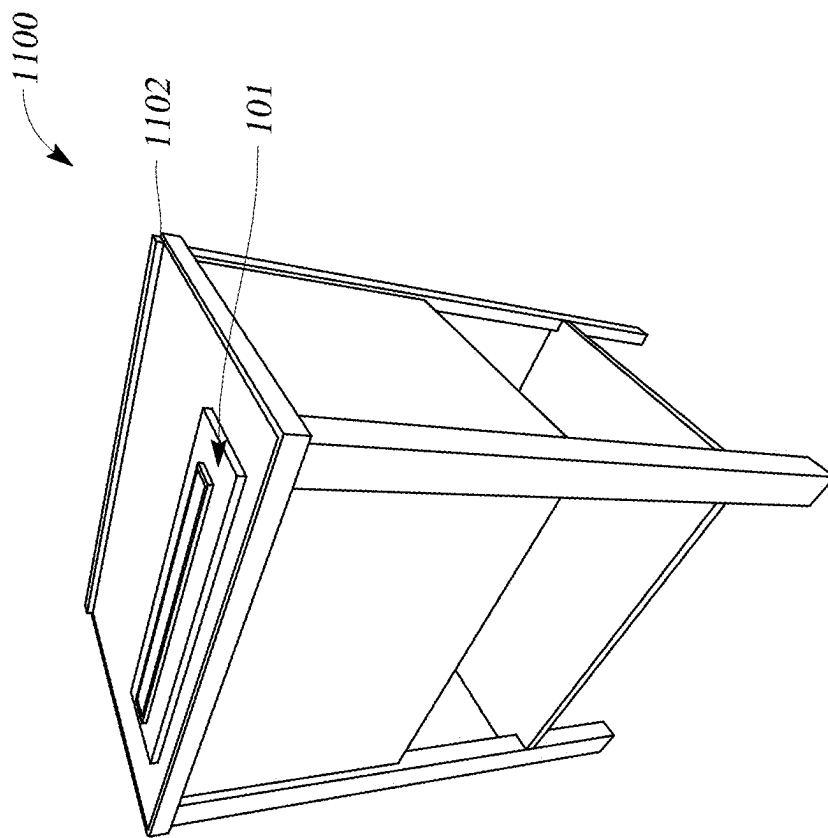


FIG. 11

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AIRFIELD SYSTEMS, DEVICES, AND METHODS**CROSS REFERENCE**

This application is a continuation of U.S. patent application Ser. No. 17/937,075 filed Sep. 30, 2022, now U.S. Pat. No. 11,788,747, and titled, "AIRFIELD SYSTEMS, DEVICES, AND METHODS," which claims priority benefit to U.S. Provisional Patent Application No. 63/263,843 filed Nov. 10, 2021, and titled, "AIRFIELD SYSTEMS," each of which are incorporated herein by reference in their entirety.

All applications for which a foreign or domestic priority claim is identified in the Application Data Sheet as filed with the present application are hereby incorporated by reference under 37 CFR 1.57.

FIELD

The disclosure relates generally to the field of airfield barriers. Specifically, the application relates to the field generating airfield barriers with purified air, devices for generating airfields, methods of using such devices, and methods for manufacturing such devices.

BACKGROUND

Infectious aerosols are generated by people who are infected with viruses or bacteria, including those having the common cold, influenza, and/or coronavirus infection. The aerosols from carriers of the infection may comprise a collection of pathogen-laden particles in air. These aerosol particles may deposit onto or be inhaled by others who are not infected causing new infections and for disease spread.

SUMMARY OF CERTAIN ASPECTS

Several embodiments disclosed herein pertain to airfield generating devices (e.g., airfield generators), methods of using the same, and methods for manufacturing the same. In several embodiments, these devices are useful in inhibiting or preventing the transmission of infectious diseases, pollutants, allergens, odors, etc. The prevention and/or reduction of transmission of infection and/or disease-causing aerosols is especially important in today's society. For example, the COVID-19 (the disease caused by the novel coronavirus SARS-CoV2) pandemic caused public activity to substantially halt in the United States and other countries around the world. The risks to health associated with SARS-CoV2 resulted in disruptions in normal daily life and caused a massive economic impact, resulting in mass layoffs and closures of businesses just a few weeks into the crisis. These shutdowns were especially detrimental to businesses where close interactions are the norm, such as restaurants, classrooms, libraries, etc. Several embodiments disclosed herein provide devices configured to address issues with the transmission of pathogens.

In several embodiments, the devices (e.g., airfield generators) disclosed herein generate an air barrier (e.g., an airfield) between subjects. In several embodiments, the air barrier comprises fast moving, clean air. In several embodiments, the air barrier separates one subject's air environment from a second subject's air environment. In several embodiments, the moving air in the generated air barrier captures and/or pushes contaminated aerosols from the first subject away from the second subject so that the second subject is

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not exposed to pathogens from the first subject. In several embodiments, the velocity of the air in the airfield is sufficiently high so as to substantially inhibit or prevent aerosols and/or pathogens from breath, sneezes, and/or coughs from passing through the airfield. In several embodiments, the velocity of the air in the airfield is sufficiently high so as to reduce to a safe and/or non-transmissible level aerosols and/or pathogens from breath, sneezes, and/or coughs from passing through the airfield. In several embodiments, the velocity of the air in the airfield is sufficiently high so as to reduce particulate levels in aerosols and/or pathogens from breath, sneezes, and/or coughs from passing through the airfield.

In several embodiments, the airfield generator may be supplied with clean air from an outside source of clean air (e.g., an air tank, etc.). Alternatively, in several embodiments, the airfield generator is adapted to generate clean air from contaminated air to generate the airfield and a clean air environment. For example, in several embodiments, the airfield generator may be equipped with one or more filters configured to remove pathogens from the air. These filters may be used to generate clean and/or pure air that is accelerated by the airfield generator to produce the airfield barrier. For example, the airfield generator may be configured to use recycled air from a room in which the airfield generator is located to generate the airfield. As will be appreciated, the airfield generator may also act as a whole room air purifier. As illustration, in several embodiments, the airfield generator may be configured to pull air from the room in which the airfield generator resides into an air intake of the airfield generator, to filter and/or purify the air, to accelerate the air to a velocity sufficient to provide an airfield, and to expel the air as an airfield through an outlet of the airfield generator. In several embodiments, the air of the airfield circulates back into the airfield generator for recycling, cleaning, and continued generation of the airfield. Alternatively or additionally, the airfield generator may be configured to work with an existing HVAC system for buildings or rooms. For example, the airfield generator may acquire air from a supply vent of an HVAC system in an existing room and may be configured to direct exhaust air (e.g., from the airfield) to an air intake vent for the HVAC system in the room.

In several embodiments, advantageously, the airfield generator is compact, modular, and/or portable. In several embodiments, the airfield generator is configured to be installed as part of a structure (and/or to be retrofitted to a structure) without effecting the normal use of the structure. In several embodiments, the airfield generator is configured to attach to and/or inhibit or prevent the transmission of pathogens across structures. In several embodiments, the structures may include tables, desks, cubbies, workstations, etc. In several embodiments, when adapted to be used with a particular structure (such as a desk, table, etc.), the airfield generator is compact enough to provide little or no interference with the space beneath the structure (e.g., the leg space under the desk or table).

Existing wind-generating units (e.g., motors, fans, etc.) that are sufficiently powerful to provide adequate velocity of air to be used as an airfield are unacceptably noisy. The noise level generated inhibits or prevents the use of such wind-generating units in situations where the use of airfield generator would be desired. For example, excessive noise in a restaurant or classroom is not desirable and inhibits or prevents subjects from engaging in conversations at normal volume levels (about 60-70 dB). In several embodiments, advantageously, the airfield generator comprises one or

more sound dampening features that absorb sound generated from the wind-generating unit (e.g., motor and/or fan) of the airfield generator. In several embodiments, the dampening feature reduces the noise level of the wind generating unit by equal to or at least about: 10 dB, 20 dB, 30 dB, 40 dB, 50 dB, or ranges including and/or spanning the aforementioned values.

In several embodiments, the airfield generator comprises a housing. In several embodiments, the airfield generator housing is configured to engage a filter system. In several embodiments, the airfield generator housing comprises a base. In several embodiments, the base comprises at least one air intake, an internal cavity providing an air passage through the base, and a filter system housing. In several embodiments, the filter system housing is provided within the air passage. In several embodiments, the airfield generator housing comprises or further comprises an outlet. In several embodiments, the outlet comprises an upwardly directed opening configured to generate an airfield. In several embodiments, the outlet is in fluid communication with the air passage of the base. In several embodiments, the airfield generator comprises a motor. In several embodiments, the motor is positioned at least partially within the internal cavity. In several embodiments, the motor is configured to generate an air flow from an ambient environment surrounding the airfield generator. In several embodiments, the motor generates airflow through the filter system and out of the airfield generator via the outlet of the housing thereby generating an airfield. In several embodiments, the airfield generated by the outlet provides a barrier (e.g., airfield) between a first side of the airfield and a second side of the airfield. In several embodiments, the airfield is configured to inhibit passage of aerosol particles through the airfield from the first side of the airfield to the second side of the airfield.

Any of the embodiments described above, or described elsewhere herein, can include one or more of the following features. No features are essential or critical.

In several embodiments, the airfield generator comprises the filter system while in other embodiments the filter system is separate from the airfield generator. In several embodiments, the filter system is configured to be engageable with the housing. In several embodiments, the filter system is configured to filter air passing into the airfield generator via the at least one air intake and through the airfield generator via the air passage.

In several embodiments, the outlet of the housing extends between a first side of the housing and a second side of the housing. In several embodiments, the outlet is a shape appropriate to generate first and second air environments that are substantially separated from one another by the airfield. In several embodiments, the outlet is a shape appropriate to generate air of sufficient velocity to provide the airfield. In several embodiments, the outlet has a dimension in one direction that is larger than its direction in a second direction. For example, in several embodiments, the outlet has a length measured in a direction proximal to one side of the housing and extending distally to a second side of the housing. In several embodiments, the outlet also has a width. In several embodiments, the length of the outlet is greater than its width. In several embodiments, the ratio of the length of the outlet to the width of the outlet is equal to or at least about: 30:1, 20:1, 15:1, 10:1, 15:2, 5:1, 5:2, and ratios between the aforementioned ratios. In several embodiments, the length of the outlet runs along a width of an object for which separate air environments are desired. For example, in several embodiments the length of the outlet is placed along the width of a table separating two equal or

non-equal portions of the table along the length of the table. In several embodiments, when users are seated at the heads of the table, the airfield provides a separation between the air environment of the subjects. In several embodiments, the outlet may comprise one or more fins (e.g., adjustable or nonadjustable fins). In several embodiments, adjustable fins may allow a user to direct the air of the airfield in a particular direction (e.g., away from a particular user, toward a vent intake of the room, etc.).

In several embodiments, the filter system comprises a plurality of filters. In several embodiments, the plurality of filters comprises at least a first filter and a second filter. In several embodiments, the first filter has a first parameter and the second filter has a second parameter. In several embodiments, the first parameter is different than the second parameter. In several embodiments, the first parameter and the second parameter comprise at least one of a filter size, a filtering capacity, or a filter shape. In several embodiments, the air intake of the airfield generator comprises a first and a second air intake. In several embodiments, the first filter is configured to engage with a first filter housing of the airfield generator housing and is configured to filter a first portion of air traveling into the base through the first air intake. For example, the first filter may engage a first filter dock of the housing. In several embodiments, the second filter is configured to engage with a second filter housing of the base, the second filter being configured to filter a second portion of air traveling into the base through the second air intake.

In several embodiments, the internal cavity of the base extends widthwise between a first side of the housing (or a corresponding first side of the base) and a second side of the housing (or corresponding second side of the base) providing a width of the internal cavity. In several embodiments, the internal cavity has a length that extends through the base from an entrance to an exit of the internal cavity. In several embodiments, the first filter is proximal to the entrance of the internal cavity. In several embodiments, the second filter is proximal to the exit of the internal cavity.

In several embodiments, the length of the outlet of the housing spans (or substantially spans) the width of the internal cavity and/or air passage of the base. In several embodiments, the length of the outlet is greater than the width of the internal cavity. In several embodiments, the length of the outlet is approximately the same size as the width of the internal cavity. In several embodiments, the ratio of the length of the outlet to the width of the internal cavity is equal to or at least about: 2:1, 3:2, 4:3, 5:4, 6:5, 1:1, or ratios between the aforementioned ratios.

In several embodiments, the second filter is a polygonal filter having a filtering portion extending along a length of the second filter between a proximal end portion and a terminal end portion. In several embodiments, the proximal end portion and the terminal end portion are a corresponding polygonal shape visible when viewed along the length of the second filter (e.g., a triangular shape, square shape, pentagonal shape, hexagonal shape, etc.). In several embodiments, the length of the second air filter is sufficient to span the width of the internal cavity and/or air passage of the housing. In several embodiments, the second filter housing is polygonal in shape (e.g., having a triangular shape, a square shape, a pentagonal shape, a hexagonal shape, etc.). In several embodiments, the polygonal shape of the second filter housing corresponds to the polygonal shape of the polygonal filter. In several embodiments, the polygonal shape of the second filter housing is apparent when viewing the base from its side. In several embodiments, the second filter housing is configured to receive the second filter

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through a filter housing aperture. In several embodiments, the filter housing aperture is polygonal. In several embodiments, the second filter may be slide into the second filter housing through the width of the base. In several embodiments, when placed in the second filter housing, the second filter spans the internal cavity such that air traveling through the second air intake is forced through the second filter and into the internal cavity.

In several embodiments, the second filter comprises at least a first side, a second side, and a third side defined by vertices of the polygonal shape, wherein the first side defines a first filtering surface of the filtering portion of the second filter, and wherein the second side defines at least a second filtering surface of the filtering portion of the second filter. In several embodiments, as air passes through the second filter, the air flows through at least the first filtering surface and/or the second filtering surface of the second filter. In several embodiments, the second filter comprises a triangular pocket filter. In several embodiments, the filter system comprises a triangular pocket filter.

In several embodiments, the housing comprises a first engagement mechanism, wherein the filter system comprises a second engagement mechanism, and wherein the first engagement mechanism is configured to removably receive the second engagement mechanism to removably engage the filter system with the housing.

In several embodiments, the motor comprises an electric motor, such as an inductive motor. The motor can comprise a fixed or variable speed motor. In some embodiments, the motor operates on AC power and in other embodiments the motor operates on DC power. In several embodiments, the motor is configured to generate an air flow of at least 370 cubic feet per minute. In several embodiments, the motor is configured to generate an air flow of equal to or at least about: 100 cubic feet per minute, 250 cubic feet per minute, 350 cubic feet per minute, 400 cubic feet per minute, 450 cubic feet per minute, 500 cubic feet per minute, 650 cubic feet per minute, 750 cubic feet per minute, 1000 cubic feet per minute, or ranges including and/or spanning the aforementioned values. For example, in several embodiments, the motor is configured to generate an air flow ranging from 100 cubic feet per minute to 1000 cubic feet per minute, from 350 cubic feet per minute to 400 cubic feet per minute, from 350 cubic feet per minute to 750 cubic feet per minute, etc.

In several embodiments, the upwardly directed opening is substantially vertical and/or is configured to direct air in a substantially vertical direction. In several embodiments, the outlet comprises a nozzle being configured to alter an air flow angle of the upwardly directed opening relative to a vertical direction. In several embodiments, the nozzle is configured to alter the air flow angle between 0 degrees and 45 degrees relative to the vertical direction.

In several embodiments, the housing is configured to seal the internal cavity.

In several embodiments, the airfield generator further comprises a sterilization system (e.g., other than the filter system). In several embodiments, the sterilization system is configured to sterilize at least one of the housing, the motor, the filter system housing or the filter system.

In several embodiments, the airfield generator further comprises at least one noise attenuation element. In several embodiments, the noise attenuation element is configured to reduce noise produced by the airfield generator.

In several embodiments, the housing is positioned on a support structure, and wherein the airfield generator is

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configured to generate the airfield such that the upwardly directed opening is angled relative to a top surface of the support structure.

In several embodiments, the airfield is generated using air from at least one of the first side of the airfield, the second side of the airfield, or both.

In several embodiments, the airfield generator is configured to reduce transmission of particulates sized 0.3 to 1 micron in the air at an efficiency of at least 75%, 80%, 90%, 95%, 97.5%, 99%, or 99.9%. In several embodiments, the airfield generator is configured to transmission of reduce particulates sized greater than 1 micron in the air at an efficiency of at least 75%, 80%, 90%, 95%, 97.5%, 99%, or 99.9%. In several embodiments, the particle reduction efficiency is accomplished at a flow rate of 100 cubic feet per minute, 250 cubic feet per minute, 350 cubic feet per minute, 400 cubic feet per minute, 450 cubic feet per minute, 500 cubic feet per minute, 650 cubic feet per minute, 750 cubic feet per minute, 1000 cubic feet per minute, or ranges including and/or spanning the aforementioned values. In several embodiments, the efficiency of reduction of particulate transmission may be measured across a given distance between a first point (where the particulate is generated) and a second point (where the amount of particulate is measured). To measure efficiency, the amount of particulate is measured at the second point in a system lacking an airfield generator. This amount of particulate is then compared to the amount of particulate measured at the second point in a second system having an airfield generator separating the first and second points. In several embodiments, the reduction in particle transmission includes particles generated from breath during respiration, talking, coughing, and/or sneezing.

In several embodiments, the airfield generator is configured to reduce incidences of infectious disease transfer, and wherein the infectious diseases is a common cold, influenza, and/or COVID.

Several embodiments pertain to an airfield generator comprising a housing comprising a base and an outlet. In several embodiments, the airfield generator comprises a first filter being configured to filter air passing through the first filter, the first filter comprises a first parameter. In several embodiments, the airfield generator comprises a second filter being configured to filter air passing through the second filter, the second filter comprising a second parameter, the second parameter being different than the first parameter. In several embodiments, the airfield generator comprises a motor being positioned at least partially within the housing. In several embodiments, the motor is configured to generate air flow from an ambient environment, through at least one of the first filter or the second filter, and through the outlet of the housing to generate an airfield, the airfield comprising air flow of filtered air traveling in an upward direction from the outlet of the housing.

In several embodiments, the first parameter and the second parameter comprise at least one of a filter size, a filtering capacity, or a filter shape. In several embodiments, the first filter is configured to engage with a first filter housing and is configured to filter air traveling into the base through a first air intake of the first filter housing. For example, the first filter may engage a first filter dock of the housing. In several embodiments, the second filter is configured to engage with a second filter housing of the base, the second filter being configured to filter air traveling into the base through a second air intake of the second filter housing.

In several embodiments, the second filter is a polygonal filter having a filtering portion extending along a length of

the second filter between a proximal end portion and a terminal end portion. In several embodiments, the second filter comprises at least a first side, a second side, and a third side defined by vertices of the polygonal shape, wherein the first side defines a first filtering surface of the filtering portion of the second filter, and wherein the second side defines at least a second filtering surface of the filtering portion of the second filter. In several embodiments, as air passes from the base to the outlet through the second filter, the air flows through at least the first filtering surface and/or the second filtering surface of the second filter. In several embodiments, the second filter comprises a triangular pocket filter. In several embodiments, the filter system comprises a triangular pocket filter.

In several embodiments, the housing comprises a first engagement mechanism, wherein the filter system comprises a second engagement mechanism, and wherein the first engagement mechanism is configured to removably receive the second engagement mechanism to removably engage the filter system with the housing.

In several embodiments, the motor comprises an inductive motor. In several embodiments, the motor is configured to generate an air flow of at least 370 cubic feet per minute.

In several embodiments, the upwardly directed opening is substantially vertical and/or is configured to direct air in a substantially vertical direction. In several embodiments, the outlet comprises a nozzle being configured to alter an air flow angle of the upwardly directed opening relative to a vertical direction. In several embodiments, the nozzle is configured to alter the air flow angle between 0 degrees and 45 degrees relative to the vertical direction.

In several embodiments, the housing is configured to seal the internal cavity.

In several embodiments, the airfield generator further comprises a sterilization system (e.g., other than the filter system). In several embodiments, the sterilization system is configured to sterilize at least one of the housing, the motor, the filter system housing or the filter system.

In several embodiments, the airfield generator further comprises at least one noise attenuation element. In several embodiments, the noise attenuation element is configured to reduce noise produced by the airfield generator.

In several embodiments, the housing is positioned on a support structure, and wherein the airfield generator is configured to generate the airfield such that the upwardly directed opening is angled relative to a top surface of the support structure.

In several embodiments, the airfield is generated using air from at least one of the first side of the airfield, the second side of the airfield, or both.

In several embodiments, the airfield generator is configured to reduce particulates sized 0.3 to 1 micron in the air at an efficiency of at least 75%, 80%, 90%, 95%, 97.5%, 99%, or 99.9%. In several embodiments, the airfield generator is configured to reduce particulates sized greater than 1 micron in the air at an efficiency of at least 75%, 80%, 90%, 95%, 97.5%, 99%, or 99.9%. In several embodiments, the particle reduction efficiency is accomplished at a flow rate of 100 cubic feet per minute, 500 cubic feet per minute, 650 cubic feet per minute, 1000 cubic feet per minute, 1500 cubic feet per minute, 2000 cubic feet per minute, 5000 cubic feet per minute, 7500 cubic feet per minute, or ranges including and/or spanning the aforementioned values.

In several embodiments, the airfield generator is configured to reduce incidences of infectious disease transfer, and wherein the infectious diseases is a common cold, influenza, and/or COVID.

Several embodiments pertain to an airfield generator comprising airfield generator comprising a housing comprising a base and an outlet. In several embodiments, the airfield generator comprises a filter being configured to filter air passing through the filter. In several embodiments, the airfield generator comprises an inductive motor being positioned at least partially within the housing, the motor being configured to generate air flow from an ambient environment, through the filter, and through the outlet of the housing to generate an airfield, the airfield comprising an airflow of filtered air traveling in an upward direction from the outlet of the housing.

In several embodiments, the filter is one of a plurality of filters. In several embodiments, the filter is a first filter and the plurality of filters comprises at least a second filter, the first filter having a first parameter, the second filter having a second parameter, and wherein the first parameter is different than the second parameter.

In several embodiments, the first parameter and the second parameter comprise at least one of a filter size, a filtering capacity, or a filter shape. In several embodiments, the first filter is configured to engage with a first filter housing and is configured to filter air traveling into the base through a first air intake of the first filter housing. For example, the first filter may engage a first filter dock of the housing. In several embodiments, the second filter is configured to engage with a second filter housing of the base, the second filter being configured to filter air traveling through a second air intake of the second filter housing.

In several embodiments, the second filter is a polygonal filter having a filtering portion extending along a length of the second filter between a proximal end portion and a terminal end portion. In several embodiments, the second filter comprises at least a first side, a second side, and a third side defined by vertices of the polygonal shape, wherein the first side defines a first filtering surface of the filtering portion of the second filter, and wherein the second side defines at least a second filtering surface of the filtering portion of the second filter. In several embodiments, as air passes through the second filter, the air flows through at least the first filtering surface and/or the second filtering surface of the second filter. In several embodiments, the second filter comprises a triangular pocket filter. In several embodiments, the filter comprises a triangular pocket filter.

In several embodiments, the housing comprises a first engagement mechanism, wherein the filter comprises a second engagement mechanism, and wherein the first engagement mechanism is configured to removably receive the second engagement mechanism to removably engage the filter with the housing.

In several embodiments, the airfield generator further comprises a cooling system to cool the motor. In several embodiments, the motor is configured to generate an air flow of at least 370 cubic feet per minute. In several embodiments, the motor is configured to generate an air flow of equal to or at least about: 100 cubic feet per minute, 250 cubic feet per minute, 350 cubic feet per minute, 400 cubic feet per minute, 450 cubic feet per minute, 500 cubic feet per minute, 650 cubic feet per minute, 750 cubic feet per minute, 1000 cubic feet per minute, or ranges including and/or spanning the aforementioned values.

In several embodiments, the upward direction is substantially vertical.

In several embodiments, the upwardly directed opening is substantially vertical and/or is configured to direct air in a substantially vertical direction. In several embodiments, the outlet comprises a nozzle being configured to alter an air

flow angle of the upwardly directed opening relative to a vertical direction. In several embodiments, the nozzle is configured to alter the air flow angle between 0 degrees and 45 degrees relative to the vertical direction.

In several embodiments, the housing is configured to seal the internal cavity.

In several embodiments, the airfield generator further comprises a sterilization system (e.g., other than the filter system). In several embodiments, the sterilization system is configured to sterilize at least one of the housing, the motor, and the filter (e.g., the first or second filter). In several embodiments, the sterilizing system may comprise, for example, a UV light.

In several embodiments, the airfield generator further comprises at least one noise attenuation element. In several embodiments, the noise attenuation element is configured to reduce noise produced by the airfield generator.

In several embodiments, the housing is positioned on a support structure, and wherein the airfield generator is configured to generate the airfield such that the upwardly directed opening is angled relative to a top surface of the support structure.

In several embodiments, the airfield is generated using air from at least one of the first side of the airfield, the second side of the airfield, or both.

In several embodiments, the airfield generator is configured to reduce particulates sized 0.3 to 1 micron in the air at an efficiency of at least 75%, 80%, 90%, 95%, 97.5%, 99%, or 99.9%. In several embodiments, the airfield generator is configured to reduce particulates sized greater than 1 micron in the air at an efficiency of at least 75%, 80%, 90%, 95%, 97.5%, 99%, or 99.9%. In several embodiments, the particle reduction efficiency is accomplished at a flow rate of 100 cubic feet per minute, 250 cubic feet per minute, 350 cubic feet per minute, 400 cubic feet per minute, 450 cubic feet per minute, 500 cubic feet per minute, 650 cubic feet per minute, 750 cubic feet per minute, 1000 cubic feet per minute, or ranges including and/or spanning the aforementioned values.

In several embodiments, the airfield generator is configured to reduce incidences of infectious disease transfer, and wherein the infectious diseases is a common cold, influenza, and/or COVID.

Several embodiments pertain to a method for reducing incidences of infectious disease transfer. In several embodiments, the method comprises obtaining an airfield generator. In several embodiments, the method comprises activating the airfield generator. In several embodiments, the method comprises causing the motor to generate an air flow from an ambient environment surrounding the airfield generator, through the filter system, and out the outlet of the housing, thereby generating an airfield. In several embodiments, the infectious disease is caused by one or more of a Rhinoviruses, Coronavirus, influenza virus types A, B, C, D, *Streptococcus pneumoniae*, *Haemophilus influenzae*, *Moraxella catarrhalis*, *Staphylococcus aureus*, other streptococci species, anaerobic bacteria, or gram negative bacteria.

Several embodiments pertain to a method for reducing symptoms of allergy. In several embodiments, the method comprises obtaining an airfield generator. In several embodiments, the method comprises activating the airfield generator. In several embodiments, the method comprises causing the motor to generate an air flow from an ambient environment surrounding the airfield generator, through the filter system, and out the outlet of the housing, thereby generating an airfield. In several embodiments, the allergy is a seasonal allergy or a food allergy.

Several embodiments pertain to a method for manufacturing an airfield generator. In several embodiments, the method comprises obtaining a housing. In several embodiments, the method comprises obtaining a filter system. In several embodiments, the method comprises obtaining a motor. In several embodiments, the method comprises assembling the housing with the motor. In several embodiments, the method comprises engaging the filter system with the housing. In several embodiments, the method comprises obtaining inserting the plurality of filters into the airfield generator.

Neither the preceding Summary nor the following Detailed Description purports to limit or define the scope of protection. The scope of protection is defined by the claims.

BRIEF DESCRIPTION OF THE DRAWINGS

The following drawings and the associated descriptions are provided to illustrate embodiments of the disclosure and do not limit the scope of the claims. Throughout the drawings, reference numbers are re-used to indicate correspondence between referenced elements.

FIGS. 1A-1H illustrate an exemplary airfield system in accordance with aspects of this disclosure.

FIG. 2A schematically illustrates a block diagram of an airfield generator in accordance with aspects of this disclosure.

FIG. 2B depicts a subsystem of airfield systems in accordance with aspects of this disclosure.

FIG. 3 depicts a flowchart illustrating a process of controlling an airfield generator in accordance with aspects of this disclosure.

FIGS. 4A-4E, 5A-5B, 6A-6C, 7, 8A-8D, and 9A-9J depict subsystems of airfield systems in accordance with aspects of this disclosure.

FIGS. 10A-10B depict an air flow path of an airfield generator in accordance with aspects of this disclosure.

FIG. 11 illustrates an exemplary airfield system in accordance with aspects of this disclosure.

DETAILED DESCRIPTION OF CERTAIN EMBODIMENTS

The disclosure describes various devices, systems, and methods for airfield systems and, in particular, airfield systems to generate separate zones of space using a filtered airfield. For example, the systems may filter an airfield using minimum efficiency reporting value (MERV) filters.

The disclosure will now be described with reference to the accompanying figures, wherein like numerals refer to like elements throughout. The following description is merely illustrative in nature and is in no way intended to limit the disclosure, its application, or uses. It should be understood that steps within a method may be executed in different order without altering the principles of the disclosure. Furthermore, embodiments disclosed herein can include several novel features, no single one of which is solely responsible for its desirable attributes or which is essential to practicing the systems, devices, and methods disclosed herein.

Generally, airfield generators of the disclosure may produce an airfield traveling at a sufficient speed (e.g., approximately 650 feet per minute and/or as disclosed elsewhere herein) across a set distance such that the airfield generator is configured to effectively increase a social distance between the two interacting people (e.g., people who are conversing, laughing, etc.) when the generator is placed in between the individuals. For example, the airfield generator

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may be configured to redirect any aerosols present in a first individual's breath (e.g., during talking, sneezing, coughing) such that the aerosols are directed away from another individual. For instance, the breath aerosol may be raised into a HVAC intake zone, rather than in the face of other people. In some instances, the airfield can include a volume of air that has been filtered through an optional filter set up, which can be utilized in multiple configurations to address different use cases. Generally, the airfield generators of the disclosure may also assist in increase air changes per hour, ACH change over, to decrease transmission of unfiltered air particulates, e.g., due to close proximity transmission.

In several embodiments, the social distance between individuals separated by the airfield generator (relative to those not separated by the airfield generator) is increased by a distance of equal to or at least about: 5 feet, 10 feet, 15 feet, 20 feet, or ranges including and/or spanning the aforementioned values. In several embodiments, the equivalent social distance is increased by a factor of equal to or at least about: 2 times (e.g., 2x), 3x, 4x, 5x, 10x, 15x, or 20x, or ranges including and/or spanning the aforementioned values.

A. OVERVIEW OF AIRFIELD SYSTEMS

FIGS. 1A-1G illustrate an exemplary airfield system 100 in accordance with aspects of this disclosure. FIGS. 1A-1G depict different views of the airfield system 100. As depicted in FIG. 1A, airfield system 100 includes an airfield generator 101 and a structure 102. The airfield generator 101 may generate an airfield 106A from outlet 104, as discussed below. For instance, airfield generator 101 may filter environmental air and output filtered air to generate the airfield 106A.

The airfield generator 101 may be positioned so that airfield 106A may separate distinct zones of space 106B and 106C, so that transmission of unfiltered air (and particulates therein) between the distinct zones of space 106B and 106C is reduced. For instance, the distinct zones of space 106B and 106C may be adjacent to the airfield generator 101 and separated by the airfield 106A for at least a defined length (e.g., at least a length of the outlet 104 of the airfield generator 101). The airfield 106A may pull ambient air in the distinct zones of space 106B and 106C into the airfield 106A increasing airflow and circulation in the distinct zones of space 106B and 106C creating an air dam.

As depicted in FIG. 1A, the airfield 106A may be generally directed in the z direction and extend along the x direction, to thereby separate the distinct zones 106B and 106C, for at least a portion along the x direction, in respective y directions. As used herein, the z direction may be a vertical direction (e.g., in a field of gravity), and the x direction and y direction may be lateral directions (referred to as front and back direction for the y direction, and left and right direction for the x direction). In some instances, the z direction may refer to a height of the airfield being generated at least partially in a direction of air flow, the x direction may refer to a length of the airfield being generated (e.g., along a major axis of the airfield), and the y direction may refer to a width or depth of the airfield (e.g., along a minor axis of the airfield).

Various embodiments of the airfield generator 101 as described herein may provide the benefits of producing a high-volume amount of compact air through the airfield 106A with the use of a generator 101 that is compact in size. The airfield 106A can be generated generally along the z direction in an upward direction.

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The outlet 104 of the airfield generator 101 may be various shapes to output the airfield 106A. FIGS. 1A-1G illustrate an outlet 104 with a generally elongated rectangular shape. However, it will be understood that the outlet 104 may be any size or shape suitable to generate an airfield. For example, the outlet 104 may comprise a generally curved shape. In some instances, the curve may be configured such that at least one of the individuals in one of the distinct zones 106B, 106C is located at a focal point of the curve. The outlet 104 may have a width 103 in the y direction. In some embodiments, the outlet 104 may have a constant width 103 across the outlet 104 in the x direction. In other embodiments, the width 103 may be different across the outlet 104 in the x direction. For example, the width 103 may be larger at the middle of the outlet 104 than the width 103 at the ends of the outlet 104 so the air speed is constant across the outlet 104 in the x direction.

In some embodiments, the outlet 104 may be a nozzle with a continuous opening. However, the outlet 104 may instead have a non-continuous opening, such as with a grate or other structure to inhibit or prevent foreign objects to enter the outlet 104, while still allowing a continuous airfield 106A to be output therefrom.

The outlet 104 may be adjustable to change one or more features of the airfield 106A (e.g., a direction, angle relative to the surface of the structure 102, size, air speed, volume of air generated, etc.). For instance, the outlet 104 may be adjustable to change an angle of air flow relative to a z axis, such as from forward to backward or backward to forward within a defined range of angles. For instance, the range may be $\pm 450^\circ$ relative to the z axis. The range may be $\pm 15^\circ$, $\pm 25^\circ$, $\pm 35^\circ$, 45° , 50° , $\pm 60^\circ$, $\pm 70^\circ$, $\pm 80^\circ$, or ranges including and/or spanning the aforementioned values. As an example, the outlet 104 may be hinged to thereby adjust the angle from the z direction at which the airfield 106A is projected into space, thereby adjusting the distinct zones of space 106B and 106C. In some embodiments, the outlet 104 may be adjustable manually or electronically via a controller 216 (described below with reference to FIG. 2A). In some embodiments, the outlet 104 may automatically move from forward to backward and backward to forward within at least a portion of the defined range of angles at a predetermined angular velocity and/or at predetermined intervals. In the case of a structure 102 with a surface (such as a table), the outlet 104 may protrude through the surface to a fixed height in the z direction from the surface. For instance, the protrusion of the outlet 104 may enable the outlet 104 to be adjusted through the defined range of angle without interfering with the surface.

Generally, the structure 102 may include various different forms of supports. As depicted in FIGS. 1A-1G, the structure 102 may be an item of furniture, such as a table. The structure 102 (e.g., table) may have a surface. The structure 102 may separate two or more groups of people in respective zones of space. However, one of skill in the art would recognize that the structure could alternatively be, for example, a counter (e.g., at a checkout/check-in of a business, at a bar, etc.), a mobile stand to support the airfield generator 101, a wall fixture, a barrier (e.g., between office desks, cubicles, etc.), a portion of an HVAC system, a ceiling, a drop ceiling, a portion of a vehicle, etc. Therefore, generally, the structure 102 may be a physical object that may fix and hold the airfield generator 101 in place, so that the airfield generator 101 may generate the airfield 106A and the distinct zones of space 106B and 106C. As an additional example, FIG. 11 depicts an alternative airfield system 1100. Airfield system 1100 may include the airfield generator 101

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with a different structure **1102**. The different structure **1102** may be an article of furniture, such as a podium (as depicted), or otherwise. In various embodiments, the structure **102** comprises, and/or the airfield generator **101** is comprised, in a desk, podium, counter, table, wall or pony wall, ceiling, floor, etc. In some embodiments, the structure **102** comprises, and/or the airfield generator **101** is comprised, in a vehicle, such as a car or airplane (e.g., to generate an airfield barrier between adjacent occupants or passengers).

While only one airfield generator **101**, one airfield **106A**, and two distinct zones of space **106B** and **106C** are depicted in FIG. 1A, one of skill in the art would recognize that multiple (e.g., two or more) airfield generators **101** may be arranged to generate multiple (e.g., two or more) airfields **106A** to generate a plurality of zones of space **106B** and **106C**. Generally, the arrangement of airfield generators **101** (and their respective airfields **106A**) may define boundaries of the plurality of zones of space **106B** and **106C**. For instance, airfield generators **101** (and their respective airfields **106A**) may be orthogonal to each other, or arranged at an acute or obtuse angle with respect to each other and spaced a part to define the boundaries of the plurality of zones of space **106B** and **106C**.

FIGS. 1B-1C depict features of various embodiments of the airfield generator **101** from below the surface of the structure **102**, from front and back views, respectively, of the airfield generator **101**. One of skill in the art would recognize that, when the structure **102** does not include a surface, the airfield generator **101** may have the same components, but the outlet **104** may be changed (e.g., shortened) as the protrusion through the surface may not be necessary. In particular, FIGS. 1B-1C depict a base **108**, a fan **109** (also called a blower), and a main filter housing **110** of airfield generator **101**. The fan **109** can comprise, for example, a centrifugal fan, axial fan, or otherwise. FIGS. 1D-1E depict features of various embodiments of the airfield generator **101** from below the surface of the structure **102**, from a left and right views of the airfield generator **101**. In particular, FIGS. 1B-1C depict a motor **114** and bypass inlets **116** of airfield generator **101**. FIGS. 1F-1G depict features of various embodiments of the airfield generator **101** from below the surface of the structure **102**. In particular, FIGS. 1F-1H depict various attachment systems **120**, **122** and **124** of airfield generator **101**.

The base **108** may have an interior volume to receive filtered air from a main filter (e.g., in the main filter housing **110**) and a rack filter **802** (e.g., via the bypass inlets **116** as illustrated in FIGS. 8A-8D) into at least one cavity (e.g., as illustrated in FIGS. 4A-4E) to pass the filtered air to the centrifugal fan **109**. The base **108** may also be configured to removably secure the centrifugal fan **109** and motor **114** to create a sealed interface therebetween during operation (e.g., from vibrations and attenuate noise). The base **108** may provide for a sealed interface to enclose and protect one or more components (e.g., the centrifugal fan **109** and the motor **114**) from the external environment. For example, the base **108**, when closed, may provide for an internal cavity that is water-proof (or at least water-resistant) to inhibit unintended fluid (e.g., liquids) from entering into the internal cavity of the base **108**.

As illustrated in FIGS. 1F-1G, the attachment systems **120** and **122** may secure the base **108** to the structure **102**. For instance, attachment systems **120** and **122** may be a bracket, which may be discontinuous (such as attachment system **120**) or continuous along a length of the base **108** (such as attachment system **122**). The attachment systems

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120 and **122** may use, e.g., fasteners to attach the base **108** to the structure **102**, but one of skill in the art would recognize that other approaches may be taken (e.g., adhesive, etc.).

In some embodiments, as shown in FIG. 1H, the attachment system **124** may secure the base **108** to the structure **102** via vibration dampeners **126**. The vibrations dampeners **126** may be straps, nylon straps, rubber, springs, wires, or any other vibration dampening connection. In some embodiments, the vibration dampeners **126** may be one or more pieces of rubber coupled to the base **108** and the structure **108**. In the embodiments, where the vibration dampeners **126** are straps or nylon straps, the vibration dampeners **126** may be secured to the base **108** via one or more connectors **128**. The connectors may be a metal plate screwed into the base **108**, or any other fastener for connecting the straps to the base **108**.

The vibration dampeners **126** may secure the base **108** to the structure **102** such that the base **108** is free floating. The vibration dampeners **126** may be coupled to the structure by connection system **130**. The connection system **130** may include a connecting portion **132** and an adjustment portion **134**. The connecting portion **132** may rotatably or movably couple the vibration dampeners **126** to the structure **102** such that when the base **108** moves or vibrates, the vibration dampeners **126** can move or vibrate relative to the structure **102** without transferring any movement or vibration to the structure **102**. In this way, when the motor **114** is powered and generating airfield **106A**, vibration or movement of the base **108** created by the motor **114** and the centrifugal fan **109** does not transfer, or is reduced from transferring, from the base **108** to the structure **102**.

The adjustment portion **134** may allow a user to change a length of the vibration dampeners **126**. The user may secure the base **108** to the structure **102** as shown in FIG. 1H so the outlet **104** is below the structure **102**, or the user may use the **134** to shorten a length of the vibration dampeners **126** so the outlet **104** may extend through an opening **136** above the structure **102**.

In some embodiments, the opening **136** in the structure **102** may be larger than the outlet **104** so that if the outlet **104** is above the structure **102**, when the base **108** vibrates or moves, the outlet **104** may vibrate or move without contacting the structure **102**.

In some embodiments, the structure **102** may be a ceiling or a drop ceiling, and the base **108** may be secured to a secondary structure so the base **108** can be hung (e.g., upside down) above the structure **102**. The secondary structure may be a surface above a drop ceiling, a portion of an HVAC system or a surface near the portion of the HVAC system, or any other structure above or near the structure **102**. The base **108** may be uncoupled to the structure **102** so the movement or vibration of the base **108** will not be transferred to the structure **102**. The vibration dampeners **126** may reduce or eliminate caused noise caused by movement or vibration that may be transferred from the base **108** to the secondary structure.

The vibration dampeners **126** may reduce or eliminate vibration or movement of the structure **102** and/or reduce or eliminate sound caused by the vibration or movement of the structure **102**.

In the embodiments where the structure **102** or the secondary structure are a portion of an HVAC system or a surface near the portion of the HVAC system, the airfield generator **101** may provide purification to the HVAC system, and the base **108** may be positioned so the airfield **106A** is directed into the HVAC system.

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With reference to FIG. 1B, the centrifugal fan 109 may receive filtered air from the at least one cavity, compress the filtered air to increase the airflow speed of the filtered air, output the compressed filtered air to the outlet 104 to thereby generate the airfield 106A. The motor 114 may control operation of the centrifugal fan 109 by rotating impellers 706 (e.g., as illustrated in FIG. 7) of the centrifugal fan 109 at fixed or various speeds via a shaft 912 (e.g., as illustrated in FIG. 9J). For instance, the motor 114 may cause the impellers 706 of the centrifugal fan 109 to rotate at a fixed rotation per minute (RPM). Alternatively, the motor 114 may cause the impellers 706 to rotate at various RPMs, such as from a first RPM to a second RPM to increase airflow speed of the airfield 106A from a first airflow speed to a second airflow speed.

Generally, the centrifugal fan 109 may extend axially along an axis of rotation of the impellers 706, with a first opening in a shroud of the centrifugal fan 109 to receive the filtered air and a second opening in the shroud to output the compressed filtered air to the outlet 104. The first and second openings may be substantially similar in length to each other and to the outlet 104. The centrifugal fan 109 may be positioned along one end of the base 108 to be secured to the base 108 by, e.g., fasteners. For instance, the centrifugal fan 109 may be adjacent to, abut, or overlap an edge of the base 108 on a front or rear of the base 108. The motor 114 may be attached to the centrifugal fan 109.

In some instances, the motor 114 can comprise an inductive or alternating-current (AC) motor. The inductive motor can advantageously increase the durability and/or power of the motor 114 to improve the filtration capability of the airfield system 100. For example, an increased power may permit the motor 114 to force an increased amount of air, relative to alternative motor designs, through higher quality filtration components. The higher quality filter components may include an increased number and/or rating of the filter. In some instances, the motor 114 may be configured to generate an airfield 106A with air passing through at a rate of about 370 cubic feet per minute. The motor 114, in some embodiments, may be configured to generate an airfield 106A with air passing through at a rate of about 275 cubic feet per minute if the air is being filtered through two separate MERV 8 filters. An inductive motor may advantageously provide a steady, reliable movement of air relative to alternative designs that may result in variability of the rate of air flow throughout use.

The main filter housing 110 may be configured to removably receive a main filter (e.g., a one or more filters) to filter a first portion of environmental air 112 and pass the filtered air to the at least one cavity via a main inlet 426 (e.g., as illustrated in FIG. 4D). Generally, the main filter housing 110 may be a cuboid shape that is generally rectangular with a height to receive the one or more filters in an ordered arrangement (e.g., as illustrated in FIG. 6A-6C). Generally, the one or more filters of the main filter may include at least one of a pre-screen, a charcoal filter, or one or more MERV filters. Details of the one or more filters and the ordered arrangement thereof are discussed below with respect to FIGS. 6A-6C.

The main inlet 426 may be positioned on an opposite end of the base 108 from the centrifugal fan 109. The main inlet 426 may be positioned on a bottom of the base 108 so that the first portion of environmental air 112 is drawn into the at least one cavity via the main inlet 426 in a vertical direction through the one or more filters of the main filter housing 110. The main inlet 426 may be formed in the bottom of the base 108 by walls 412-416 (e.g., as illustrated

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in FIG. 4B), cover 418 (e.g., as illustrated in FIG. 4C), and rack holder 420 (e.g., as illustrated in FIG. 4D) creating a seal between the sealed interior volume of the base 108 and the main filter housing 110.

The bypass inlets 116 may be openings in the walls 412-416 of the base 108 on respective lateral (e.g., left and right) sides. The bypass inlets 116 may receive a rack filter 802 (e.g., as illustrated in FIGS. 8A-8D) in rack holder 420 to filter a second portion of environmental air 118 and pass the filtered air to the at least one cavity. For instance, the bypass inlets 116 may be positioned on an opposite end of the base 108 from the centrifugal fan 109. As an example, the bypass inlets 116 may be generally triangular, so that generally triangular rack filters 802 may be inserted through the bypass inlets 116. The generally triangular rack filters 802 can be configured to create an airtight or substantially airtight seal with the structure of the bypass inlets 116. Moreover, the bypass inlets 116 may be positioned in a corner opposite the centrifugal fan 109, as the centrifugal fan 109 may (in the case of no bypass inlets 116 and rack filter 802) create a dead zone of filtered air in the at least one cavity due to vortices in the circular motion from the main inlet 426 to the first opening of the centrifugal fan 109. In this manner, the bypass inlets 116 and the rack filter 802 may increase the volumetric flow rate of filtered air, while avoiding a dead zone of filtered air in the at least one cavity. The rack filter 802 may be a MERV 13 level triangulated pocket filter. In some embodiments, the bypass inlets 116 may also draw air over the motor 114, thereby cooling the motor 114.

In some embodiments, the main filter housing 110 may be omitted (e.g., as illustrated in FIG. 4E), so that unfiltered air is received in the at least one cavity. In this case, the airfield 106A may still operate to reduce transmission of particulates as the airfield 106A may have an airspeed high enough to redirect unfiltered air from one zone of space to not enter another zone of space. For instance, this may reduce construction and operational cost of the airfield generator 101, while still providing a reduction in transmission of particulates between each zone of space. For instance, as a result of filtering the environmental air using only the bypass inlets 116 with the rack filter 802 and generating an airfield 106A, each zone of space may have reduced transmission of unfiltered air and reduce the effective social distance between different groups in each zone.

In some embodiments, the main filter housing 110 may be omitted for a polygonal filter 502 (e.g., as illustrated in FIG. 5A). In this case, the environmental air may be filtered but not as thoroughly as if the main filter housing 110 was used. For instance, this may reduce construction and operational cost of the airfield generator 101, while still providing filtered air to the at least one cavity. The airfield 106A may have an airspeed high enough to redirect unfiltered air from one zone of space to not enter another zone of space. Therefore, in this case, the particulates may be both filtered out of the air to be used as the airfield 106A (thereby reducing particulates in the environment) and redirected so as not to interact with a different zone of space. For instance, as a result of filtering the environmental air using the polygonal filter 502 and the bypass inlets 116 with the rack filter 802 and generating an airfield 106A, each zone of space may have reduced transmission of unfiltered air and reduce the effective social distance between different groups in each zone.

In some embodiments, the main filter housing 110 may be used to provide a higher level of filtration of the unfiltered air than using the polygonal filter 502. While this may cost

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more to construct and operate than the previous two embodiments, the increase in filtration may enable indoor activities with reduced transmission of particles between the zones of space. For instance, as a result of filtering the environmental air using the main filter housing **110** and the bypass inlets **116** with the rack filter **802** and generating an airfield **106A**, each zone of space may have reduced transmission of unfiltered air and reduce the effective social distance between different groups in each zone.

In some embodiments, the main filter housing **110** and/or the polygonal filter **502** may be removably engaged with the base **108**, so that a user of the airfield generator **101** may remove the main filter housing **110** and/or the polygonal filter **502**. For instance, the main filter housing **110** and/or the polygonal filter **502** may be interchangeable to interface with the main inlet **426**, or omitted entirely. Each of the main filter housing **110** and the polygonal filter **502** may have engagement mechanisms that correspond to an engagement mechanism on the base **108**. For instance, the user may modify the configuration depending on application. Moreover, the one or more filters of the main filter housing **110** may be replaced with a same or different filters, depending on application. For example, for a first application if the user wants a high level of airflow or high air speed through the outlet **104**, and a high level of filtration of air is not important for the first application, the user may replace the one or more filters of the main filter housing **110**, the polygonal filter **502**, and/or the rack filter **802** with alternative one or more filters of the main filter housing **110**, the polygonal filter **502**, and/or the rack filter **802** with a lower MERV rating to increase the level of airflow or air speed of the airfield generator **101**. For a second application, if a high level of filtration is important, and the level of airflow or air speed through outlet **104** is not important for the second application, the user may replace the one or more filters of the main filter housing **110**, the polygonal filter **502**, and/or the rack filter **802** with alternative one or more filters of the main filter housing **110**, the polygonal filter **502**, and/or the rack filter **802** with a higher MERV rating to increase the level of filtration of the airfield generator **101**. Furthermore, the one or more filters of the main filter housing **110**, the polygonal filter **502**, and the rack filter **802** may be removable to clean the various filters to ensure proper filtration.

In some embodiments, the airfield generator may have a housing, a filter system, and a motor. The housing may include: a base comprising an air intake, an internal cavity providing an air passage through the base, and a filter housing, the filter housing being within the air passage; and an outlet. The outlet can comprise an opening (e.g., an upwardly directed opening) configured to generate an airfield, the outlet being in fluid communication with the air passage of the base. The outlet can extend between a first side of the housing and a second side of the housing. The filter system may be engageable with the housing. The filter system may be configured to filter air passing through the airfield generator via the air passage. The motor may be positioned at least partially within the internal cavity. The motor may be configured to generate an air flow from an ambient environment surrounding the airfield generator, through the filter system, and out of the airfield generator via the outlet of the housing thereby generating an airfield. The airfield generated by the outlet may provide a barrier between a first side of the airfield and a second side of the airfield. The airfield may be configured to inhibit passage of aerosol particles through the airfield from the first side of the airfield to the second side of the airfield. Moreover, the housing may include a first engagement mechanism. The

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filter system may include a second engagement mechanism. The first engagement mechanism may be configured to removably receive the second engagement mechanism to removably engage the filter system with the housing.

In some embodiments, the filter system may include a plurality of filters. At least one of the filters may be insertable into the filter housing to filter the air through the air intake and at least another filter may be insertable into the filter system to filter a separate portion of air. In some embodiments, the plurality of filters may include at least a first filter having a first parameter and at least a second filter having a second parameter. The first parameter may be different than the second parameter. The first parameter and the second parameter comprise at least one of a filter size, a filtering capacity, or a filter shape. A filter size may indicate a weight or volume of the filter (e.g., in standard sizes or as physical attributes). Filter capacity may indicate a property of the filter to remove a certain percentage of particulates at various cubic feet per minute. For example, a filter capacity may be a filter's ability to capture particles of various sizes, such as at least 0.3 microns, at least 1 microns, at least 3 microns, etc. For instance, filter capacity may be indicated by a MERV rating, such as MERV 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, or 16. It will be understood that any component of any filter or filter system described herein may comprise any type of filter described herein or any combination thereof.

In some embodiments, the first filter may be configured to fit within the housing and may be configured to filter air traveling into the base through the air intake. In some embodiments, the second filter may be configured to fit within the filter housing of the base and be configured to filter air from the air passage prior expulsion through the outlet as the airfield.

In some embodiments, the first or second filter may be a polygonal filter having a filtering portion extending along a length of the first or second filter between a proximal end portion and a terminal end portion. The proximal end portion and the terminal end portion may be a corresponding polygonal shape visible when viewed along the length of the second filter. The first or second filter may comprise at least a first side, a second side, and a third side defined by vertices of the polygonal shape, where the first side defines a first filtering surface of the filtering portion of the first or second filter, and wherein the second side defines at least a second filtering surface of the filtering portion of the first or second filter. As air passes from the base to the outlet through the first or second filter, the air flows through at least the first filtering surface and the second filtering surface of the first or second filter. The first or second filter comprises a pocket filter with a polygonal shape, such as a triangular pocket filter. For example, the polygonal shape may comprise any one of a triangle, rectangle, pentagon, hexagon, or any of shape as desired.

FIG. 2 schematically illustrates a block diagram of an airfield generator **200** in accordance with aspects of this disclosure. For instance, the airfield generator **200** may be a part of the airfield system **100**, discussed above. As depicted in FIG. 2, the airfield generator **200** may include one or more of a main filter **204**, a rack filter **206**, at least one cavity **208**, a fan **210** (also called a blower), an outlet **212**, a controller **216**, sensor(s) **218**, sanitation system **220**, or any combination thereof. The airfield generator **200** may take in environmental air **202** and output an airfield of filtered air **214**, so as to generate at least two zones of space separated by the airfield of filtered air **214**. In operation, respective portions of environmental air **202** may be received by and filtered by

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the main filter **204** and the rack filter **206**, respectively. The filtered air may be directed into the at least one cavity **208** to be gathered in a sealed interior and received by the fan **210**. Fan **210** may compress the received filtered air and output the compressed air to the outlet **212**. The outlet **212** may output the airfield of filtered air **214**, so as to generate at least two zones of space separated by the airfield of filtered air **214**.

The controller **216** may control the fan **210**, based on data from the sensor(s) **218** and/or user inputs. For instance, the controller **216** may cause the fan **210** of the airfield generator **200** to turn on or turn off in response to a user input or a signal from one of the sensor(s) **218**. For instance, a first sensor of the sensor(s) **218** may monitor a temperature of a motor of the fan **210** (such as motor **114**), a second sensor of the sensor(s) **218** may monitor a current or voltage draw of the motor, a third sensor of the sensor(s) **218** may monitor a volumetric flow rate of the filtered air or of airfield of filtered air **214**, and/or a fourth sensor of the sensor(s) **218** may monitor a filter status of the main filter **204** and/or the rack filter **206**. The controller **216** may receive data from the sensor(s) **218** and determine (based on arbitrarily complex conditions, generally referred to as “error conditions” indicating “errors”) to issue alerts (such as replace filter(s)) or turn on or off the fan **210**.

As shown in FIG. 2B, the sensors **218** may be a kill switch and the outlet **212** may include a grate **219**. The grate **219** may have one or more portions that interact with the sensors **218** when the grate **219** is placed on the outlet **212** so the sensors **218** may detect whether the grate **219** is on the outlet **212**. In this way, when the sensors **218** do not detect the grate **219** is on the outlet **212**, the controller **216** may receive data from the sensors **218** indicating that there is no grate **219** on the outlet **212** and the controller **216** may turn off the fan **210** or cut off power to the fan **210**. When the sensors **218** detect the grate **219** is on the outlet **212**, the controller may receive data from the sensors **218** indicating that the grate is on the outlet **212** and the controller may turn on the fan **210** or send power to the fan **210**. During operation of the airfield generator **200**, removal of the grate can result in stoppage of the fan **210**, such as due to a hardwired electrical connection or a command from the controller **216**.

Moreover, the controller **216** may control the sanitization system **220** to perform additional air sanitization. For instance, the controller **216** may periodically (such as every set period of time) or continuously, cause the sanitization system **220** to perform air sanitization. The sanitization system **220** may be a UV sterilization system within a sealed enclosure of the airfield generator **200**. For instance, the sanitization system **220** may include one or more UV sterilization systems, and the one or more UV sterilization systems may be in the at least one cavity **208**, adjacent to the fan **210**, adjacent to the main filter **204**, and/or adjacent to the rack filter **206** to perform air sanitization in addition to the main filter and the rack filter. One of skill the art would recognize that the one or more UV sterilization systems may direct UV light to one or several of the at least one cavity **208**, the fan **210**, the main filter **204**, and the rack filter **206** at a same time, therefore reducing cost while ensuring filtered environmental air (or unfiltered portion of environment air when the main filter or polygonal filter is not included) is sanitized.

Generally, the controller **216** may have a user interface. The user interface may display information to users (e.g., status, data, etc.) of the airfield generator **200** and receive user inputs to control operations of the airfield generator **200** or multiple airfield generators **200**. One of skill in the art

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would recognize that the controller **216** may be a computer, micro-controller, etc. that executes software using a memory (storing data and instructions on a non-transitory compute readable medium) and a processor to execute the instructions to perform the various operations described herein.

B. EXAMPLE CONTROL PROCESS

FIG. 3 depicts a flowchart illustrating a process **300** of controlling an airfield generator in accordance with aspects of this disclosure. For instance, the airfield generator may be a part of the airfield system **100** or **200**, discussed above. The operations of the process **300** may be performed by a controller, such as controller **216**. As depicted in FIG. 3, the operations of the process **300** may start by receiving an instruction to turn on (Block **302**). For instance, the controller may receive a user input or sensor signal indicating an instruction to turn on.

The controller may then proceed to cause a motor of a fan to generate an airfield out of an outlet (Block **304**). For instance, the controller may transmit an instruction to the motor **114** to turn on and cause impellers **706** to rotate at an RPM to thereby draw environmental air **112** and **118** through various filters, such as the rack filter **802** and main filter of the main filter housing **110**, compress the filtered are, and output the airfield via the outlet **104**.

The controller may then proceed to monitor sensors and control a sanitization system (Block **306**). For instance, the controller may monitor the sensor(s) **218** to determine whether errors conditions are satisfied or not and control the UV sterilization systems to perform additional sterilization (on filtered or unfiltered air when the main filter or the triangular filter is not included).

The controller may then proceed to, if an error is detected, issue an alert and/or turn off (Block **308**). For instance, the controller may determine an error condition is satisfied based on sensor data and determine whether to issue an alert or turn off based on policies. For instance, if an error condition is not considered dangerous but preferred to be resolved (e.g., replace filters), the controller may issue an alert. On the other hand, if a motor temperature or an absence of the grate **219** (or some arbitrary complex conditional) indicates a dangerous circumstance, the controller may determine an instruction to turn off the motor.

The controller may then proceed to, responsive to an instruction to turn off, cause the motor of the fan to cease generating the airfield out of outlet (Block **310**). For instance, the controller may receive a user input to turn off or determine a dangerous condition (and thereby determine an instruction to turn off), and send an instruction to the motor to turn off.

C. EXAMPLE SUBSYSTEMS

FIGS. 4A-4E, 5A-5B, 6A-6C, 7, 8A-8D, and 9A-9D depict subsystems of airfield systems **100**, **200**, or **300** in accordance with aspects of this disclosure. For ease of reference, the following description will refer to features of airfield system **100**, but one of skill in the art would recognize that the features are applicable to airfield systems **200** or **300** as well.

FIG. 4A depicts the first portion of environmental air **112**, the second portion of environmental air **118** (on one lateral side), and the airfield **106A** of the airfield system **100**, without the structure **102**. Generally, the volumetric flow rate of the first portion of environmental air **112** and the second portion of environmental air **118** (from both lateral

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sides) may correspond to the volumetric flow rate of the airfield **106A**. Moreover, due to the relative differences in inlet areas of the bypass inlets **116** and the main inlet **426**, the volumetric flow rate of the first portion of environmental air **112** and the volumetric flow rate of the second portion of environmental air **118** (from both lateral sides) may not be substantively similar. Instead, the volumetric flow rate of the first portion of environmental air **112** may be a substantial proportion of the volumetric flow rate of the airfield **106A**. Therefore, the main filter may filter a substantial portion of the volumetric flow rate of the airfield **106A**. As the main filter can be composed of larger filters in both thickness and geometric shape (thereby, increasing surface area for filtering) and more filter layers (generally), the effective MERV level of the main filter may be higher, and in some embodiments substantially higher, than the rack filter. Therefore, the effective MERV level of the airfield system **100** may be greatly improved, with respect to the rack filter **802** alone, as a MERV level of system is based on (upstream versus downstream) rate of filtering at a specific volumetric flow rate. Therefore, in certain embodiments, certain versions of the main filter may be used to achieve a certain MERV level, while certain versions of the main filter, the polygonal filter **502**, or no main filter are used, based on circumstances to achieve different MERV levels.

FIG. 4B depicts the base **108** without cover **418** (see FIG. 4C) and rack holder **420** (see FIG. 4D). In particular, FIG. 4B illustrates the walls **412-416**, a first cavity **406** and a second cavity **408**, in relation to the centrifugal fan **109** to define the sealed interior volume of the base **108**. Generally, the walls **412-416** include a first and second side walls **412** and **414**, which may have the bypass inlets **116**, and an end wall **416**. The end wall **416** may be on an opposite end of the base **108** from the centrifugal fan **109**. The first and second side walls **412** and **414** and the end wall **416** may define the structure of the base **108** and ensure sealing to the centrifugal fan **109**, the cover **418**, and the rack holder **420**. The first and second side walls **412** and **414** may ensure sealing between the interior volume along an interface with the shroud of the centrifugal fan **109** and the first and second side walls **412** and **414**.

The second cavity **408** may generally correspond to the main inlet **426** and the bypass inlets **116**. For instance, the second cavity may accommodate the rack filter **802** and allow the first portion of environmental air **112** to mix with the second portion of environmental air **118** before they enter the first cavity **406**. The first cavity **406** may be provided to allow for filtered air to enter the first opening in the shroud of the centrifugal fan **109**. Therefore, the first cavity **406** may have a defined gap between the first opening in the shroud and the second cavity **408**.

FIG. 4C depicts the cover **418** covering the first cavity **406**. The cover **418** may seal the interior volume along an interface with the shroud of the centrifugal fan **109** and the walls **412-416**, for instance the first and second side walls **412** and **414**. The cover **418** may extend from the centrifugal fan **109** to the main inlet **426**.

FIG. 4D depicts the rack holder **420**. The rack holder **420** may include structure **422** and support rails **424**. The structure **422** may support the support rails **424** and ensure sealing around the bypass inlets **116** and the walls **412-416**. Generally, the structure **422** and the support rails **424** may define the bypass inlets **116** to accommodate the rack filter **802** and provide structural support to fix the rack filter **802** during operation of the airfield system **100**.

FIG. 4E depicts the main inlet **426**. In particular, the main inlet **426** may be formed in the structure **422** of the rack

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holder **420**. Notably, the main inlet **426** may be unblocked by the rack filter **802** within the interior volume. The rack filter **802** may be configured to reduce or avoid vortices in the circular motion from the main inlet **426** to the first opening of the centrifugal fan **109**, such as by the rack filter **802** having a triangular shape. Other shapes are contemplated too, such as rectangular, pentagonal, hexagonal, etc. Moreover, the main inlet **426** may be spaced apart (in the z direction) from the rack filter **802** in accordance with a height of the walls **412-416** and the support rails **424**, so as to not interfere with the vortices in the circular motion from the main inlet **426** to the first opening of the centrifugal fan **109**.

FIG. 5A depicts the polygonal filter **502** covering the main inlet **426**. The polygonal filter **502** may be a triangulated pocket filter or other polygonal shape. The pocket filter may be configured (e.g., by having a triangulated shape) to allow the filter to expand through pockets thereof to increase surface area to increase filtering surface area, versus a fixed shape filter of similar dimensions. The polygonal filter **502** may consist of MERV 13 filter formed into a triangular shape that generally extends along an axial direction of impellers **706**, so that unfiltered air may pass through the filter into the second cavity **408**.

The polygonal filter **502** may include a structure **502B** and a triangular filter having axial filter portion(s) **502A** and an end filter portion **502C**. The structure **502B** may provide a rigid exterior portion of the polygonal filter **502** with one or more openings to accommodate the axial filter portion(s) **502A** and the end filter portion(s) **502C**, to thereby filter environmental air passing through the one or more openings. The structure **502B** may be made of cardboard, plastic, metal, or other suitable materials, or combinations thereof. The axial filter portion(s) **502A** and the end filter portion(s) **502C** may be unitary in construction (e.g., a filter formed into a triangular cylinder shape with proximal and terminal ends made of filter material), to seal each of the one or more openings of the polygonal filter **502** and to filter air moving therethrough. In this manner, construction of the filter may be easier (by avoiding internal adhesion for each filter portion to each opening and coordination thereof) but may be more costly as portions of filter not adjacent to the one or more openings may provide only partial filtering efficiency (e.g., as they are adjacent to the structure **502B**). Alternatively, the axial filter portion(s) **502A** and the end filter portion(s) **502C** may be separate pieces of filter fixed to the structure **502B** to seal each of the one or more openings of the polygonal filter **502** and to filter air moving therethrough. In this manner, less filter material may be used, but construction complexity may be increased.

In some embodiments, each axial filter portion **502A** may correspond to an axially extending opening on a surface of the structure **502B** of the polygonal filter **502**. The axial filter portion(s) **502A** (and the corresponding openings associated therewith) may form a filtering portion of each face of the polygonal filter **502**. The structure **502B** may form a non-filtering portion of each face of the polygonal filter **502**. Generally, the filtering portion of each face may be smaller, equal to, or larger than a non-filtering portion of each face of the polygonal filter **502**. In particular, a ratio of the filtering portion of each face to the non-filtering portion of each face is equal to or at least about: 30:1, 20:1, 15:1, 10:1, 15:2, 5:1, 5:2, ratios between the aforementioned ratios. As noted above, each axial filter portion **502A** may expand through (e.g., in a radially outward manner) or expand away (e.g., in a radially inward manner) from its respective axially extending opening on the surface of the structure **502B** of

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the polygonal filter **502**. In this manner, each axial filter portion **502A** may increase an effective filter surface area, to thereby increase filtering capacity polygonal filter **502**.

The end filter portion(s) **502C** may correspond to proximal and terminal open ends of the polygonal filter **502**. The end filter portion **502C** may provide additional filtering surface area for the polygonal filter **502**, as environmental air may pass through the end filter portion **502C** separately from the axial filter portion **502A**.

In some embodiments, the polygonal filter **502** may include a first side, a second side, and a third side defined by vertices of the polygonal shape, where the first side defines a first filtering surface using a first axial filter portion **502A**, and where the second side defines at least a second filtering surface using a second axial filter portion **502A**. The third side may or may not include additional filter portions and may be designed to face the main inlet **426**. In this manner, the first and second sides may provide at least a certain amount of filtering, while the third side may provide an outlet of the polygonal filter **502** to the main inlet **426**. In the case that the third side also has additional filter portions, the third side may filter the air filtered by the first and second sides before entering the main inlet **426**, thereby increasing an effective MERV rating of the polygonal filter **502** (but also increasing a pressure requirement on the motor **114**).

In some embodiments, the filter **502** may comprise a triangulated pocket filter. The triangulated pocket filter may advantageously provide for a higher efficiency filtration system relative to alternative designs (e.g., a pleated filter system). In some embodiments, the filter **502** may comprise any polygonal shape suitable. For example, the polygonal shape may comprise any one of a triangle, rectangle, pentagon, hexagon, or any of shape as desired. As a particular example, the filter **502** may be an isosceles triangle with the first and second sides having a same length, and the third side having a length corresponding to the main inlet **426** (e.g., to cover the main inlet **426** and provide for engagement mechanisms of filter **502** corresponding to the engagement mechanism on the base **108**).

FIG. 5B depicts a main filter **504** covering the main inlet **426** to filter the first portion of environmental air **112**. The main filter **504** may correspond to the main filter housing **110** with one or more filters, discussed above. As depicted in FIG. 5B, there are none of the one or more filters included therein, so the interior may be seen. Details of the main filter **504** are discussed below with respect to FIGS. 6A-6C.

FIG. 6A depicts an outside of the main filter **504**. As depicted in **600A**, the main filter **504** includes an exterior grate **604**, a pre-screen **606**, and a door **608** with a hinge **610**. FIG. 6B depicts the door **608** opened by the hinge **610**, to expose a support **612** for exterior grate **604**, a first support **613**, a first filter **614**, a second support **615**, and a second filter **616**. FIG. 6C depicts an internal separator screen **618**.

The exterior grate **604** may retain the pre-screen **606** and provide a first structural interface with the environment so that pre-screen **606** and other filters are not dislodged. The exterior grate **604** may be metal or plastic and provide a large surface area for first portion of environmental air **112** to pass through the exterior grate **604**. The pre-screen **606** may be a charcoal filter to pre-filter unfiltered air. The pre-screen **606** may filter out larger particulates than the first filter **614** or second filter **616**.

The door **608** and the hinge **610** may operate together to hold the exterior grate **604**, the pre-screen **606**, the first filter **614**, and the second filter **616**, in place (when the door is closed), and provide access to the pre-screen **606**, the first filter **614**, and the second filter **616** to replace each of the

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pre-screen **606**, the first filter **614**, and the second filter **616** as operational use indicates necessary.

The support **612** for the exterior grate **604** may support the exterior grate **604** and define an opening of the door **608** when the hinge **610** opens the door **608**. The exterior grate **604** may be retained by structure (see, e.g., lip of main filter **504** above exterior grate **604**) and supported by the support **612** on at least one side.

The first support **613** and the second support **615** may define slots (of predetermined size) to support to first filter **614** and the second filter **616**. For instance, the first support **613** and the second support **615** may be spaced apart by a set distance, such as a standard size of filters for the first filter **614** and the second filter **616**. The first support **613** and the second support **615** may inform a user where to insert the first filter **614** and the second filter **616** and guide the insertion and removal of the first filter **614** and the second filter **616**. The first support **613** and the second support **615** may extend a length of first filter **614** and the second filter **616** from the opening of the door **608** to an opposite side of the main filter **504**, so as support or retain the first filter **614** and the second filter **616**.

The first filter **614** and the second filter **616** may be a same or different MERV levels. The first filter **614** and the second filter **616** may be a same or different thicknesses. The first filter **614** and the second filter **616** may be a same lateral size (e.g., a same cross-section), so that the unfiltered air passes through both. Similarly, the pre-screen **606** may be a same lateral size (e.g., a same cross-section), as the first filter **614** and the second filter **616**. For instance, the first filter **614** may be a MERV 8 filter and the second filter **616** may be a MERV 13 filter. One of skill in the art would recognize that various combinations of different MERV levels are possible, such as a MERV 8 and a MERV 8 filter, a MERV 13 and a MERV 13 filter, etc.

The internal separator grate **618** may be positioned behind support **612** for the exterior grate **604** to provide a zone of space for pre-filtered air to gather before passing through the first filter **614** and the second filter **616**. The internal separator grate **618** may be metal or plastic and provide a large surface area for first portion of environmental air **112** to pass through the internal separator grate **618**.

FIG. 7 depicts noise attenuation materials **702** and **704** of the at least one cavity in view of the impellers **706** of the first opening in the shroud of the centrifugal fan **109**. As discussed above, impellers **706** may compress the air (a mix of filtered and/or unfiltered, depending on configuration) to generate the airfield **106A** out of the outlet **104**. The noise attenuation materials **702** and **704** may line internal surface surfaces of the base **108** in the first cavity **406** and the second cavity **408**. The noise attenuation materials **702** and **704** may comprise rubber that is adhesively bound to portions of the first cavity **406** and the second cavity **408**.

FIGS. 8A-8D depict various rack filters, including a rack filter **802**, a rack filter **804**, a rack filter **806**, and a rack filter **808**. The rack filter **802** may include a triangulated pocket filter to allow the filter to expand through pockets thereof to increase surface area to increase filtering surface area, versus a fixed shape filter of similar dimensions. The rack filter **802** may consist of MERV 13 filter (or other MERV level filter) formed into a triangular shape that generally extends along an axial direction of impellers **706**, so that unfiltered air may pass through the bypass inlets **116** and through the filter into the second cavity **408**.

The rack filter **802** may include a structure **802B** and a triangular filter having axial filter portion(s) **802A** filtering environmental air coming into the structure **802B** by end

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openings **802C**. The structure **802B** may provide a rigid exterior portion of the rack filter **802** with one or more openings to accommodate the axial filter portion(s) **802A**, to thereby filter environmental air passing through the end openings **802C**, through the axial filter portion(s) **802A**, through the one or more openings, and into the second cavity **408**. The structure **802B** may be made of cardboard, plastic, metal, or other suitable materials, or combinations thereof. The axial filter portion(s) **802A** may be unitary in construction (e.g., a filter formed into a triangular cylinder shape with proximal and terminal ends left open to form the end openings **802C**), to seal each of the one or more openings of the rack filter **802** and to filter air moving therethrough. In this manner, construction of the filter may be easier (by avoiding internal adhesion for each filter portion to each opening and coordination thereof) but may be more costly as portions of filter not adjacent to the one or more openings may provide only partial filtering efficiency (e.g., as they are adjacent to the structure **802B**). Alternatively, the axial filter portion(s) **802A** may be separate pieces of filter fixed to the structure **802B** to seal each of the one or more openings of the rack filter **802** and to filter air moving therethrough. In this manner, less filter material may be used, but construction complexity may be increased. In some embodiments, the axial filter portion(s) **802A** may include charcoal, activated charcoal and/or activated carbon.

In some embodiments, each axial filter portion **802A** may correspond to an axially extending opening on a surface of the structure **802B** of the rack filter **802**. The axial filter portion(s) **802A** (and the corresponding openings associated therewith) may form a filtering portion of each face of the rack filter **802** that has an opening (e.g., at least one face has an opening, but one, two, or three (or more when the polygonal shape is not a triangle) may also have openings). The structure **802B** may form a non-filtering portion of each face of the rack filter **802** that has an opening. The structure **802B** may also form non-filtering portions on any face that does not have an opening. Generally, the filtering portion of each face of the rack filter **802** may be smaller, equal to, or larger than a non-filtering portion of each face of the rack filter **802**. In particular, a ratio of the filtering portion of each face to the non-filtering portion of each face is equal to or at least about: 30:1, 20:1, 15:1, 10:1, 15:2, 5:1, 5:2, and ratios between the aforementioned ratios. As noted above, each axial filter portion **802A** may expand through (e.g., in a radially outward manner) or expand away (e.g., in a radially inward manner) from its respective axially extending opening on the surface of the structure **802B** of the rack filter **802**. In this manner, each axial filter portion **802A** may increase an effective filter surface area, to thereby increase filtering capacity rack filter **802**.

The end openings **802C** may correspond to proximal and terminal open ends of the rack filter **802**. The end openings **802C** may form a part of an air intake of the airfield generator in conjunction with the bypass inlets **116**, so environmental air may pass through the end openings **802C** and then through the axial filter portion(s) **802A**.

In some embodiments, the rack filter **802** may include a first side, a second side, and a third side defined by vertices of the polygonal shape. In some embodiments, the first side defines a first filtering surface using a first axial filter portion **802A**, where the second side defines at least a second filtering surface using a second axial filter portion **802A**, and where the third side defines at least a third filtering surface using a third axial filter portion **802A**. In this manner, the first, second, and third sides may provide respective amounts of filtering (based on airflow geometry). In some embodi-

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ments, the first side defines a first filtering surface using a first axial filter portion **802A**, where the second side defines at least a second filtering surface using a second axial filter portion **802A**, and the third side defines non-filtering surface (e.g., with no opening in structure **802B**). In this manner, the first and second sides may provide respective amounts of filtering (based on airflow geometry). In some embodiments, the first side defines a first filtering surface using a first axial filter portion **802A**, and the second side and the third side define non-filtering surfaces (e.g., with no opening in structure **802B**). In this manner, the first and second sides may provide respective amounts of filtering (based on airflow geometry).

In some embodiments, the rack filter **802** may comprise a triangulated pocket filter. The triangulated pocket filter may advantageously provide for a higher efficiency filtration system relative to alternative designs (e.g., a pleated filter system). In some embodiments, the rack filter **802** may comprise any polygonal shape suitable. For example, the polygonal shape may comprise any one of a triangle, rectangle, pentagon, hexagon, or any of shape as desired. As a particular example, the rack filter **802** may be a right triangle with the first and second sides facing walls **412-416**, and the third side forming a hypotenuses therebetween. Generally, as discussed herein, the rack filter **802** may be inserted into the bypass inlets **116** of airfield generator **101** and secured in place to the support rails **424** (e.g., by a foam sealing or other sealing material (not depicted) that surrounds the structure **802B** between the structure **802B** and the structure of the bypass inlets **116**, where the sealing inhibits or prevents air from flowing into the base **108** between the structure **802B** and the structure of the bypass inlets **116** and does not block the end openings **802C**).

The rack filter **804**, the rack filter **806**, and the rack filter **808** may be alternative embodiments of the rack filter **802**. For ease of reference, only differences between each will be discussed, while similar structural and functional features are applicable to each.

The rack filter **804** may include a structure **804B** and a triangular filter having axial filter portion(s) **804A** filtering environmental air coming into the structure **804B** by end openings **804C**. In particular, the axial filter portion(s) **804A** may be more rigid than the axial filter portion(s) **802A** of rack filter **802**, so that each axial filter portion **804A** may expand through (e.g., in a radially outward manner) or expand away (e.g., in a radially inward manner) from its respective axially extending opening on the surface of the structure **802B** to a smaller degree than the axial filter portion(s) **802A** of the rack filter **802**. In this manner, wear and tear (due to movement of the axial filter portion(s) **804A** when pressure changes occur) may be reduced, but an increase in filtering surface may not be as large.

The rack filter **806** may include a structure **806B** and a triangular filter having axial filter portion(s) **806A** filtering environmental air coming into the structure **806B** by end openings **806C**. The rack filter **808** may include a structure **808B** and a triangular filter having axial filter portion(s) **808A** filtering environmental air coming into the structure **808B** by end openings **808C**. In particular, both the rack filter **806** and the rack filter **808** may have axial filter portions **806A**, **808A** and respective openings on two filtering surfaces, whereas the rack filters **802**, **804** may have axial filter portions **802A**, **804A** on a single surface thereof. Additionally, both the rack filter **806** and the rack filter **808** may have multiple (e.g., two or more) axial filter portions **806A**, **808A** on each filtering surface extending axially down the surface separated from each other by portions of

structure **806B**, **808B**, whereas rack filter **802**, **804** may have continuous axial filter portions **802A**, **804A**.

Moreover, the rack filters **806**, **808** may differ in certain respects. For instance, the rack filter **806** and the rack filter **808** may have a same or different number of axial filter portions **806A**, **808A** and respective openings on two filtering surfaces. The sizes (e.g., length and width of openings of the number of axial filter portions **806A**, **808A**) may be a same width or different. The separation spacing between the openings of the number of axial filter portions **806A**, **808A** may be a same separation spacing or different.

FIGS. **9A-9J** depict components of a thermal control system, such as a cooling system **902**, for airfield generators. For instance, the airfield system **100**, **200**, **300**, in any of the embodiments disclosed herein, may comprise the cooling system **902** to dissipate any heat being output from one or more components of the system and/or to reduce a temperature of one or more components. As described above the bypass inlets **116** may draw air over the motor **114** and/or draw hot air away from the motor **114**, thereby cooling the motor.

A schematic overview of the cooling system **902** is shown in FIG. **9J**. The system can include one or more heat exchanges (also called heat exchangers). In some instances, the cooling system **902** may comprise a first heat exchange **904** (as illustrated in FIG. **9A-9C**), a pump **908** (as illustrated in FIGS. **9E-9H**), and a second heat exchange **910** (as illustrated in FIG. **9I**). The first heat exchange **904** may be positioned at least partially around the motor **114** to function as a heat sink, so that heat from the motor **114** may be transferred to a coolant of the cooling system **902**. The pump **908** may cause coolant within the cooling system **902** to circulate between the first heat exchange **904** and the second heat exchange **910**. The second heat exchange **910** may be positioned, for example, in the first cavity **406** (as shown in FIG. **9J**) to transfer heat from the coolant of the cooling system **902** to filtered air as it enters the first opening in the shroud of the centrifugal fan **109**.

In some embodiments, the first heat exchange **904** may have a first inlet portion **904A**, a first outlet portion **904B**, and a first heat transfer portion **904C** connecting the first inlet portion **904A** and the first outlet portion **904B**. The first inlet portion **904A** may be connected to the pump **908** to receive accelerated coolant that was cooled by the second heat exchange **910**. The first outlet portion **904B** may be connected to the second heat exchange **910** to thereby provide heated coolant. The first heat transfer portion **904C** may be a coil wound around (in a first direction (e.g., clockwise) or a second direction (e.g., counterclockwise)) an outside surface of the motor **114** one or more times (e.g., a plurality of times successively along an axially direction of the motor **114**), so that heat from the motor **114** may be transferred to the coolant passing through the heat transfer portion **904C**.

As shown in FIGS. **9B** and **9C**, the first heat transfer portion **904C** may be separated from the outside surface of the motor **114** by a distance **904D** so the motor **114** can vibrate without contacting the first heat transfer portion **904C**. The distance **904D** can be an air gap and/or void. The first heat transfer portion **904C** may be coupled to the base **108** so the first heat transfer portion **904C** does not contact the outside surface of the motor **114**.

In some embodiments, the first heat transfer portion **904C** can include an interior screen **904C-1**, coil **904C-2**, a first cover **904C-3**, and a second cover **904C-4**. The interior screen **904C-1** may be a thin sheet of material between the motor **114** and the coil **904C-2** and may increase the heat

transfer from the motor **114** to the coil **904C-2**. The interior screen **904C-1** may include copper, aluminum, and/or any other material suitable for heat exchange.

The coil **904C-2** may be connected to the first inlet portion **904A** and the first outlet portion **904B** and may be wound around the interior screen **904C-1** one or more times. The coil **904C-2** may be tubing and may transport coolant around the motor **114** from the first inlet portion **904A** to the first outlet portion **904B**. As the coolant passes through the coil **904C-2** the coolant may draw heat away from the motor **114** and transport the heat away from the motor **114**.

The first cover **904C-3** may be wrapped around the outside of the coil **904C-2** and may keep heat from the motor **114** near the coil **904C-2** and away from the motor **114** to increase or maximize an amount of heat the coolant can transport away from the motor **114**. The first cover **904C-3** may include aluminum, copper, and/or any other material that may keep heat near the coil **904C-2**.

The second cover **904C-4** may be a thin sheet of material wrapped around the first cover **904C-3**. The second cover **904C-4** may be coupled to the first cover **904C-3** via a magnet or other securement mechanism. The magnet may be a magnet just strong enough to couple the second cover **904C-4** to the first cover **904C-3** such that the magnet does not affect the motor **114**. The second cover **904C-4** may be an insulator designed to keep heat from escaping outside of the first cover **904C-3**. The second cover **904C-4** may include Teflon or any other suitable insulating material.

The second heat exchange **910** may have a second inlet portion **910A**, a second outlet portion **910B**, and a second heat transfer portion **910C** connecting the second inlet portion **910A** and the second outlet portion **910B**. The second inlet portion **910A** may be connected to the first heat exchange **904** (e.g., the first outlet portion **904B**) to receive heated coolant that was heated by the first heat exchange **904**. The second outlet portion **910B** may be connected to the pump **908** to thereby provide cooled coolant to the pump. The second heat transfer portion **910C** may traverse the first cavity one or more times. For instance, the second heat transfer portion **910C** may traverse the first cavity one time, two times, three times, or generally, a plurality of times. In this manner, the second heat transfer portion **910C** may transfer heat from the coolant to filtered air as it enters the first opening in the shroud of the centrifugal fan **109**. In some embodiments, the second heat exchange **910** may also have a plurality of protrusions **910C**. The plurality of protrusions **910C** may increase a surface area of the second heat exchange **910**, to increase heat transference from the coolant to the filter air. For instance, the plurality of protrusions **910C** may be fins to interact with the filtered air without substantially constricting airflow into the first opening in the shroud of the centrifugal fan **109**.

The pump **908** may have a third inlet portion **908A**, a third outlet portion **908B**, an impeller **908C**, and an actuator system **908D**. The third inlet portion **908A** may be connected to the second outlet portion **910B** of the second heat exchange **910** to receive cooled coolant from the second heat exchange **910**. The third outlet portion **908B** may be connected to the first inlet portion **904A** of the first heat exchange **904** to provide accelerated, cooled coolant to the first heat exchange **904**. The impeller **908C** may accelerate the coolant received by the third inlet portion **908A** in accordance with the actuator system **908D**.

The actuator system **908D** may cause the impeller **908C** to rotate to thereby accelerate the coolant. For instance, the actuator system **908D** may be an inductive coil (e.g., a 120 volt alternating current coil) that is placed adjacent (e.g.,

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substantially aligned with and near to) a magnet **908C-1** of the pump **908**. The magnet **908C-1** may be attached by a spindle **908C-2** (e.g., made of plastic) to the impeller **908C**, to thereby cause the impeller **908C** to rotate in accordance with rotation of the magnet. The inductive coil may cause the magnet **908C-1** to rotate in accordance with electricity being applied to the inductive coil. In this manner, the coolant circuit (e.g., sequentially through each of the pump **908**, the first heat exchange **904**, the second heat exchange **910**, and back to the pump **908**) may be sealed from external actuation. The impeller **908C** may be located within a coolant housing **908E** within a path of the coolant.

In some embodiments, the coolant may be driven by a heat engine effect (e.g., thermodynamic gradient causing circulation). In various systems, the impeller **908C** and the actuator system **908D** may be omitted if the motor **114** is regulated to avoid outputting substantial heat.

Alternatively, the actuator system **908D** may be selectively activated to induce the motion of the impeller **908C** if the motor **114** is not currently outputting substantial heat. For instance, the controller **216** may monitor a temperature of the motor **114** (or an indication thereof, such as current drawn thereby) and control the actuator system **908D** to reduce circulation time, to thereby increase heat transference. Alternatively, the actuator system **904D** may be constantly activated (when the generator **101** is turned on), as the heat transference away the motor **114** may be necessary to regulate the temperature thereof. In this case, the controller **216** may be used to constantly activate the actuator system **904D**, or the actuator system **904D** may be directly hardwired to an electricity source of the generator **101** without the controller **216** controlling the actuator system **904D**.

In other embodiments, the first heat exchange **904** may instead have a different configuration. For instance, as illustrated in FIG. 9D, the first heat exchange **904** instead be a third heat exchange **906**. The third heat exchange **906** may include a fourth inlet portion **906A**, a fourth outlet portion **906B**, and a third heat exchange portion **906C**. The fourth inlet portion **906A** may correspond to the first inlet portion **904A** of the first heat exchange **904**, as described above. The fourth outlet portion **906B** may correspond to the first outlet portion **904B** of the first heat exchange **904**, as described above. The third heat exchange portion **906C** may be a coil wound around an outside surface of the motor **114**, so that heat from the motor **114** may be transferred to the coolant passing through the heat transfer portion **906C**.

The third heat exchange portion **906C** may include a plurality radial portions **906C-1**, a plurality of axial portions **906C-2**, and a plurality of circumferential portions **906C-3**. Each axial portion **906C-2** may extend from a proximal end portion of the motor **114** to a terminal end portion of the motor **114**. Each radial portion **906C-1** may extend radially outward then extend radially inward while extending circumferentially about a circumference defined substantially by the outer surface of the motor **114**. Each circumferential portions **906C-3** may extend circumferentially about the circumference while extending in a first axial direction then in a second axial direction opposite the first axial direction. For instance, each circumferential portions **906C-3** may connect two axial portions **906C-2** at the proximal end portion of the motor **114**, while each radial portion **906C-1** may connect two axial portions **906C-2** at the terminal end portion of the motor **114**.

The third heat exchange **906** may include a plurality of struts **906D**. Each strut **906D** may extend between adjacent portions of the third heat exchange **906** to increase heat

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transference between the motor **114** and the coolant. For instance, some struts **906D** may extend circumferentially about the circumference between two or more axial portions **906C-2**. Other struts **906D** may extend circumferentially about the circumference between end portions of a circumferential portion **906C-3** or a radial portion **906C-1**.

Generally, the coolant may be any suitable fluid to transfer heat. For instance, the coolant may be a non-hydrous fluid. The coolant may be non-toxic. The coolant may have a boiling point above temperatures the motor **114** may achieve. The first heat exchange **904** second heat exchange **910** and/or third heat exchange **906** may be any appropriate material, such as copper, aluminum, etc.

Therefore, the cooling system **902** may capture heat from the motor **114** (e.g., in an enclosed environment) and increase efficiency and safety of the airfield generator **101**. For instance, when the airfield generator **101** is placed in confined parameters (such as under counters or podiums), the heat from the motor **114** may be transferred to the filtered air (which is at ambient temperatures) and distributed away from the system in the airfield **106A** generated by the airfield generator **101**.

D. AIR FLOW PATH

FIGS. **10A** and **10B** depict an air flow path of airfield systems of the disclosure. With reference to FIG. **10A**, a first portion of environmental air **1012** may be drawn into a main filter housing **1010** through a main inlet **1026**. The first portion of environmental air **1012** may be drawn through one or more filters **614**, **616** (shown in FIG. **6B**) to filter the first portion of environmental air **1012** to make a first portion of filtered air **1014**. The first portion of filtered air **1014** may be drawn through a main outlet **1028** out of the main filter housing **1010**.

With reference to FIG. **10B**, the first portion of filtered air **1014** from the main outlet **1028** may be drawn through a secondary inlet **1030** into a base **1008**. The base **1008** may include a rack filter **1016**. A second portion of environmental air **1013** may be drawn into the base **1008** through one or more bypass inlets **116** (shown in FIGS. **1D** and **1E**) and through a rack filter **1016**. The rack filter **1016** may be one or more of the rack filters **802-808**. The rack filter **1016** may filter the second portion of environmental air **1013** to make a second portion of filtered air **1015**. The second portion of filtered air **1015** may exit the rack filter **1016** through a first side **1020**, a second side **1022**, and a third side **1024**. The second portion of filtered air **1015** that exits the rack filter **1016** through the second side **1022** and the third side **1024** may be drawn through a back portion **1019** of the base **1008**. The second portion of filtered air **1015** that passes through the back portion **1019** may make an airflow of an airfield **1006** laminar. The second portion of filtered air **1015** and the first portion of filtered air **1014** may combine into a filtered air flow **1026**. The filtered air flow **1026** may be drawn into a fan housing **1028** and forced through an outlet **1030** by fan **1029** to make the airfield **1006**. The outlet may be generally vertical and/or upwardly facing (e.g., in a directional generally perpendicular to and/or away from the floor).

The first portion of environmental air **1012**, the second portion of environmental air **1013** from a first of the one or more bypass inlets **116**, and the second portion of environmental air **1013** from a second of the one or more bypass inlets **116** may each have different flow rates. The airfield **1006** may have a flow rate higher than the first portion of environmental air **1012**, the second portion of environmental air **1013** from the first of the one or more bypass inlets **116**,

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and/or the second portion of environmental air **1013** from the second of the one or more bypass inlets **116**. For example, the first portion of environmental air **1012** may have a flow rate of less than or equal to about 230 ft/min, the second portion of environmental air **1013** from the first of the one or more bypass inlets **116** may have a flow rate of less than or equal to about 630 ft/min, and the second portion of environmental air **1013** from the second of the one or more bypass inlets **116** may have a flow rate of less than or equal to about 550 ft/min and the airfield **1006** may have a flow rate of at least about 900 ft/min.

E. FURTHER ASPECTS

Generally, airfield systems of the disclosure, including airfield systems **100**, **200**, or **300**, may be manufactured and/or assembled. For instance, a method to manufacture the airfield system **100** may include: obtaining a housing, a filter system, a motor of an airfield system, where the filter system is engageable with the housing; obtaining a plurality of filters; assembling the housing with the motor; engaging the filter system with the housing; and inserting the plurality of filters.

Moreover, airfield systems of the disclosure, including airfield systems **100**, **200**, or **300**, may reduce the incidences of infectious disease transfer. For instance, a method to reduce the incidences of infectious disease transfer may include: receiving an instruction to operate an airfield generator of airfield systems **100**, **200**, or **300**; and causing the motor to generate an air flow from an ambient environment surrounding the airfield generator, through the filter system, and out the outlet of the housing, thereby generating an airfield. The infectious diseases may be a common cold, influenza, and/or COVID. The infectious disease may be caused by one or more of a Rhinoviruses, Coronavirus, influenza virus types A, B, C, D, *Streptococcus pneumoniae*, *Haemophilus influenzae*, *Moraxella catarrhalis*, *Staphylococcus aureus*, other streptococci species, anaerobic bacteria, or gram negative bacteria.

Further, airfield systems of the disclosure, including airfield systems **100**, **200**, or **300**, may reduce symptoms of allergy. For instance, a method to reduce symptoms of allergy may include: receiving an instruction to operate an airfield generator of airfield systems **100**, **200**, or **300**; and causing the motor to generate an air flow from an ambient environment surrounding the airfield generator, through the filter system, and out the outlet of the housing, thereby generating an airfield. The allergy may be a seasonal allergy or a food allergy.

Airfield generators, of airfield systems **100**, **200**, or **300**, may generate the airfield using air from at least one of the first side of the airfield, the second side of the airfield, or both. The airfield generators may reduce particulates sized 0.3 to 1 micron in the air at an efficiency of at least 75%, and particulates sized greater than 1 micron in the air at an efficiency of at least 90%, at a specific cubic feet per minute. The airfield generators may reduce incidences of infectious disease transfer. For instance, the infectious diseases may be a common cold, influenza, and/or COVID.

Many other variations than those described herein will be apparent from this disclosure. For example, certain acts, events, or functions of any of the algorithms described herein can be performed in a different sequence, can be added, merged, or left out altogether (for example, not all described acts or events are necessary for the practice of the algorithms). Moreover, acts or events can be performed concurrently, for example, through multi-threaded process-

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ing, interrupt processing, or multiple processors or processor cores or on other parallel architectures, rather than sequentially. In addition, different tasks or processes can be performed by different machines and/or computing systems that can function together.

It is to be understood that not necessarily all such advantages can be achieved in accordance with any particular example of the examples disclosed herein. Thus, the examples disclosed herein can be embodied or carried out in a manner that achieves or optimizes one advantage or group of advantages as taught herein without necessarily achieving other advantages as may be taught or suggested herein.

The various illustrative logical blocks, modules, and algorithm steps described in connection with the examples disclosed herein can be implemented as electronic hardware, computer software, or combinations of both. To clearly illustrate this interchangeability of hardware and software, various illustrative components, blocks, modules, and steps have been described above generally in terms of their functionality. Whether such functionality is implemented as hardware or software depends upon the particular application and design constraints imposed on the overall system. The described functionality can be implemented in varying ways for each particular application, but such implementation decisions should not be interpreted as causing a departure from the scope of the disclosure.

The various illustrative logical blocks and modules described in connection with the examples disclosed herein can be implemented or performed by a machine, such as a general purpose processor, a digital signal processor (DSP), an application specific integrated circuit (ASIC), a field programmable gate array (FPGA) or other programmable logic device, discrete gate or transistor logic, discrete hardware components, or any combination thereof designed to perform the functions described herein. A general purpose processor can be a microprocessor, but in the alternative, the processor can be a controller, microcontroller, or state machine, combinations of the same, or the like. A processor can include electrical circuitry or digital logic circuitry configured to process computer-executable instructions. In another example, a processor can include an FPGA or other programmable device that performs logic operations without processing computer-executable instructions. A processor can also be implemented as a combination of computing devices, for example, a combination of a DSP and a microprocessor, a plurality of microprocessors, one or more microprocessors in conjunction with a DSP core, or any other such configuration. A computing environment can include any type of computer system, including, but not limited to, a computer system based on a microprocessor, a mainframe computer, a digital signal processor, a portable computing device, a device controller, or a computational engine within an appliance, to name a few.

The steps of a method, process, or algorithm described in connection with the examples disclosed herein can be embodied directly in hardware, in a software module stored in one or more memory devices and executed by one or more processors, or in a combination of the two. A software module can reside in RAM memory, flash memory, ROM memory, EPROM memory, EEPROM memory, registers, hard disk, a removable disk, a CD-ROM, or any other form of non-transitory computer-readable storage medium, media, or physical computer storage known in the art. An example storage medium can be coupled to the processor such that the processor can read information from, and write information to, the storage medium. In the alternative, the storage medium can be integral to the processor. The storage

medium can be volatile or nonvolatile. The processor and the storage medium can reside in an ASIC.

The apparatuses and methods described herein may be implemented by one or more computer programs executed by one or more processors. The computer programs include processor-executable instructions that are stored on a non-transitory tangible computer readable medium. The computer programs may also include stored data. Non-limiting examples of the non-transitory tangible computer readable medium are nonvolatile memory, magnetic storage, and optical storage.

Some embodiments have been described in connection with the accompanying drawings. The figures are drawn and/or shown to scale, but such scale should not be interpreted as limiting, since dimensions and proportions other than what are shown are contemplated and are within the scope of the disclosed invention. Distances, angles, etc. are merely illustrative and do not necessarily bear an exact relationship to actual dimensions and layout of the devices illustrated. Components can be added, removed, and/or rearranged. Further, the disclosure herein of any particular feature, aspect, method, property, characteristic, quality, attribute, element, or the like in connection with various embodiments can be used in all other embodiments set forth herein. Additionally, any methods described herein may be practiced using any device suitable for performing the recited steps.

F. TERMINOLOGY

Terms of orientation used herein, such as “top,” “bottom,” “horizontal,” “vertical,” “longitudinal,” “lateral,” and “end” are used in the context of the illustrated embodiment. However, the present disclosure should not be limited to the illustrated orientation. Indeed, other orientations are possible and are within the scope of this disclosure. Terms relating to circular shapes as used herein, such as diameter or radius, should be understood not to require perfect circular structures, but rather should be applied to any suitable structure with a cross-sectional region that can be measured from side-to-side. Terms relating to shapes generally, such as “circular” or “cylindrical” or “semi-circular” or “semi-cylindrical” or any related or similar terms, are not required to conform strictly to the mathematical definitions of circles or cylinders or other structures, but can encompass structures that are reasonably close approximations. The terms “up” and “down” (and related terms) are with reference to the pull of Earth’s gravity.

The terms “approximately,” “about,” and “substantially” as used herein represent an amount close to the stated amount that still performs a desired function or achieves a desired result. For example, in some embodiments, as the context may dictate, the terms “approximately,” “about,” and “substantially” may refer to an amount that is within less than or equal to 10% of the stated amount. The term “generally” as used herein represents a value, amount, or characteristic that predominantly includes or tends toward a particular value, amount, or characteristic. As an example, in certain embodiments, as the context may dictate, the term “generally parallel” can refer to something that departs from exactly parallel by less than or equal to 20 degrees.

Conditional language used herein, such as, among others, “can,” “might,” “may,” “for example,” and the like, unless specifically stated otherwise, or otherwise understood within the context as used, is generally intended to convey that certain examples include, while other examples do not include, certain features, elements and/or states. Thus, such

conditional language is not generally intended to imply that features, elements and/or states are in any way required for one or more examples or that one or more examples necessarily include logic for deciding, with or without author input or prompting, whether these features, elements and/or states are included or are to be performed in any particular example. The terms “comprising,” “including,” “having,” and the like are synonymous and are used inclusively, in an open-ended fashion, and do not exclude additional elements, features, acts, operations, and so forth. Also, the term “or” is used in its inclusive sense (and not in its exclusive sense) so that when used, for example, to connect a list of elements, the term “or” means one, some, or all of the elements in the list. Further, the term “each,” as used herein, in addition to having its ordinary meaning, can mean any subset of a set of elements to which the term “each” is applied.

Disjunctive language such as the phrase “at least one of X, Y, or Z,” unless specifically stated otherwise, is otherwise understood with the context as used in general to present that an item, term, etc., may be either X, Y, or Z, or any combination thereof (for example, X, Y, and/or Z). Thus, such disjunctive language is not generally intended to, and should not, imply that certain examples require at least one of X, at least one of Y, or at least one of Z to each be present.

Unless otherwise explicitly stated, articles such as “a” or “an” should generally be interpreted to include one or more described items. Accordingly, phrases such as “a device configured to” are intended to include one or more recited devices. Such one or more recited devices can also be collectively configured to carry out the stated recitations. For example, “a processor configured to carry out recitations A, B and C” can include a first processor configured to carry out recitation A working in conjunction with a second processor configured to carry out recitations B and C.

G. SUMMARY

Various embodiments and examples of airfield systems, devices, and methods have been disclosed. Although the systems, devices, and methods have been disclosed in the context of those embodiments and examples, and the above detailed description has shown, described, and pointed out novel features as applied to various examples, it will be understood that various omissions, substitutions, and changes in the form and details of the disclosed technology can be made without departing from the spirit of the disclosure. As will be recognized, the embodiments described herein can be embodied within a form that does not provide all of the features and benefits set forth herein, as some features can be used or practiced separately from others. This disclosure expressly contemplates that various features and aspects of the disclosed embodiments can be combined with, or substituted for, one another. Accordingly, the scope of this disclosure should not be limited by the particular disclosed embodiments described above, but should be determined only by a fair reading of the claims that follow.

What is claimed is:

1. An airfield generator comprising:

a motor configured to drive a blower;

an internal cavity configured to convey an airflow generated by the blower, wherein the blower is positioned in the internal cavity and the motor is positioned outside of the internal cavity; and

a cooling system configured to cool the motor, the cooling system comprising:

a first heat exchange positioned around the motor of the airfield generator;

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- a second heat exchange associated with the internal cavity of the airfield generator, wherein the first heat exchange is in fluid communication with the second heat exchange; and
- a pump configured to circulate a heat exchange fluid through the cooling system between the first heat exchange and the second heat exchange,
- wherein the first heat exchange is configured to transfer heat from the motor of the airfield generator to the heat exchange fluid, and the second heat exchange is configured to transfer heat from the heat exchange fluid to a portion of the airflow in the internal cavity of the airfield generator.
2. The airfield generator of claim 1, further comprising one or more sensors configured to detect a temperature of the motor of the airfield generator or the heat exchange fluid.
3. The airfield generator of claim 2, further comprising a controller configured to control a speed of the pump based on the temperature of the motor or the heat exchange fluid.
4. The airfield generator of claim 1, wherein the first heat exchange is separated from an outside surface of the motor.
5. The airfield generator of claim 1, wherein the first heat exchange comprises a coil around the motor.
6. The airfield generator of claim 1, wherein the first heat exchange comprises an interior screen, a first cover, and a second cover, wherein the interior screen is positioned between the motor of the airfield generator and a first heat transfer portion of the first heat exchange, wherein the first cover is positioned over the first heat transfer portion, and wherein the second cover is positioned over the first cover.
7. The airfield generator of claim 6, wherein the interior screen is configured to increase heat transfer from the motor.
8. The airfield generator of claim 1, wherein the second heat exchange comprises a plurality of protrusions.
9. The airfield generator of claim 1, wherein the pump comprises an impeller.
10. The airfield generator of claim 1, wherein the second heat exchange traverses the internal cavity a plurality of times.
11. An airfield generator comprising:
- a blower;
 - a motor configured to drive the blower;
 - an internal cavity configured to convey an airflow generated by the blower; and
 - a cooling system configured to cool the motor, the cooling system comprising:
 - a first heat exchange positioned around the motor;

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- a second heat exchange positioned in the internal cavity;
 - pipings fluidly connecting the first heat exchange and the second heat exchange; and
 - a pump configured to convey a heat exchange fluid through the piping between the first heat exchange and the second heat exchange,
- wherein the first heat exchange is configured to transfer heat from the motor to the heat exchange fluid, and the second heat exchange is configured to transfer heat from the heat exchange fluid to a portion of the airflow, thereby heating the portion of the airflow.
12. The airfield generator of claim 11, further comprising one or more sensors configured to detect a temperature of the heat exchange fluid.
13. The airfield generator of claim 12, further comprising a controller configured to control a speed of the pump based on the temperature of the motor of the heat exchange fluid.
14. The airfield generator of claim 11, wherein the first heat exchange is separated from an outside surface of the motor.
15. The airfield generator of claim 11, wherein the first heat exchange comprises coil around the motor of the airfield generator.
16. The airfield generator of claim 11, wherein the first heat exchange comprises an interior screen, a first cover, and a second cover, wherein the interior screen is positioned between the motor of the airfield generator and a first heat transfer portion of the first heat exchange, wherein the first cover is positioned over the first heat transfer portion, and wherein the second cover is positioned over the first cover.
17. The airfield generator of claim 16, wherein the interior screen is configured to increase heat transfer from the motor to the first heat transfer portion.
18. The airfield generator of claim 11, wherein the second heat exchange comprises a plurality of protrusions that are configured to increase heat transfer from the second heat exchange.
19. The airfield generator of claim 11, wherein the pump comprises an impeller.
20. The airfield generator of claim 11, wherein the second heat exchange traverses the internal cavity a plurality of times.

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